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Graph Neural Networks for Multivariate Time Series Activity Recognition

A Controlled Benchmark of Temporal Graph Neural Networks and
Graph-Construction Strategies against Sequence-Model Baselines

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Abstract

Classifying multivariate time series (MTS) is a core machine-learning problem in which several variables are recorded simultaneously and are typically not independent. Standard sequence models such as Long Short-Term Memory (LSTM) networks and Temporal Convolutional Networks (TCNs) model the temporal evolution of each variable but treat the variable set as an unstructured vector at each time step, relying on internal capacity to learn inter-variable dependencies implicitly. Temporal Graph Neural Networks (TGNNs) instead encode variables as nodes and their pairwise relationships as graph edges, enabling explicit spatial message passing at every time step alongside temporal modelling. Doing so, however, requires an adjacency matrix, and on body-worn sensors there is none to supply, so the graph must be constructed before any training begins. This thesis asks (i) whether TGNNs outperform established sequence models in accuracy and in computational cost, and whether any such advantage is conditional on which graph is supplied, (ii) which properties of a constructed graph govern performance, how much each is worth, and whether those quantities survive an order-of-magnitude increase in channel count, and (iii) whether the domain knowledge encoded in a predefined sensor graph contributes anything beyond the aggregation that graph imposes.

This thesis conducts a controlled, reproducible benchmark on human activity recognition from body-worn inertial sensors, a domain in which the relational structure between variables is physically grounded in human anatomy. The benchmark spans two phases, a low-dimensional phase using the University of East Anglia (UEA) multivariate archive datasets BASICMOTIONS (6-variate) and EPILEPSY (3-variate), and a high-dimensional phase using the UCI Daily and Sports Activities (DSA) dataset (45-variate, 19 activities, 8 subjects, subject-independent leave-one-subject-out cross-validation). Five TGNN architectures (GConvLSTM, GConvGRU, A3TGCN, DCRNN, and a hybrid model) are compared against LSTM, TCN and CNN-BiGRU baselines under a fixed protocol with five random seeds per configuration.

Moreover, in the low-dimensional phase the TGNN edge over the sequence baselines is *conditional on the graph*. With a naive predefined graph the strongest baseline (CNN-BiGRU) beats the TGNNs on EPILEPSY by 5.4 points, and the TGNNs recover parity with it—at roughly a fifth of its parameters—only once a well-chosen graph is used. The graph-construction strategy accordingly matters far more than the choice of temporal operator. Data-driven correlation graphs are the strongest family on EPILEPSY (95.7% mean accuracy, best configuration 98.7%), while a plain binary predefined graph is the weakest. In the high-dimensional phase the *granularity* of the predefined graph is decisive. A compact five-node anatomical (coarse) graph and a learned adaptive graph both reach $\sim 83\%$ – 84% accuracy, whereas a 45-node sensor-channel (fine) graph collapses to $\sim 68\%$ —below the no-graph baselines. The best TGNN configurations (coarse and adaptive graphs) exceed all sequence baselines on DSA, with the top model reaching 86.7% against a strongest-baseline (CNN-BiGRU) of 82.9%. Notably, the data-driven correlation graph that *won* at low dimensionality does not transfer. On DSA it falls to 76.1% to 81.8%. Because that graph spans the same 45 nodes as the fine graph while carrying more edges and scoring substantially higher, the pair separates two factors that are usually confounded, namely node *granularity*, worth 14.5% of accuracy, and edge *provenance*—whether weights are estimated from data or assigned by rule—worth 11.0%. A random-topology control then isolates the contribution of domain knowledge itself, in both roles a structural prior could play. As a *predefined* graph, random topology matched in node and edge count reaches 84.1% against anatomy’s 83.7%. As an initialisation for the learned adjacency it reaches 85.9% against 86.7%. Both differences are under one percentage point and smaller than the run-to-run variation of the experiment, and the same control repeated on a second architecture agrees to within 0.1%. The coarse graph’s advantage over the no-graph baselines therefore decomposes into 3.8% for aggregating 45 channels into 5 nodes and -0.4% for making that aggregation anatomically correct. Taken together, the evidence indicates that structural prior knowledge does not provide a measurable advantage. What matters is the graph’s granularity and where its edge weights come from, rather than whether it encodes domain structure or is learned end-to-end—a distinction worth 1.1%, an order of magnitude less than granularity. All comparisons are tested with paired non-parametric statistics, reported in full in the appendix.

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Chapter 1

Introduction

1.1 Motivation

A multivariate time series is a recording in which several signals are measured at the same time and are indexed by a common clock. Data of this shape is produced across a wide range of domains. Wearable inertial sensors record acceleration and rotation at several points on the body, electroencephalography records electrical potential at many scalp positions, road networks record speed and flow at thousands of loop detectors, and industrial plants record temperature, pressure and flow at co-located instruments. What these settings share is that the recorded variables are not independent of one another. They are coupled by physical proximity, by mechanical linkage, or by a common underlying process, and that coupling carries information about what the system is doing [1, 2].

Human Activity Recognition (HAR) from body-worn sensors is a mature and practically important instance of this problem [3, 4]. Accelerometers, gyroscopes and magnetometers are attached to the torso and limbs, and the resulting streams are used to infer what the wearer is doing. The coupling between the streams is unusually explicit here, because it is anatomy. The two legs of a walking person move in near anti-phase, the arms swing against the torso, and a sensor on the right wrist is mechanically related to one on the right upper arm in a way it is not related to one on the left ankle.

Why the problem matters. Reliable activity recognition underpins several deployed applications. In clinical rehabilitation, exercise adherence is monitored by classifying prescribed movements from a wrist or thigh sensor, so that a therapist can see whether a programme was actually followed between appointments. In elderly care, continuous activity classification supports fall detection and flags departures from a person’s normal routine, which is used as an early indicator of functional decline. In sports science, segmenting a training session into its constituent activities allows load and technique to be quantified without manual annotation of video. The errors that matter differ by application. A missed fall is far more costly than a spurious one, whereas a rehabilitation monitor that over-reports adherence gives a clinician a false picture of recovery, so per-class behaviour rather than overall accuracy is frequently the quantity of interest. Improvements also have to be obtained under a hard constraint, since these models run on battery-powered wearables where parameter count and inference cost are not free.

The classification task. Throughout this thesis the task is single-label multiclass classification of a fixed-length window. Formally, the input is a window $X \in \mathbb{R}^{T \times C}$ holding C simultaneously recorded channels over T time steps, and the output is one label $y \in \{1, \dots, K\}$ drawn from a fixed set of K activities. In plain terms, a few seconds of sensor recording goes in and the name of one activity comes out. Each window carries exactly one label, windows are classified independently of one another, and no partial or overlapping labels are assigned. This is the standard formulation in wearable HAR and is what makes results across the three datasets used here directly comparable.

How the literature models such data. The methods that address this task fall into a small number of broad categories. Classical distance and feature-based methods classify a window either by comparing it to labelled examples under an elastic distance, most prominently one-nearest-neighbour with Dynamic Time Warping (DTW), or by extracting hand-designed statistical features and applying a conventional classifier [5]. *Recurrent neural networks*, chiefly the Long Short-Term Memory (LSTM) network [6] and the Gated Recurrent Unit (GRU), process the window step by step and carry state through gated memory. *Temporal convolutional models* replace recurrence with stacked dilated causal convolutions, which gives a long receptive field with parallel training [7], and the same convolutional idea underlies strong general-purpose time-series classifiers such as InceptionTime [8] and ROCKET [9]. *Attention-based models* built on the Transformer [10] relate every pair of time steps directly rather than through a recurrence. *Graph neural networks* form the final category and are the subject of this thesis.

What the first four categories have in common, and what motivates the fifth, is their treatment of the channel dimension. However carefully they model time, they consume the C channels at each step as an unstructured vector. Any relationship between channels has to be rediscovered inside the network’s weights, even when that relationship is already known before training begins.

Graph Neural Networks (GNNs) make the relationship explicit instead [11]. Each variable becomes a node, each pairwise relationship becomes an edge with a weight, and message passing propagates information along those edges.

Combining a graph convolution with a recurrent or attention-based temporal encoder gives a Temporal Graph Neural Network (TGNN), which performs spatial message passing at every time step while modelling temporal dynamics. TGNNs are therefore a natural candidate whenever the variables correspond to spatially distributed sensors with a physically meaningful structure, which is exactly the wearable HAR setting.

Why a graph, stated plainly. The argument for a graph in this setting rests on three facts, and it is worth putting them in one place because the rest of the thesis depends on the third.

1. The channels are *not* independent. A sensor on the right wrist and one on the right upper arm are mechanically coupled, the two legs of a walking person move in near anti-phase, and that coupling carries information about the activity.
2. A sequence model has no mechanism for representing that. It receives the C channels at each time step as an unordered vector, so a relationship between two channels can only be recovered implicitly, from the weights, and only if the training data is sufficient to recover it.
3. A graph supplies the relationship directly. Writing the coupling into an adjacency matrix means message passing propagates information along it at every time step, so the model is told which channels are related rather than having to infer it.

The third fact is where the difficulty is. Supplying the relationship requires knowing it, and writing it down as a specific matrix. That is the step this thesis takes as its subject.

1.2 Problem Statement

A temporal graph neural network cannot be run at all until an adjacency matrix is supplied, and on body-worn sensors there is none to supply. **The graph must therefore be constructed, that construction is a free choice made once and before any training, and no study has measured what the choice is worth.** It matters because every result a temporal graph model reports is conditional on a decision its authors typically describe in a single sentence.

Three consequences follow. Methodologically, a design that varies the graph and the architecture together cannot attribute an observed gain to either, so published improvements from temporal graph models are hard to interpret and hard to reproduce. Practically, a practitioner deploying a wearable classifier has to make the choice and has no evidence base for it. Scientifically, the claim that structural prior knowledge is valuable is assumed rather than measured, because the usual comparison of a predefined graph against no graph at all confounds the value of the domain knowledge in the edges with the value of merely having a graph at all.

1.3 Research Questions

The problem above is addressed through three research questions. Each is answered by a specific set of experiments, and the mapping between them is given in Table 5.2.

RQ1

Does a graph inductive bias improve on sequence models for multivariate activity recognition, in classification accuracy and in computational cost, and is any such improvement *conditional on which graph is supplied*?

RQ2

Which properties of a constructed graph govern performance, how much is each of them worth, and do those quantities survive an order-of-magnitude increase in the number of channels?

RQ3

Does the domain knowledge encoded in a predefined sensor graph contribute anything beyond the aggregation that graph imposes, or can the same structure be obtained without it?

The three divide the problem rather than repeat it, and they do so along the line the problem statement draws. RQ1 asks whether the graph mechanism is worth having at all and at what cost, but it does not stop at a ranking. Its second clause is what distinguishes it from the benchmark question the literature already asks, because if the answer depends on the graph then no unconditional ranking of graph models against sequence models is meaningful, and the remaining two questions become necessary rather than optional.

RQ2 takes the conditionality established by RQ1 and asks what it is conditional *on*. It seeks measured quantities rather than a ranking of strategies, so that a practitioner is told what a design decision is worth rather than which recipe happened to win. Its final clause concerns dimensionality, and it belongs to RQ2 rather than to a question of its own, because the property that governs performance and the property that governs whether a conclusion transfers turn out to be the same property. Separating them into two questions would obscure that they have one answer.

RQ3 asks what neither of the others can. Every comparison available to RQ1 and RQ2 is a comparison between two *constructions*, and every such comparison changes more than one thing at once, so none of them can isolate the

contribution of the domain knowledge itself as distinct from the contribution of merely having a graph. Answering it requires a control rather than a comparison, in which the construction is held fixed and only the knowledge is removed. RQ3 is therefore the question this study is designed around and the one that most sharply separates it from the literature of Chapter 2, in which the value of a structural prior is assumed rather than measured.

1.4 Contributions

There is one central goal. **To measure what the graph-construction choice is worth, by making it the only thing that varies.** Everything below exists to serve that goal, and each contribution is stated with the specific thing it makes possible, because a list of apparatus is not a contribution unless it licenses a conclusion that was not previously available.

1. **A controlled two-phase benchmark.** Five TGNN architectures are compared against three sequence baselines under one identical pipeline, on three datasets spanning 3 to 45 channels, 4 to 19 classes and 80 to 9120 instances, under both a fixed archive split and eight-fold subject-independent cross-validation. Preprocessing, optimiser, schedule, metrics and random seeds are held fixed throughout, so a difference between two conditions is attributable to the condition rather than to the setup. *What this makes possible*—a difference between two conditions becomes evidence about the condition, which is the precondition for every claim that follows.
2. **A factorial study of graph construction.** Graph construction is treated as the controlled variable rather than as a hyperparameter. The design spans predefined domain-knowledge graphs at two granularities, data-driven graphs built from three similarity statistics and five threshold policies, and adaptive graphs learned end-to-end with and without initialisation from a prior. Every pair of conditions differs in exactly one property. This is the design that allows node *granularity* to be separated from edge *provenance*, two factors that the usual predefined-versus-learned framing confounds. *What this makes possible*—answering RQ2 with two measured quantities rather than with a ranking of strategies.
3. **A random-topology control.** A graph matched to the anatomical graph in node count, edge count and training procedure, but with its edges placed at random, isolates the contribution of the domain knowledge itself rather than of merely having a graph. It is evaluated in both roles a structural prior can play, as a predefined graph and as an initialisation for the learned adjacency, and is replicated on a second architecture. *What this makes possible*—answering RQ3, which no comparison between constructions can answer, because every such comparison changes more than one thing at once.
4. **Interpretable analyses of the learned graph.** The adaptive adjacency is reconstructed from trained checkpoints and compared against the hand-designed graph through edge-set Jaccard similarity, which allows the learned topology to be reported alongside its accuracy rather than treated as opaque. Each architecture’s behaviour is attributed to an identifiable property of its recurrent cell. *What this makes possible*—the model-side reading of RQ1, in which the architecture is shown to be the smaller axis of variation, and the observation that two models reach the same accuracy from opposite topologies, which is what turns a number into an explanation.
5. **Statistical treatment appropriate to the protocol.** Every reported comparison is subjected to paired non-parametric significance testing across the subject-independent folds, with multiplicity correction and an effect size reported alongside each test. *What this makes possible*—separating an effect this design could not resolve from one that is genuinely absent, which the central negative finding of the thesis depends on.

The comparison is deliberately broad on every axis the research questions touch. Phase 1 evaluates 40 distinct graph modes for each architecture and Phase 2 a further 21, spanning the full range of ways an adjacency can be obtained, from no graph at all through to a randomised control. That breadth is what allows the study to report which design choices matter, rather than only which configuration achieved the highest score.

1.5 The Two Phases, and the Outline of This Thesis

The study runs in two phases, and because the terms are used throughout they are defined here rather than in passing. **Phase 1** is the low-dimensional phase, using two single-unit UEA archive datasets, BASICMOTIONS with six channels and EPILEPSY with three, each on the archive’s own fixed train/test split. Every channel comes from one wrist, so a graph over them relates *the axes of one sensor* and has very little to encode. **Phase 2** is the high-dimensional phase, using DSA, with 45 channels from five physically separated body-worn units and nineteen activities across eight subjects, evaluated leave-one-subject-out. A graph over those relates *body segments* and can encode genuine anatomy. The distinction is not one of size alone, and it is why a conclusion established in one phase is not automatically available in the other.

Chapter 2 reviews activity recognition, the sequence models that dominate time-series classification, graph neural networks and their temporal extensions, and closes on how existing work chooses a graph, which is where the gap addressed here lies. Chapter 3 states the requirements the research questions place on the data, introduces the three datasets against those requirements, and reports the exploratory analysis that motivates the graph designs. Chapter 4 defines the preprocessing, the five graph-construction strategies and the random-topology control, the model families and the training

objective. Chapter 5 sets out the protocol that holds everything except the graph fixed, the full experiment matrix, the mapping from experiments to research questions, and the statistical procedure. Chapter 6 reports the outcomes as eight named experiments. Chapter 7 interprets them against the research questions and states the threats to validity. Chapter 8 concludes and sets out the continuation the findings point to.

Chapter 2

Related Work

2.1 Human Activity Recognition and Its Datasets

Activity recognition from body-worn inertial sensors has been studied for two decades and is surveyed comprehensively by Bulling et al. [3], who establish the now-standard pipeline of windowing, feature extraction and window-level classification, and by Wang et al. [4] for the deep-learning era. Four application areas recur, and they differ in what a correct prediction is worth. *Rehabilitation monitoring* classifies prescribed exercises from a wrist or thigh sensor so that adherence between appointments can be verified rather than self-reported. *Elderly care* runs continuous classification to detect falls and to flag departures from a person’s habitual routine, which is used as an early marker of functional decline. *Sports analytics* segments a training session into its constituent activities so that load and technique can be quantified without hand-annotating video. *Context-aware mobile computing* uses coarse activity state to drive interface and power decisions on a phone or watch. The error that matters differs accordingly, since a missed fall is far costlier than a spurious one, whereas an over-reported adherence figure misleads a clinician, which is why per-class behaviour rather than overall accuracy is often the operative quantity.

Evaluation rests on a small number of public datasets, and their design shapes what can be concluded. Single-device datasets such as UCI HAR [12] record a smartphone at the waist and therefore offer little spatial structure between channels, since all channels come from one physical location. Multi-device datasets are the ones relevant here. PAMAP2 [13] places three inertial units on the wrist, chest and ankle, and the UCI Daily and Sports Activities dataset (DSA) [14] places five units on the torso and the four limbs while recording nineteen activities from eight subjects. DSA is used in Phase 2 of this thesis precisely because its five physically separated units give an anatomical graph that is genuinely meaningful rather than merely plausible. For the low-dimensional phase the thesis draws BASICMOTIONS and EPILEPSY from the University of East Anglia (UEA) multivariate archive [1], which standardised evaluation across dozens of multivariate datasets and remains the reference benchmark for multivariate time-series classification.

An evaluation choice recurs across this literature and is adopted here. Because subjects differ systematically in body mechanics, a random split leaks subject-specific patterns between training and test and reports optimistic numbers. Subject-independent, or leave-one-subject-out, evaluation is the protocol that reflects the deployment setting, in which a model meets wearers it was not trained on.

2.2 Sequence Models for Time-Series Classification

Time-series classification has a long history in which distance-based methods set the reference. One-nearest-neighbour with Dynamic Time Warping remained the standard baseline for years, and the large comparative study of Bagnall et al. [5] established both its strength and the conditions under which it is beaten. The multivariate setting adds the requirement to model dependence across channels as well as through time.

Within the deep-learning response, three families dominate. Recurrent networks, in particular the LSTM [6], model temporal dependence through gated memory and remain a strong HAR baseline. Temporal Convolutional Networks [7] replace recurrence with dilated causal convolutions, achieving long effective receptive fields with parallel training. Hybrid convolutional–recurrent models, of which the DeepConvLSTM architecture of Ordóñez and Roggen [15] is the canonical wearable-HAR instance, pair a convolutional feature extractor with a recurrent encoder and form a strong baseline in their own right. Purely convolutional general-purpose classifiers such as InceptionTime [8] and the random-kernel method ROCKET [9] are competitive across the UEA archive, and attention-based models derived from the Transformer [10] relate time steps directly. The review of Ismail Fawaz et al. [2] finds no single dominant method across all datasets.

Three of these architectures serve as the baselines of this thesis, and the selection is deliberate rather than exhaustive. The LSTM is the exact structural counterpart of the graph-recurrent cells under test, which replace its linear maps with graph convolutions, so the LSTM-versus-GConvLSTM contrast isolates the graph mechanism. The TCN isolates the temporal-convolution pathway. CNN–BiGRU is the standard wearable-HAR reference. What unites all three, and what motivates the graph models, is that none of them contains any mechanism for representing which channels are related. Spatial structure must be recovered implicitly from the weights rather than supplied.

2.3 Graph Neural Networks and Their Temporal Extensions

Graph neural networks generalise convolution to data defined on a graph. Spectral formulations and their localised polynomial approximation [16] led to the Graph Convolutional Network [11], which popularised message passing with the symmetric-normalised adjacency, and attention-based [17] and diffusion-based operators subsequently broadened the family. Applied to time series a graph convolution alone is insufficient, because it models the relation between channels while leaving temporal structure unaddressed.

Temporal graph neural networks resolve that asymmetry by interleaving graph convolution with temporal modelling, and supply the five architectures compared in this thesis. Structured recurrent networks [18] replace the linear maps inside an LSTM or GRU cell with graph convolutions, yielding GConvLSTM and GConvGRU. The Diffusion Convolutional Recurrent Neural Network [19] models spatial dependence as a diffusion process on the graph within a GRU. Attention-based temporal graph convolution (A3TGCN) [20] augments graph-convolutional GRUs with a temporal attention layer, and the spatio-temporal graph convolutional network of Yu et al. [21] interleaves graph and temporal convolutions without recurrence.

One observation about this literature matters here. These operators were developed predominantly for spatio-temporal forecasting of traffic and weather, where the graph is given in advance by physical infrastructure such as a road network. Their application to multivariate time-series classification over sensor graphs, where no such infrastructure exists, is comparatively under-explored, and that difference is taken up in the next section.

2.4 Graph Models Applied to Human Activity and to Time Series

Graph models have been applied to human activity, but overwhelmingly in a setting that differs from this one in the decisive respect. In *skeleton-based* action recognition the input is a sequence of body-joint coordinates recovered from video or a depth camera, and the graph is simply the skeleton, with joints as nodes and bones as edges. ST-GCN [22] established the approach and 2s-AGCN [23] extended it by making the adjacency partly learnable rather than fixed to the skeleton. Both report substantial gains over sequence models on the same data.

That success is instructive precisely because it does not transfer directly. The skeleton graph is not constructed, it is observed, since the human body supplies an unambiguous node set and edge list. Sensor-based HAR, the setting of this thesis, has no such object. A set of wearable channels has no canonical adjacency, so the graph must be constructed before any of the operators of Section 2.3 can be applied, and how it is constructed becomes a free choice rather than a given. Work on graph models for sensor-based HAR is correspondingly thinner than the skeleton literature, and where it exists the graph is normally fixed once from sensor placement and not varied.

For general multivariate time series the graph is likewise unavailable, and the response there has been to learn it. Graph WaveNet [24] introduced a self-adaptive adjacency trained end-to-end, MTGNN [25] added a graph-learning module with sparsity control, and StemGNN [26] works in the spectral domain. These are the direct antecedents of the adaptive strategy evaluated in this thesis (Section 4.3.3). Two things should be noted about them. They target forecasting rather than classification, and in each the learned graph is a component of a proposed architecture rather than the variable under study, so none reports what is lost or gained by supplying a different graph to the same model. That is the comparison this thesis makes.

2.5 How Existing Work Chooses a Graph

Every operator in the previous section presupposes an adjacency matrix, and on wearable sensors none is given. Across the literature three strategies recur for supplying one.

- **Predefined domain-knowledge graphs** take edges from external knowledge, such as sensor co-location, human anatomy or a road network [19, 21]. They require no data to construct and are interpretable, but they are only as good as the assumed structure.
- **Data-driven graphs** estimate edges from statistical dependence in the training data, using thresholded Pearson correlation, DTW similarity, or mutual k -nearest-neighbour relations in a similarity space. They adapt to the observed signal but depend on a threshold or neighbourhood-size choice.
- **Adaptive learned graphs** treat the adjacency itself as a parameter optimised end-to-end with the classifier, as popularised by Graph WaveNet [24] and extended by MTGNN [25]. They require no prior assumption but forfeit interpretability unless the learned graph is subsequently examined.

The methodological limitation of this literature is not that the strategies are unknown, but that they are hardly ever compared under conditions that permit attribution. A strategy is typically selected once and reported alongside a proposed architecture, so the graph choice and the architecture choice vary together. Comparisons are usually performed on a single dataset, so it is not known whether a conclusion survives a change in dimensionality. Most importantly, when a predefined graph is compared against no graph at all, the comparison confounds two distinct benefits, that of aggregating channels into a structured node set and that of the specific domain knowledge in the edges. A predefined graph that beats a no-graph baseline establishes only that *some* graph helps, not that the anatomy did.

Chapter 3

Data

The research questions place requirements on the data before any dataset is named, and the three used here were chosen against those requirements rather than for availability. Stating them first makes clear why these three and not others.

1. **A physically grounded graph has to exist at all.** The recordings must come from body-worn inertial sensors, since otherwise there is no domain knowledge to encode in an adjacency and RQ3 cannot be posed.
2. **Dimensionality has to vary, and substantially.** RQ2 asks whether the quantities it measures survive an order-of-magnitude increase in channel count, which is unanswerable unless the datasets actually differ by close to that much. These span 3 to 45.
3. **At least one dataset must carry real anatomy.** A graph over the three axes of one wrist sensor relates axes rather than body parts, so its anatomical content is nominal. Testing whether anatomical knowledge is worth anything needs physically separated units.
4. **At least one dataset must record several subjects.** Deployment means meeting wearers the model was never trained on, so the decisive comparisons have to be evaluated subject-independently, which requires enough subjects to hold one out at a time.
5. **The data must be public and benchmarked,** so that the numbers are comparable with published work rather than only internally consistent.

Requirements 1 and 5 are satisfied by all three datasets and requirement 2 by the set as a whole. Requirements 3 and 4 are satisfied by DSA alone, which is why DSA carries the central experiments while the two UEA datasets establish the low-dimensional reference point against which transfer is judged.

The two phases. The three datasets are organised into the two phases defined in Section 1.5 (Table 3.1). **Phase 1** is BASICMOTIONS and EPILEPSY, both a single wrist-worn unit on the archive’s fixed split, where a graph relates the axes of one sensor. **Phase 2** is DSA, five separated units and eight subjects evaluated leave-one-subject-out, where a graph relates body segments. Reading a Phase-1 result as though it were a Phase-2 result is precisely the error the transfer clause of RQ2 exists to detect, which is why the two are named separately everywhere in this thesis rather than pooled.

Table 3.1: The three datasets. Phase 1 uses two single-unit UEA datasets, in which the graph relates the axes of one sensor. Phase 2 uses the multi-unit DSA dataset, in which the graph relates physically separated body segments. Instances are fixed-length windows. Sequence length is the number of time steps per window after preprocessing.

Dataset	Phase	Channels	Seq. length	Instances	Classes	Evaluation protocol
BASICMOTIONS	1	6	100	80	4	fixed archive train/test split
EPILEPSY	1	3	206	275	4	fixed archive train/test split
DSA	2	45	125	9 120	19	8-fold leave-one-subject-out

3.1 Phase 1: UEA Single-Unit Datasets

The two Phase-1 datasets satisfy requirements 1, 2 and 5 and are used together rather than interchangeably, because they play different roles. BASICMOTIONS is the saturated implementation check and EPILEPSY is the discriminative one, and each is described below with the role it serves.

3.1.1 BasicMotions

BASICMOTIONS is drawn from the University of East Anglia (UEA) multivariate archive [1]. Four students wore a smartwatch on the wrist while performing four activities, each repeated five times, and the resulting recordings were

segmented into 80 windows of 100 time steps. The device provides six channels, a three-axis accelerometer and a three-axis gyroscope, so every channel originates from a single physical location. The archive supplies a fixed split of 40 training and 40 test windows, which is used unchanged here so that results are comparable with published numbers on the same benchmark.

The dataset is included as a controlled low-dimensional reference point rather than as a discriminative test. With four well-separated activities and six channels the task is close to saturated for modern methods, which makes it useful for verifying that every architecture and graph strategy is correctly implemented, and useful as the least structurally informative setting in the study. Because all six channels sit on one wrist, any graph over them relates sensor *axes* rather than body segments, and the anatomical justification for an edge is correspondingly weak. Class counts are given in Table 3.2, and the four classes are exactly balanced.

3.1.2 Epilepsy

EPILEPSY is also drawn from the UEA archive [1] and records a three-axis accelerometer worn on the dominant wrist by six participants. Each participant performed four activities, one of which is a simulated epileptic seizure enacted according to a medical protocol, and the recordings were segmented into 275 windows of 206 time steps. The archive split provides 137 training and 138 test windows.

EPILEPSY plays a different role from BASICMOTIONS. It is the more discriminative of the two Phase-1 datasets, in that methods separate clearly on it, and it is the sparsest setting in the study with only three channels. It also carries the only non-uniform class distribution among the three datasets, ranging from 60 SAWING windows to 74 WALKING windows (Table 3.2). The imbalance is mild, at a ratio of roughly 1.23 to one, but it is the reason macro-F1 is reported alongside accuracy throughout (Section 5.2). The seizure class is also the clinically consequential one, so per-class behaviour is examined for this dataset specifically in Section 6.1.3.

Table 3.2: Phase-1 class labels and window counts. BASICMOTIONS is exactly balanced. EPILEPSY is mildly imbalanced, with a majority-to-minority ratio of 1.23, which is why macro-F1 is reported alongside accuracy.

Dataset	Class label	Train	Test	Total
BASICMOTIONS	BADMINTON — racquet play	10	10	20
	RUNNING — running on the spot	10	10	20
	STANDING — standing still	10	10	20
	WALKING — normal gait	10	10	20
	<i>total</i>	40	40	80
EPILEPSY	EPILEPSY — enacted seizure	34	34	68
	RUNNING — running	36	37	73
	SAWING — sawing a plank	30	30	60
	WALKING — normal gait	37	37	74
	<i>total</i>	137	138	275

3.2 Phase 2: UCI Daily and Sports Activities

DSA [14] is the dataset on which the central questions of this thesis are answered, and it is included because it supplies the two properties Phase 1 lacks. It has genuinely separated sensor units, so an anatomical graph is physically meaningful, and it has multiple subjects, so subject-independent evaluation is possible.

Eight subjects, four female and four male, each performed nineteen activities for five minutes. Five MTx inertial units were worn, on the torso, the right arm, the left arm, the right leg and the left leg. Each unit records nine channels, being three-axis acceleration, three-axis angular velocity from a gyroscope, and three-axis magnetic field, giving $5 \times 9 = 45$ channels in total. Recordings are sampled at 25 Hz and segmented into five-second windows of 125 time steps, producing $8 \text{ subjects} \times 19 \text{ activities} \times 60 \text{ windows} = 9120$ instances. The nineteen classes are exactly balanced at 480 windows each (Table 3.3).

Two properties make DSA the decisive dataset here. First, the physical placement maps directly onto body segments joined by anatomical structure, so a predefined graph encodes real mechanical relationships rather than a plausible guess, which is what RQ3 requires, since domain knowledge that is only nominal cannot be tested for its worth. Second, the same 45 channels can be organised either as 45 channel-level nodes or as 5 body-segment nodes carrying nine features each. The same underlying data therefore supports two graph granularities, and comparing them isolates node granularity from every other factor, which is what RQ2 requires. Because subjects differ systematically in body mechanics, evaluation uses eight-fold leave-one-subject-out cross-validation, which is the protocol that reflects deployment on an unseen wearer.

Table 3.3: The nineteen DSA activity classes, grouped by the movement type they represent. Every class contributes exactly 480 windows (60 per subject across eight subjects), so the dataset is perfectly balanced and accuracy is an unbiased summary. Several groups are mutually confusable, in particular the three static postures and the three treadmill-walking variants, which is what makes the task non-trivial at 19 classes.

Group	Activity classes	Windows each
Static postures	sitting, standing, lying on back, lying on right side	480
Stairs	ascending stairs, descending stairs	480
Elevator	standing in an elevator, moving in an elevator	480
Walking	walking in a car park, treadmill flat, treadmill inclined	480
Running	running on a treadmill	480
Gym machines	stepper, cross trainer, rowing	480
Cycling	horizontal cycling, vertical cycling	480
Whole body	jumping, playing basketball	480
<i>19 classes, 8 subjects, 60 windows per subject per class</i>		9 120

3.3 Exploratory Analysis

Before any model is fitted, the three datasets are examined directly. This section reports what the signals look like, how the classes are distributed, and how strongly the channels are related. The last of these is not merely descriptive, since the inter-channel dependence structure is exactly what a graph is supposed to encode, and it turns out to explain several later results.

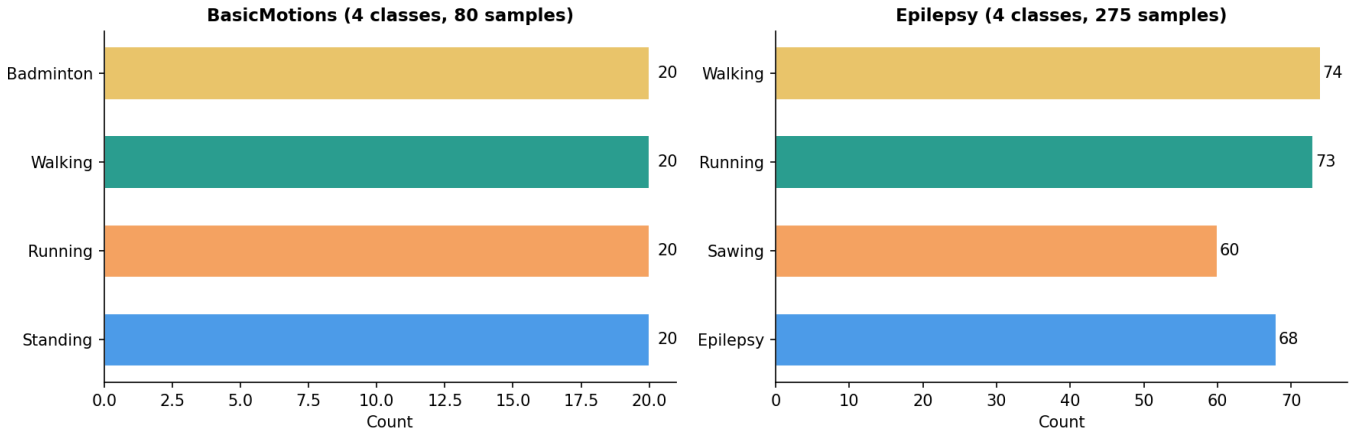


Figure 3.1: Class distributions for the three datasets. BASICMOTIONS (20 windows per class) and DSA (480 windows per class) are exactly balanced by construction. EPILEPSY is mildly imbalanced, from 60 SAWING windows to 74 WALKING windows. No resampling or class weighting is applied anywhere in this thesis, and macro-F1 is reported alongside accuracy so that the EPILEPSY imbalance cannot distort a conclusion.

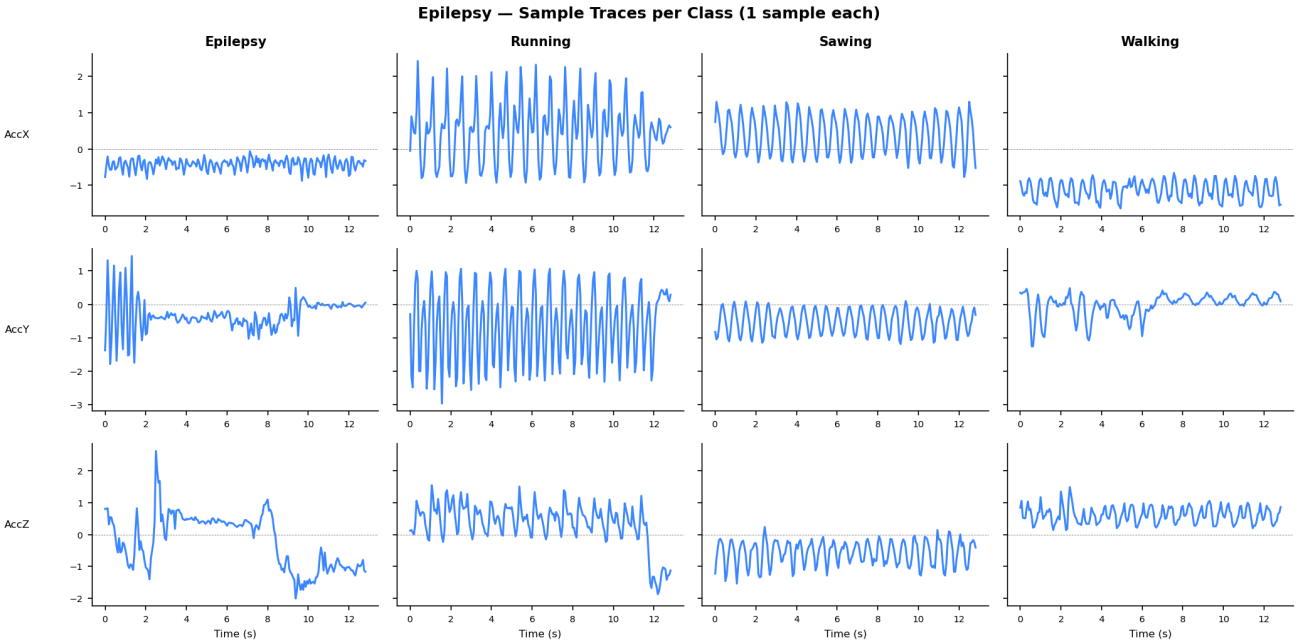
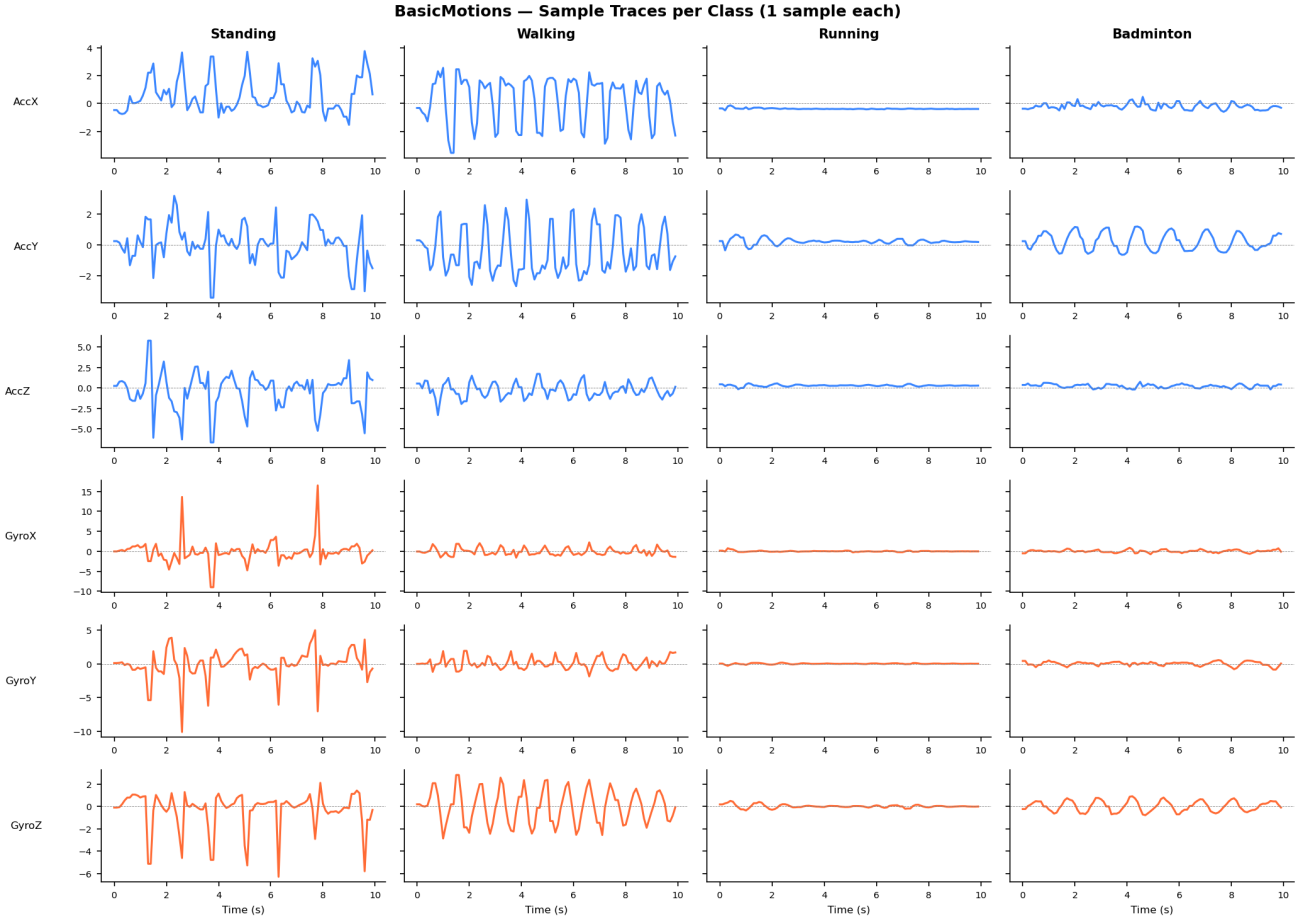


Figure 3.2: Representative raw windows from the two Phase-1 datasets, one window per class, before z-score normalisation. Each line is one channel. BASICMOTIONS separates on gross amplitude and periodicity, STANDING being nearly flat and BADMINTON high-amplitude and irregular, which is consistent with the near-ceiling accuracies reached there. EPILEPSY is finer, since WALKING and RUNNING differ mainly in the frequency and amplitude of a similar periodic pattern.

DSA — the torso unit (1 of 5), all 9 of its channels, on 6 of the 19 activities
One 5-second window per subject (125 steps at 25 Hz), three subjects per panel, raw sensor units before z-score normalisation

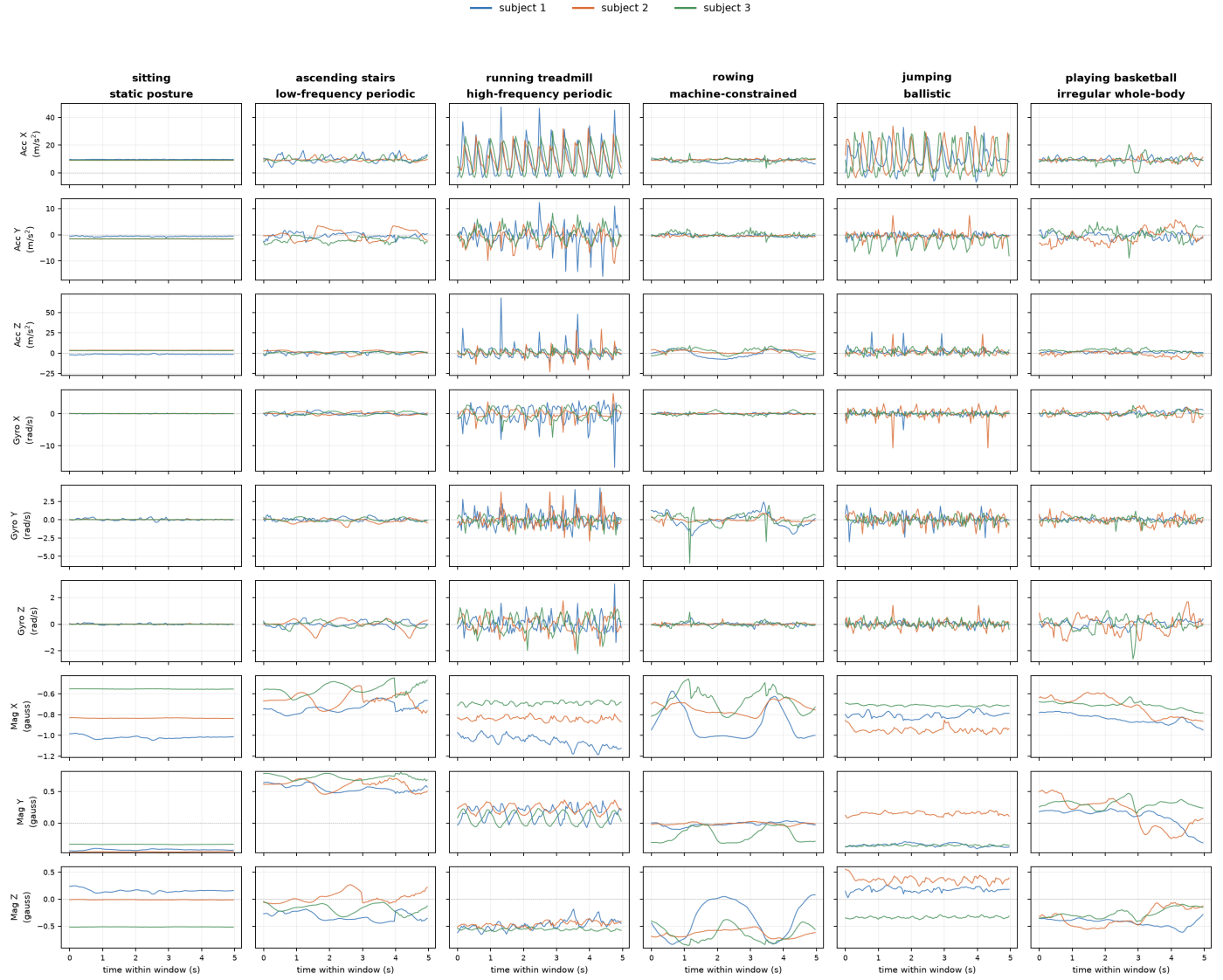


Figure 3.3: DSA — the torso unit, which is one of the five, showing all nine of its channels on six of the nineteen activities. Rows are the channels, grouped by modality, and columns are activities. Each panel draws one five-second window (125 steps at 25 Hz) from each of three subjects, so the three traces show between-subject variation rather than repeated windows from one person, which is the variation the subject-independent protocol of Section 3.2 requires the models to survive. Values are raw sensor units, before the z-score normalisation of Section 4.2, and each row shares one vertical scale across all six columns so that amplitudes are comparable left to right. The six activities were chosen to span the movement types the dataset contains rather than to be representative in any statistical sense, running from a static posture through periodic gait to ballistic and irregular whole-body motion. Two things the panel makes visible carry into later chapters. The three modalities occupy different numerical ranges, with the accelerometer near gravity and the magnetometer within roughly ± 1 , which is why per-channel normalisation is required rather than optional. And the magnetometer rows separate the three subjects by an offset that persists across every activity, which is a sensor-orientation difference rather than a movement one, and is part of what makes leave-one-subject-out harder than a random split. Subjects 1, 2 and 3 are drawn because subjects 0 and 5 carry zero-padded windows on the stair activities (Section 4.2).

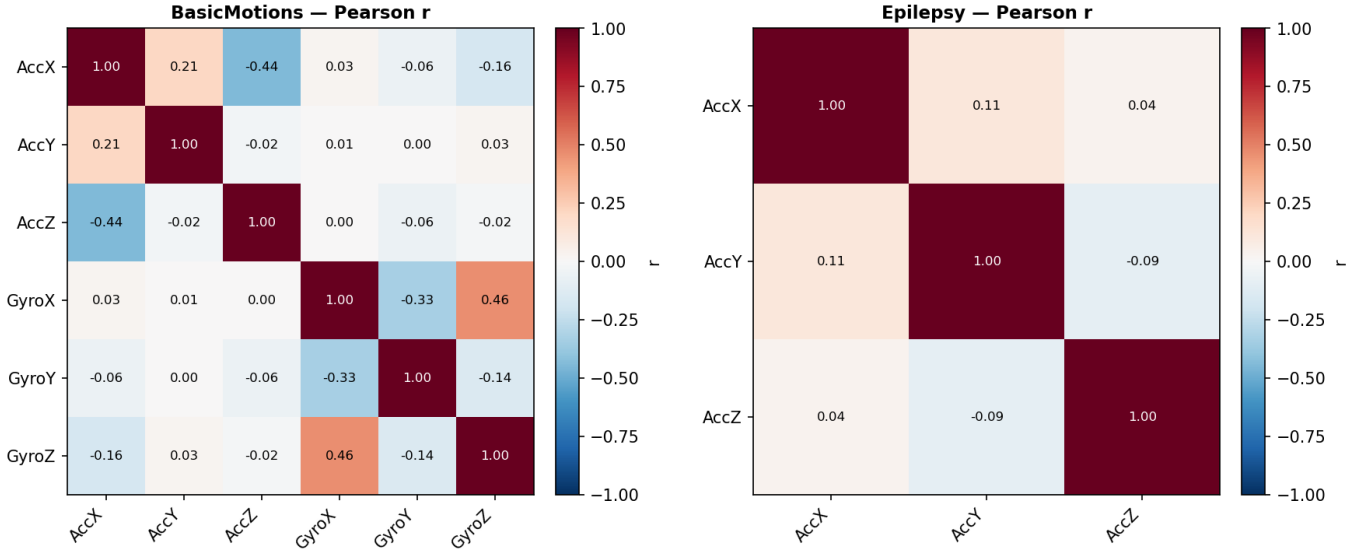


Figure 3.4: Inter-channel Pearson correlation computed on the training split of each dataset. This is the empirical dependence structure that a graph is intended to represent. EPILEPSY’s three accelerometer axes are almost uncorrelated, with a maximum absolute coefficient of 0.11, whereas BASICMOTIONS’s six channels show clear intra-modality blocks and DSA shows strong block structure aligned with its five sensor units. The EPILEPSY panel is the reason a fixed correlation threshold of 0.3 produces an almost empty graph on that dataset, and therefore the reason the data-driven threshold policies of Section 4.3.2 are needed.

Three observations from this analysis carry into the rest of the thesis. First, the datasets differ enormously in how much relational structure they contain, from three nearly uncorrelated axes in EPILEPSY to forty-five channels with strong block structure in DSA, which is precisely the range RQ2 requires. Second, the DSA correlation blocks coincide with the five physical units, which is the empirical justification for the coarse anatomical graph defined in Section 4.3.1. Third, absolute correlation magnitudes vary by an order of magnitude between datasets, so no single fixed threshold can define a sensible data-driven graph for all of them, and threshold policy must itself be treated as an experimental variable.

Chapter 4

Methodology

Section 3.3 established that the three datasets differ by an order of magnitude in how much inter-channel dependence they contain, and that no single fixed rule turns that dependence into a graph across all of them. This chapter therefore treats the rule itself as the object to be varied.

The chapter has one organising principle. *Graph construction is the only thing allowed to vary*, so it is described at length in Section 4.3, and everything else is described in order to establish that it is held fixed.

Section 4.1 fixes the notation, and in particular separates the channel count C from the node count N , a distinction the granularity experiments turn on entirely. Section 4.2 gives the data pipeline every dataset passes through, together with the graph preprocessing every adjacency passes through regardless of where it came from, which is part of what makes the comparison controlled rather than part of any one strategy. Section 4.3 is the substance of the chapter, defining the five construction strategies and the random-topology control as a taxonomy whose names are then used without variation for the rest of the thesis. Section 4.4 defines the models, comprising the two graph convolution operators, the four recurrent graph cells built on them, the classifier wrapper shared by all of them, the hybrid that blends a prior with a learned graph, and the three sequence baselines that receive no graph at all. Section 4.5 states the training objective and the sparsity penalty applied to learned adjacencies.

Read together, Sections 4.2, 4.4 and 4.5 describe what is constant, and Section 4.3 describes what changes. An observed difference between two conditions is therefore attributable to the graph rather than to anything surrounding it, which is the property the whole study rests on.

4.1 Problem Formalisation and Notation

A single instance is a window $X \in \mathbb{R}^{T \times C}$ containing C simultaneously recorded channels over T time steps, and its label is one class $y \in \{1, \dots, K\}$. A graph model additionally receives a time-invariant adjacency matrix $A \in \mathbb{R}^{N \times N}$ defined over N nodes. The distinction between C and N is essential to this thesis and is not a matter of notation only. In a *channel-level* graph each channel is its own node, so $N = C$ and each node carries a scalar feature per time step. In the *coarse* DSA graph each of the five sensor units is one node, so $N = 5$ while $C = 45$, and each node carries a nine-dimensional feature vector per time step. Comparing these two cases at fixed C is what the granularity experiments of Section 6.3 do.

Given X and A , a model produces a predicted class-probability distribution $\hat{y} = \text{softmax}(z) \in \mathbb{R}^K$ over the K classes, where z are the output logits and $\hat{y}_k$ is the predicted probability that the window belongs to class k . The predicted label is $\arg \max_k \hat{y}_k$. Symbols are used consistently throughout the thesis as listed in Table 4.1, and in particular distinct letters are reserved for instances, subjects, nodes and channels, which are quantities that must not be conflated.

Table 4.1: Notation used throughout. Distinct symbols are reserved for instances, subjects, nodes, channels, time steps and classes.

Symbol	Quantity	Meaning
M	instances	number of fixed-length windows in a split
S	subjects	number of distinct people recorded (DSA only, $S = 8$)
T	time steps	length of one window after preprocessing
C	channels	number of recorded sensor signals
N	nodes	number of vertices of the graph, $N = C$ for a channel-level graph and $N = 5$ for the coarse DSA graph
K	classes	number of activity labels
A	adjacency	$N \times N$ matrix of edge weights
$\hat{A}$	normalised adjacency	A after self-loops and symmetric normalisation, Eq. 4.1
d	rank	truncation rank of the SVD used to seed an adaptive graph
H	hidden size	width of the recurrent hidden state, $H = 64$ throughout

4.2 Preprocessing

All three datasets pass through one pipeline, applied identically to every model and every graph strategy. The steps are given in order.

1. **Loading and orientation.** UEA datasets are read with the `aeon` library, which returns arrays in the channel-first layout (M, C, T) . These are transposed to the time-first layout (M, T, C) that the recurrent models consume. DSA is read from the UCI text files and assembled into the same (M, T, C) layout together with a subject identifier per window.
2. **Missing values and zero padding.** The two UEA datasets are complete. DSA is complete in the sense that no value is absent, but 278 of its 9120 windows, or 3.0 %, end in a run of 25 time steps that is exactly zero on all 45 channels at once, which is the final second of a five-second window. The pattern is systematic rather than scattered. It affects only subjects 0 and 5, only on ascending stairs, descending stairs and jumping, and where it occurs it covers every one of that subject’s 60 windows for the activity. It arises from the segmentation of source recordings that are shorter than the others rather than from any step in this pipeline. No imputation is applied, because a zero run is what the source provides and substituting an estimate would invent signal that was never recorded. The affected windows are retained so that the class balance and the fold sizes stay exactly as Section 3.2 describes them, and the consequence for the two affected folds is stated in Section 7.4.
3. **Segmentation.** BASICMOTIONS and EPILEPSY are supplied by the archive as fixed-length windows and are used unchanged, at $T = 100$ and $T = 206$ respectively. DSA is segmented into non-overlapping five-second windows at 25 Hz, giving $T = 125$. Windows do not overlap, so no information is shared between two instances.
4. **Label encoding.** Each activity name, which the raw files store as text, is mapped to an integer *class index* in $\{0, \dots, K - 1\}$ by an encoder fitted on the training split only. These class indices are what the loss of Eq. 4.3 consumes.
5. **Normalisation.** Each channel is z-score normalised, $x' = (x - \mu_c)/\sigma_c$, with μ_c and σ_c estimated per channel on the training split only and then applied verbatim to validation and test data. Normalisation is required here rather than optional, because the channels are recorded in different physical units, with acceleration in m/s^2 , angular velocity in rad/s and magnetic field in gauss, so without normalisation the large-magnitude channels would dominate the gradient. A variance floor of 10^{-8} guards against division by zero on constant channels. Estimating the statistics on training data alone is what prevents test-set leakage.
6. **Splitting.** Phase 1 uses the archive’s fixed train/test split unchanged. Phase 2 uses eight-fold leave-one-subject-out cross-validation. In each fold one subject is held out entirely for testing and a second held-out subject provides the validation set used for early stopping, with the remaining six subjects used for training. This yields 6840 training, 1140 validation and 1140 test windows per fold. No subject ever appears in more than one of the three sets within a fold, which is what makes the evaluation subject-independent.
7. **Batching and graph attachment.** Sequence baselines consume the (T, C) tensors directly. Graph models additionally require an adjacency, and in every dataset here the topology is shared by all instances, so a custom collate function batches the node-feature sequences while attaching the single shared adjacency once per batch rather than duplicating it per instance.

4.2.1 Graph Preprocessing

Every adjacency, however it was obtained, passes through one identical preprocessing sequence before it reaches a model. This step deserves to be stated explicitly, because it is applied to all strategies alike and is therefore part of what makes the comparison controlled rather than part of any one strategy.

The sequence has four stages. First, a raw adjacency A is produced by one of the strategies of Section 4.3. Second, a self-loop is added to every node, $\tilde{A} = A + I$. Third, the degree $\tilde{d}_i = \sum_j \tilde{A}_{ij}$ is computed and the matrix is symmetrically normalised,

$$\hat{A}_{ij} = \frac{\tilde{A}_{ij}}{\sqrt{\tilde{d}_i \tilde{d}_j}}, \quad \text{equivalently} \quad \hat{A} = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}. \quad (4.1)$$

Fourth, $\hat{A}$ is converted to the sparse edge-index and edge-weight pair that the message-passing implementation consumes, where the edge index lists the (i, j) pairs with non-zero weight and the edge weight lists the corresponding $\hat{A}_{ij}$.

Each stage has a specific effect on message passing. The self-loop guarantees that a node’s own features survive aggregation, since without it a node’s updated representation would be built purely from its neighbours and its own signal would be discarded. The symmetric normalisation scales each message by the inverse square root of both endpoints’ degrees, which prevents a high-degree node from dominating the aggregate simply by having many neighbours, and it bounds the spectral radius of $\hat{A}$ to $[-1, 1]$, which is a stability precondition for the Chebyshev polynomial filters of Section 4.4.1 [11, 16]. This normalisation is the reason the numerical weights of a predefined graph are not free parameters. Given a binary topology, Eq. 4.1 determines every weight, so no predefined graph in this thesis contains a hand-tuned strength.

4.3 Graph-Construction Strategies

Graph construction is the controlled variable of the study. Five strategies are compared, and because the distinction between them is central to every result that follows, they are defined here as a taxonomy and these names are used without variation throughout the thesis.

Predefined (domain knowledge)

The edge set is written down in advance from knowledge of where the sensors sit on the body. No training data is consulted. Weights follow from Eq. 4.1.

Data-driven

The edge set and its weights are estimated from the training split by a statistical dependence measure, namely signed Pearson correlation, DTW similarity, or mutual k -nearest neighbours. No domain knowledge is used, and the graph is fixed once estimated.

Adaptive (learned)

The adjacency is a free parameter of the model, optimised end-to-end together with the classifier and initialised at random. No prior graph is supplied.

Adaptive, seeded

The same learner as above, but initialised from a prior graph rather than at random. The prior may be predefined or data-driven, and it enters through a truncated singular value decomposition.

Random-topology control

A graph matched to the coarse predefined graph in node count, edge count and preprocessing, but with its edges placed at random. It is not a candidate strategy but a control, used to determine what part of a predefined graph's effect is attributable to its topology.

Two distinctions in this taxonomy are worth stating plainly, because they are routinely blurred. The first is that *predefined* and *data-driven* are separate categories, not variants of one. Both produce a graph that is fixed before training, which is what makes them superficially similar, but they differ in *provenance*, meaning where the edge weights come from. A predefined graph's weights follow from a structural rule, while a data-driven graph's weights are estimated from observations. The second is that *adaptive* and *learned* are used as synonyms throughout this thesis, both denoting an adjacency optimised by gradient descent. What varies among adaptive conditions is not whether the graph is learned but what it is initialised from, which is why the seeded variants are named by their seed graph.

4.3.1 Predefined Graphs

A predefined graph is specified by its node set and its edge list. Both are given explicitly below, since a verbal description such as the channels are connected does not determine a graph.

BasicMotions has six nodes, one per channel, ordered ACCX, ACCY, ACCZ, GYROX, GYROY, GYROZ. Two rules generate the edges. Channels of the same modality are mutually connected, giving the three accelerometer edges and the three gyroscope edges, and channels of different modality are connected when they share an axis, giving ACCX–GYROX, ACCY–GYROY and ACCZ–GYROZ. This yields 9 undirected edges out of the 15 that a complete graph on six nodes would have, so the graph is **not** fully connected. The six absent edges are the cross-modality, cross-axis pairs such as ACCX–GYROY, which are omitted because no physical argument connects them.

Epilepsy has three nodes, one per accelerometer axis. The same-modality rule applies here too, and because all three channels share a modality the rule connects all of them, giving 3 undirected edges. On three nodes this happens to coincide with the complete graph. This is a consequence of the dataset having a single modality with three axes, not a design choice, and it is the only one of the four predefined graphs in this thesis that is fully connected.

DSA is defined at two granularities, which is what makes it the dataset that separates granularity from every other factor.

- **Coarse**, 5 nodes, one per sensor unit, being Torso, Right Arm, Left Arm, Right Leg and Left Leg. The Torso is a central hub joined to all four limbs, and two bilateral edges join arm to arm and leg to leg, giving 6 undirected edges out of 10 possible. Each node carries the nine channels of its unit as a feature vector.
- **Fine**, 45 nodes, one per channel. Channels within the same unit are fully connected, giving five nine-node cliques, and channels in anatomically adjacent units are connected when they measure the same physical quantity, for example $\text{Torso}_{\text{AccX}} \leftrightarrow \text{RightArm}_{\text{AccX}}$. This yields 234 undirected edges out of 990 possible, so the fine graph occupies 23.6% of the possible edges.

Neither DSA graph is fully connected. The fine graph deliberately reuses the coarse graph's body-segment adjacency rather than introducing new anatomical claims, so that the two differ in granularity alone. It is a conservative prior,

asserting that co-located channels are related and that the same quantity measured on adjacent segments is related, rather than asserting a measured physical coupling for every channel pair.

One property of the Phase-1 graphs is worth recording, because Section 6.3.3 later turns on it. In both graphs every node has the same degree, so Eq. 4.1 assigns every edge an identical weight, 0.250 on BASICMOTIONS and 0.333 on EPILEPSY (Fig. 4.1). Neither predefined graph therefore carries any information in its weights, only in its edge set.

A consequence of Eq. 4.1 matters for the DSA results too. Because the fine graph’s nodes have nearly equal degree, its 234 edges take only *three* distinct weights spanning 18 % end to end, whereas the coarse graph’s take two values differing by 29 %. The fine graph therefore carries almost no information in its weights, a point returned to in Section 6.3.3. Appendix A.2 lists every value.

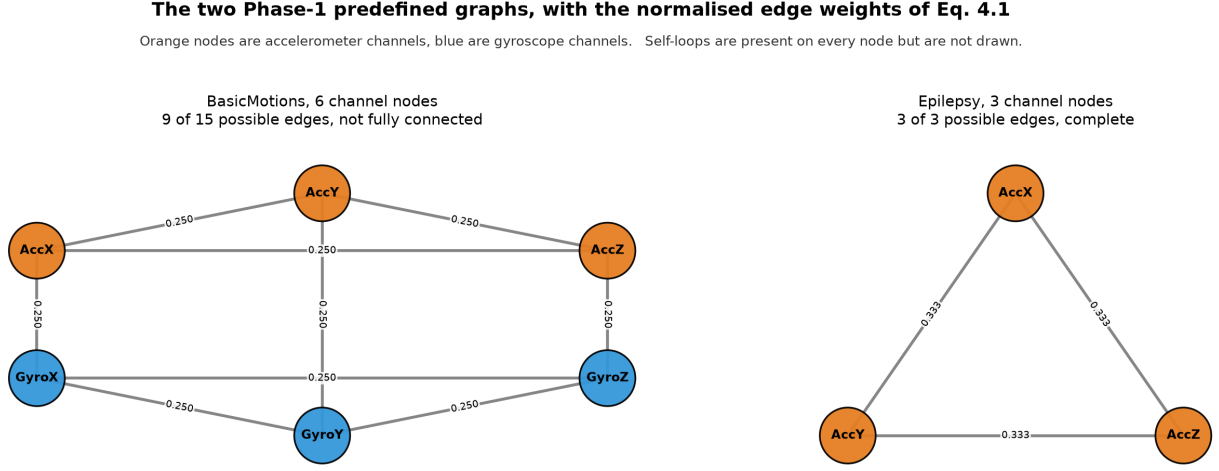


Figure 4.1: The two Phase-1 predefined graphs, drawn to the same conventions as DSA in Fig. 4.3. Node labels are channel names, edge labels are the weights that follow from Eq. 4.1, and self-loops are present on every node but are not drawn. BASICMOTIONS carries 9 of the 15 edges a complete graph on six nodes would have, so it is *not* fully connected, and the six absent pairs are the cross-modality cross-axis ones such as ACCX–GYROY. EPILEPSY is complete, but only because its three channels share a single modality, so the same-modality rule happens to connect all of them. Every edge in either graph carries the same weight, 0.250 for BASICMOTIONS and 0.333 for EPILEPSY, because within each graph every node has equal degree.

4.3.2 Data-Driven Graphs

A data-driven graph is estimated from the training split of the fold being evaluated, never from validation or test data. Three families are used.

The **signed correlation** graph retains an edge when the absolute Pearson coefficient between two channels satisfies $|r_{ij}| \geq \tau$, and uses the signed coefficient as the weight, so that anti-correlated channels are joined by a negative edge. Five threshold policies are evaluated, being a fixed $\tau = 0.3$, two data-driven thresholds at $\tau = \text{mean}|r|$ and $\tau = \text{mean}|r| + \text{std}|r|$, a top-50 % variant that keeps the strongest half of the pairs, and a keep-all variant that thresholds nothing. Threshold policy is treated as an experimental variable rather than a fixed choice because, as Fig. 3.4 shows, the magnitude of correlation varies by an order of magnitude between datasets, so a threshold that is sensible on DSA empties the graph on EPILEPSY.

The **DTW-similarity** graph converts per-channel prototype distances under Dynamic Time Warping into weights through a Gaussian kernel. DTW is included alongside correlation because the two measure genuinely different things. Pearson correlation captures synchronous linear covariation and is blind to relationships that are lagged or time-warped, which is exactly what occurs between body segments during periodic gait, where the legs move in anti-phase and the arms lag the torso. DTW aligns the sequences before comparing them, so disagreement between the two induced graphs is informative rather than redundant.

The **mutual k -nearest-neighbour** graph retains an edge between two channels only when each ranks the other among its k nearest neighbours in correlation or DTW similarity space. The mutual condition matters, because a one-sided kNN rule allows a high-variance channel to become a hub that every other channel points to. Fixing a per-node degree budget instead of a global threshold gives a graph whose density does not depend on the overall scale of the similarity values. The value $k = 3$ is used, being the smallest value that leaves the graph connected across all folds while remaining far sparser than the anatomical fine graph, which is deliberate, since it allows the DTW graph to test whether estimated weights help even at *lower* edge density than a rule-based alternative.

4.3.3 Adaptive Graphs, With and Without Seeding

In the adaptive strategies the adjacency is itself a parameter. Following Graph WaveNet [24], two node-embedding matrices $E_1, E_2 \in \mathbb{R}^{N \times d}$ produce, on each forward pass,

$$\tilde{A} = \text{softmax}(\text{ReLU}(E_1 E_2^\top)), \quad (4.2)$$

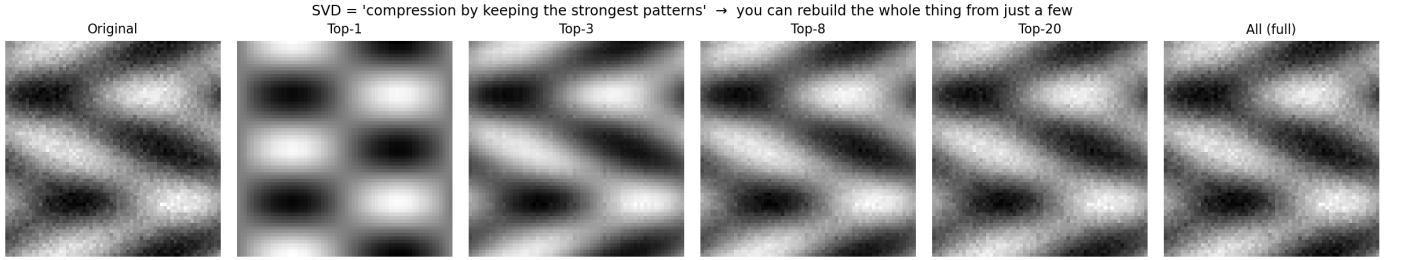
where the rectifier zeroes negative compatibilities and the row-wise softmax normalises the out-edges. The result is then symmetrised, passed through the graph preprocessing of Section 4.2.1, and sparsified by discarding weights below 0.05. The L_1 penalty of Section 4.5 prevents the learned graph from densifying.

Seeding a learned graph from a prior, and why the SVD step matters. The learner of Eq. 4.2 does not hold an adjacency directly. It holds the two embedding tables E_1 and E_2 whose product forms one. A fixed prior graph therefore cannot simply be handed to it, and converting one into that parameterisation is a graph preprocessing step in its own right, sitting between the strategies of Section 4.3 and the model.

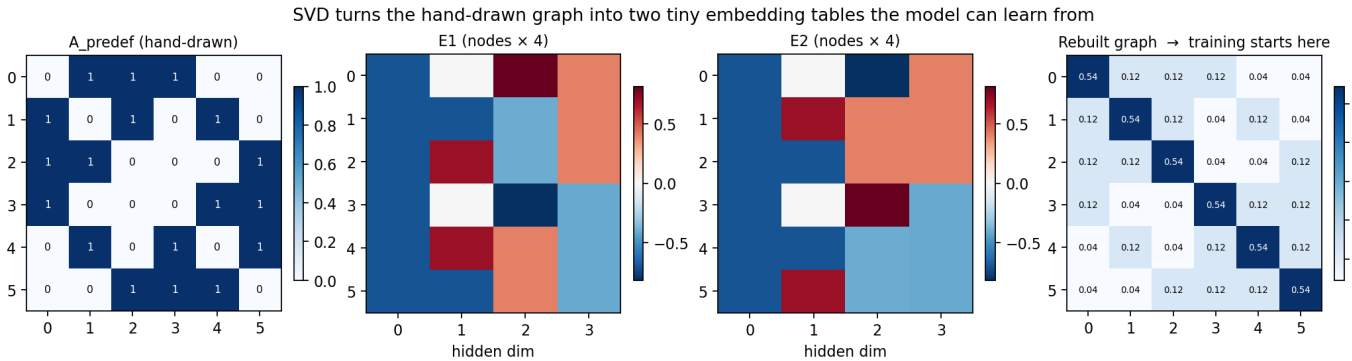
The conversion is done with a truncated *singular value decomposition* (SVD). An SVD factorises any matrix into a set of independent patterns ordered by how much of the matrix each one accounts for, so that keeping only the strongest d of them reconstructs the matrix approximately from far fewer numbers. Applied to an adjacency it answers the question the learner needs answered, namely which d node-embedding directions best reproduce this graph (Fig. 4.2).

Concretely, the embeddings are initialised from the decomposition $A = U \Sigma V^\top$ of a seed graph through $E_1 = U_d \Sigma_d^{1/2}$ and $E_2 = V_d \Sigma_d^{1/2}$, so that $E_1 E_2^\top = A_d$, the rank- d truncation of the seed. Training then begins from that reconstruction rather than from noise. By the Eckart–Young–Mirsky theorem [27], A_d is the best rank- d approximation of A obtainable by any factorisation of this form, so d bounds how much of the prior can survive into the learned graph rather than merely setting the model’s capacity. A complementary *elementwise* variant replaces the factorisation with a full $N \times N$ parameter matrix and thereby removes the SVD step entirely, which gives a direct ablation, since the two variants differ only in where the optimiser starts.

This step is consequential rather than a technicality. Removing the SVD seeding while changing nothing else costs 6.98% of accuracy on DSA (Section 6.3.6), which is the largest single effect among all the learning comparisons in this thesis. Where the optimiser starts inside the low-rank parameterisation constrains what it can subsequently reach.



(a) What a truncated SVD does to any matrix. Keeping only the strongest d patterns reconstructs the original approximately, and the reconstruction sharpens as d grows.



(b) The same operation applied to an adjacency, on a six-node example. The hand-drawn graph is factorised into the two embedding tables the learner parameterises, and their product is the graph training starts from.

Figure 4.2: The SVD seeding step, first in general and then on a graph. Panel (a) shows why the operation is described as keeping the strongest patterns, since a rank-1 truncation retains only the coarsest structure while rank-20 is already close to the original. Panel (b) applies exactly that to an adjacency matrix, which is the use made of it here. The reconstruction is deliberately approximate, because the truncation rank d is what bounds how much of the prior graph can survive into the learned one, by the Eckart–Young–Mirsky theorem. Colour encodes matrix entry value throughout, and the two embedding tables in panel (b) use a diverging scale because their entries may be negative.

After training, $\tilde{A}$ is reconstructed from the checkpointed embeddings and compared against the predefined graph by

edge-set Jaccard similarity. This is what allows a learned topology to be reported alongside its accuracy rather than treated as an opaque parameter.

4.3.4 The Random-Topology Control

Comparing an anatomical graph against *no* graph confounds two effects, the benefit of aggregating channels into a smaller structured node set and the benefit of connecting those nodes according to body structure. A predefined graph that beats a no-graph baseline demonstrates only that some graph helps. It does not establish that the domain knowledge in the edges is what helped. Answering RQ3 as posed, which asks whether the domain *knowledge* contributes anything beyond the aggregation the graph imposes, requires separating the two, and that requires a control rather than a further comparison.

The control is a graph with the same node set and the same number of edges as the coarse anatomical graph, being 5 nodes and 6 undirected edges, but with the edges placed uniformly at random among the available node pairs subject to the graph remaining connected. This is the degree- and density-matched randomisation standard in network science [28], transplanted to this setting. Every property the anatomical graph supplies is held fixed except the one under test, namely which nodes are joined. The random graph then receives identical treatment, with the same normalisation, the same self-loops and no learning, so any accuracy difference is attributable to topology alone.

Because a single random draw could be unrepresentatively good or bad, five independent random graphs are generated, using graph seeds {42, 123, 456, 789, 1024}, and each is trained to completion across all eight folds. This produces a distribution over random topologies rather than a point estimate, which makes the comparison against anatomy a comparison of expectations. The control is evaluated in both roles a structural prior can play, first as a predefined graph, which isolates the value of anatomical structure, and second as the seed for an adaptive graph, which isolates whether the learner’s benefit depends on the prior it starts from. Section 6.3.4 reports both.

4.4 Model Architectures

Every model in the study is built from the same three layers of machinery: a graph convolution operator, a recurrent cell that calls it, and one classifier wrapper shared across all of them. The baselines of Section 4.4.5 are the same wrapper with the graph removed, which is what makes the comparison in RQ1 an ablation of the graph mechanism rather than a contest between unrelated architectures.

4.4.1 Graph Convolution Operators

A temporal graph neural network replaces the linear maps inside a recurrent cell with a graph convolution, so that message passing across the sensor graph occurs at every time step rather than once. Two convolution operators are used.

Chebyshev spectral convolution. Following Defferrard et al. [16], spatial convolution is a K -order polynomial filter of the graph, evaluated through the Chebyshev three-term recurrence on the normalised adjacency $\hat{A}$, which serves as a proxy for the scaled Laplacian under the first-order approximation of Kipf and Welling [11]. The recurrence is $T_0 = I$, $T_1 = \hat{A}$ and $T_k = 2\hat{A}T_{k-1} - T_{k-2}$, and the convolution of node features X is $Z = \sum_{k=0}^{K-1} T_k(\hat{A}) X W_k + b$. With $K = 2$, used throughout, the filter holds one weight matrix for a node’s own features and one for its one-hop neighbourhood aggregate, giving separate learnable projections for self- and neighbour-information.

Diffusion convolution. DCRNN [19] instead models information flow as a truncated bidirectional random walk, $Z = \sum_k A^k X W_k^{\text{fwd}} + \sum_k (A^\top)^k X W_k^{\text{bwd}}$, learning separate forward and backward weights per hop. For a symmetric sensor graph, where $A = A^\top$, the two walks coincide and the directional machinery is inactive.

4.4.2 Recurrent Graph Cells

Four recurrent graph cells are compared, and they differ in ways that later explain their behaviour.

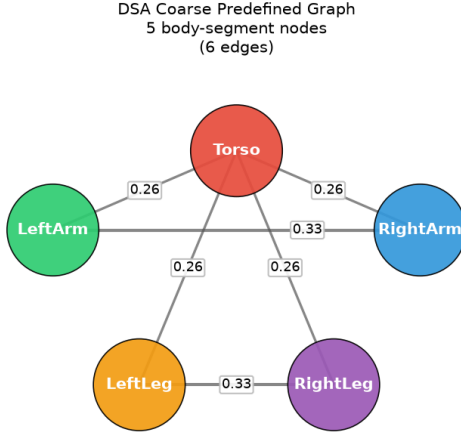
GConvLSTM [18] retains standard LSTM gating but computes all four gates from a single Chebyshev convolution over the concatenation $[x_t, h_{t-1}]$, giving $\{i_t, f_t, o_t\} = \sigma(\Theta \star_g [x_t, h_{t-1}])$ and $g_t = \tanh(\cdot)$, with $c_t = f_t \odot c_{t-1} + i_t \odot g_t$ and $h_t = o_t \odot \tanh(c_t)$. The consequence, and the motivation for the design, is that the forget gate of node i becomes informed by the current state of node j whenever the edge (i, j) exists, so temporal memory and spatial structure are fused in a single update.

GConvGRU uses the GRU’s two gates, reset r and update z , and requires two convolutions rather than one, because the reset gate must modulate h_{t-1} **before** the candidate state is formed.

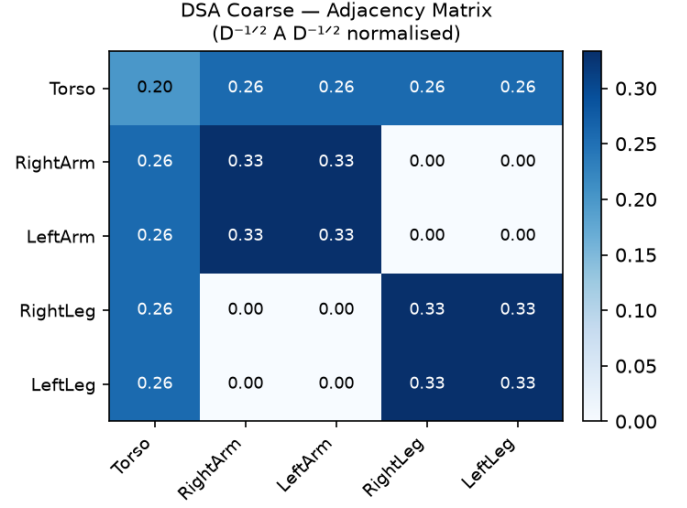
A3TGCN [20] wraps GConvGRU with a per-node sigmoid attention gate $\text{gate} = \sigma(W h_t)$, which suppresses or amplifies each node’s contribution at each time step. This is useful when only some channels are discriminative for the activity currently being performed.

DCRNN shares the GRU gate structure but substitutes diffusion convolution for the Chebyshev operator.

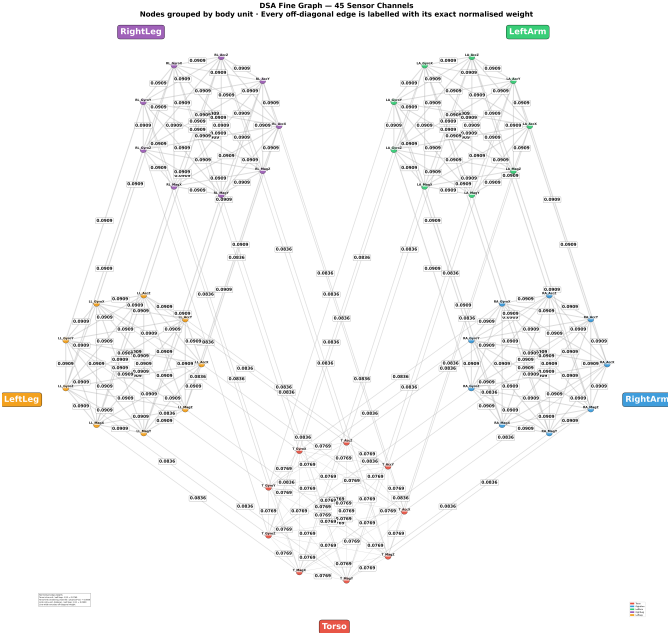
Phase 2 (DSA) — Coarse Predefined Graph
Anatomical adjacency: Torso hub + bilateral symmetry (Arms / Legs)



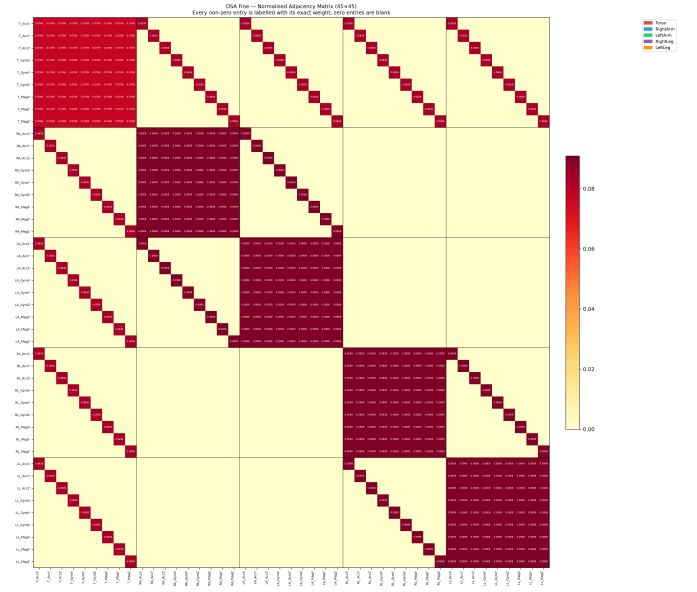
(a) Coarse, 5-node network



(b) Coarse, adjacency matrix



(c) Fine, 45-node network



(d) Fine, adjacency matrix

Figure 4.3: The two predefined DSA graphs. The coarse graph has 5 body-segment nodes and 6 undirected edges with the Torso as hub. The fine graph has 45 channel nodes and 234 undirected edges, whose five dense intra-unit blocks are visible as the block diagonal of the adjacency heatmap. The heatmaps show 513 non-zero entries for the fine graph, being 234 undirected links counted twice plus 45 self-loops. All weights follow from Eq. 4.1.

4.4.3 The TGNN Classifier

One wrapper turns any of these cells into a classifier, and it is identical across every graph strategy. A linear projection first lifts each node’s input features to hidden dimension $H = 64$. The time axis is then unrolled explicitly, which is what allows the adjacency to be supplied at every step, through stacked cells with inter-layer LayerNorm and dropout. Node features are mean-pooled to a per-step graph representation. A temporal attention pooling then forms the sequence vector as a learned convex combination over time, $\alpha_t = \text{softmax}_t(\text{MLP}(h_t))$ and $z = \sum_t \alpha_t h_t$, so that the model can weight the most discriminative temporal segments rather than relying on the final step alone. A two-layer head produces the logits.

The adjacency is retrieved through a single accessor that returns either the fixed buffer, for predefined and data-driven graphs, or a forward pass of the adaptive learner. Switching graph strategy therefore changes one line and nothing else, which is the implementation property that makes the benchmark controlled.

4.4.4 Hybrid-TGNN

To test whether a structural prior and a learned graph are complementary rather than competing, the Hybrid model uses an effective adjacency that blends the two through a single learnable scalar, $A_{\text{eff}} = \alpha \hat{A}_{\text{predef}} + (1 - \alpha) \hat{A}_{\text{learned}}$ with $\alpha = \sigma(\theta_\alpha)$. Initialising $\theta_\alpha = 2$, so that $\alpha \approx 0.88$, makes the model begin by trusting the domain graph and allows gradient descent to move toward the learned graph only if the data support it. The converged value of α is therefore an interpretable readout of how much the prior is worth to the model.

4.4.5 Baseline Models

The baselines treat the multivariate window as an unstructured sequence, with no mechanism encoding inter-channel structure, so they form the reference against which the graph inductive bias is measured in RQ1.

LSTM [6] is the primary recurrent baseline and the exact structural counterpart of GConvLSTM and GConvGRU, which replace its linear maps with graph convolutions. This makes the LSTM-versus-GConvLSTM contrast the cleanest available test of what the graph adds. Its forget-gate biases are initialised to one [29] so that the cell defaults to remembering, and a LayerNorm [30] on the final state stabilises training at the small BASICMOTIONS batch sizes.

TCN [7] stacks weight-normalised [31] causal dilated convolutions with residual connections. With kernel size three and dilations 1, 2, 4 the receptive field is $1 + 2(1 + 2 + 4) = 15$ steps, so it extracts local temporal motifs before global average pooling.

CNN-BiGRU [15] is the dominant wearable-HAR architecture, extracting local features with convolutions and modelling global order with a bidirectional GRU. It is the strongest baseline in this study on both EPILEPSY and DSA, measured by mean accuracy.

Scope of the baseline set. The baselines are deliberately restricted to recurrent and convolutional sequence models, because each is a direct structural counterpart of a component under test, which keeps the comparison a controlled ablation of the graph mechanism rather than an open architecture race. Purely attention-based sequence models are consequently outside this scope, since they neither share a component with the TGNN cells nor isolate the graph, and including them would change the question from what the graph contributes to which architecture is best.

4.5 Training Objective

All models minimise the multiclass cross-entropy between the predicted class-probability distribution and the one-hot encoded true label. For a batch of B windows,

$$\mathcal{L}_{\text{CE}} = -\frac{1}{B} \sum_{b=1}^B \sum_{k=1}^K y_{b,k} \log \hat{y}_{b,k}, \quad \hat{y}_b = \text{softmax}(z_b), \quad (4.3)$$

where $z_b \in \mathbb{R}^K$ are the logits for window b , $y_{b,k}$ is one when window b has class k and zero otherwise, and the reduction over the batch is a mean rather than a sum so that the gradient scale does not depend on batch size. The loss is unweighted, meaning no per-class weighting is applied, because BASICMOTIONS and DSA are exactly balanced and EPILEPSY’s imbalance is mild (Section 3.3). Macro-F1 is reported alongside accuracy so that the consequences of this choice remain visible.

Models with a learned adjacency add an L_1 penalty on that adjacency,

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \frac{1}{N^2} \sum_{i,j} |A_{ij}|, \quad \lambda = 10^{-3}, \quad (4.4)$$

which keeps the learned graph sparse and therefore interpretable, rather than allowing it to collapse into a dense and uninformative matrix. The penalty is normalised by N^2 so that λ carries the same meaning at five nodes and at forty-five. Gradients are clipped to norm one. The adaptive embeddings, the hybrid blend parameter θ_α , the recurrent weights and the classification head are all optimised jointly in a single backward pass.

Chapter 5

Experimental Design

Chapter 4 defined five ways to construct a graph and one control that removes the domain knowledge from a graph while leaving everything else about it intact. A comparison between them is informative only if nothing else varies alongside them. This chapter states the protocol that holds everything else fixed, the complete set of experiments performed, the mapping from those experiments to the research questions, and the statistical procedure applied to the outcomes. Fig. 5 gives the design as a whole.

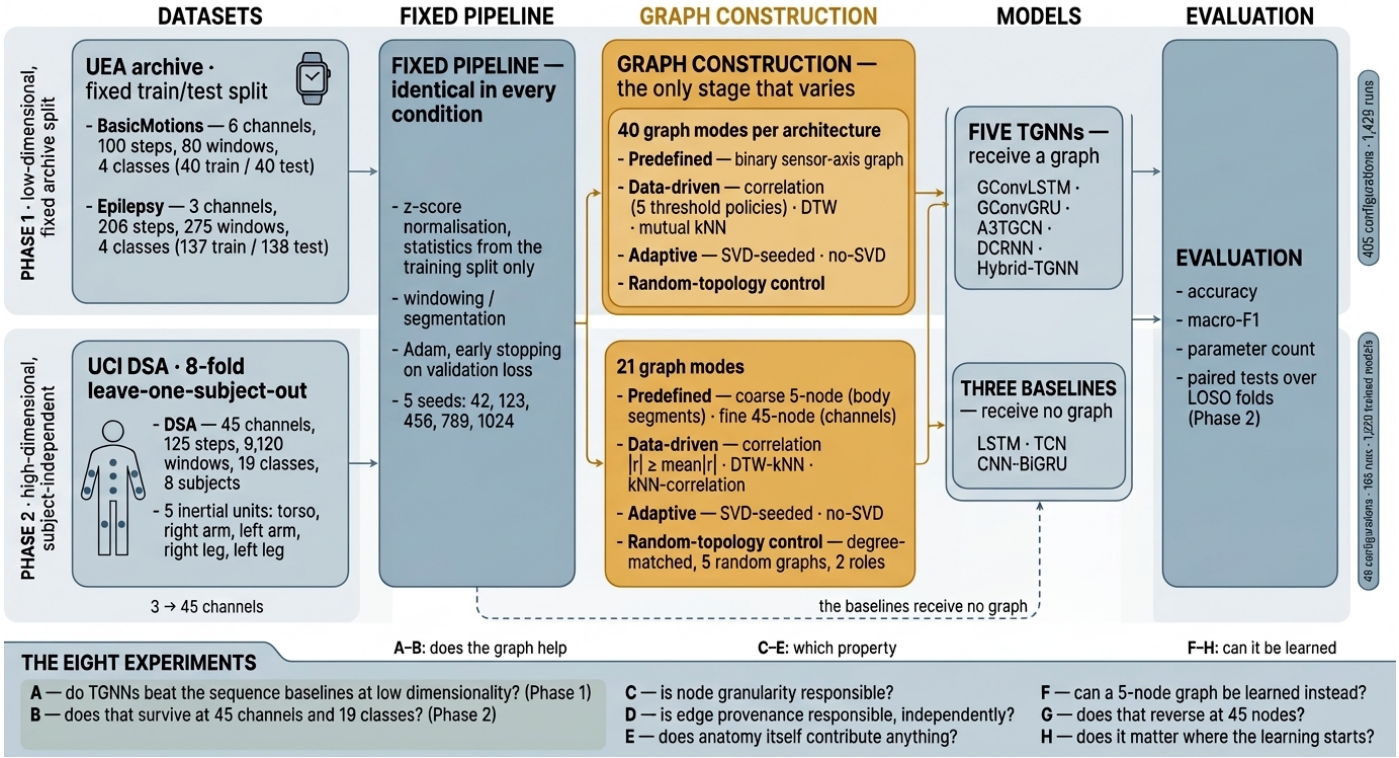


Figure 5.1: The experimental framework of this thesis.

5.1 Protocol

A configuration is one combination of dataset, model and graph strategy, and it is trained once per seed. Five seeds {42, 123, 456, 789, 1024} are used, and results are reported as the mean and standard deviation across them. Every combination evaluated in this thesis is enumerated in Table 5.1, together with the arithmetic that produces the reported totals.

Five is a deliberate choice. Run-to-run variation from initialisation and batch ordering is of the same order as several of the differences under study, with seed standard deviations here between 1 % and 4 %, so single-run comparisons would not be interpretable. Five seeds hold that noise an order of magnitude below the effects being measured while leaving budget for the breadth of graph conditions the research questions actually turn on.

Phase 1 uses the archive’s fixed train/test split. Phase 2 uses eight-fold leave-one-subject-out cross-validation and aggregates first over folds, then over seeds. Optimisation uses Adam [32] with early stopping on validation loss, a maximum budget of 150 epochs and patience-based stopping. Early stopping monitors validation loss rather than accuracy because loss is the smoother signal at nineteen classes, where accuracy moves in discrete steps of 1/1140 per evaluation.

5.2 Metrics

Three quantities are reported for every configuration.

Accuracy is the proportion of correctly classified windows. It is the primary metric and is an unbiased summary on BASICMOTIONS and DSA, both of which are exactly balanced by construction.

Macro-F1 is the unweighted mean of the per-class F1 scores. It is reported alongside accuracy for two reasons. On EPILEPSY the classes are mildly imbalanced, so accuracy alone could in principle be inflated by performance on the larger classes. On DSA the classes are balanced but several are mutually confusable, and macro-F1 penalises a model that achieves good overall accuracy by sacrificing one difficult class entirely. Because macro-F1 weights every class equally regardless of frequency, it is the stricter of the two measures, and on DSA it runs consistently 2 to 5 points below accuracy.

Computational cost is reported as the number of trainable parameters, which RQ1 requires explicitly, since a model that matches a baseline’s accuracy at a fifth of its size is not equivalent to it. The proposal also listed wall-clock training time. That measure is not reported, because the runs behind this study were executed across several machines and several revisions of the training harness and their recorded times are therefore not comparable with one another.

For DSA the learned-adjacency drift $\|A_{\text{final}} - A_{\text{init}}\|_F$ is additionally tracked, to quantify how far an adaptive graph moves from its initialisation.

5.3 The Experiment Matrix

Table 5.1 enumerates every experiment performed. The arithmetic that produces the totals is given explicitly, and the distinction between the three countable quantities is maintained throughout the thesis. A configuration is one (dataset, model, graph mode) combination. A run is one configuration trained with one seed. A trained model is one run on one fold, so in Phase 1 a run is one trained model, whereas in Phase 2 a run is eight.

Table 5.1: The complete experiment matrix. Graph modes are the distinct adjacency specifications defined in Section 4.3. Phase-1 counts are per dataset and are summed over the two datasets in the totals row. Configurations carrying five seeds are those on which statistical claims rest, while single-seed configurations are the random-topology control, where replication is provided by five independent random graphs rather than by five training seeds.

Model	Phase 1 (fixed split)		Phase 2 (8-fold LOSO)	
	Graph modes	Runs	Graph modes	Runs
A3TGCN	40	280	20	60
GConvGRU	40	280	4	20
GConvLSTM	40	280	14	30
DCRNN	40	280	4	20
Hybrid-TGNN	40	279	4	20
LSTM	1 (no graph)	10	1 (no graph)	5
TCN	1 (no graph)	10	1 (no graph)	5
CNN-BiGRU	1 (no graph)	10	1 (no graph)	5
Total	41 distinct	1 429	21 distinct	165

Phase 1 comprises 405 configurations, of which 256 were run at five seeds and 149 at one, giving $256 \times 5 + 149 \times 1 = 1\,429$ runs, and one trained model per run. Phase 2 comprises 49 configurations, of which 29 were run at five seeds and 20 at one, giving $29 \times 5 + 20 \times 1 = 165$ runs, and eight trained models per run for a total of $165 \times 8 = 1\,320$ trained models. The 1429 Phase-1 runs and 1320 Phase-2 models together constitute the evidence base of this thesis.

Which combinations are excluded, and why. The grid is intentionally not uniform. The five families that carry population-level claims, being the no-graph baselines together with the predefined coarse, predefined fine, adaptive and data-driven correlation graphs, are complete at five seeds for all five architectures in Phase 2, and every result reported as statistically significant rests on that balanced core. The remaining families answer mechanism questions, which are settled by varying the graph while holding the architecture fixed, so they are run in depth on one architecture. Adding architectures there would multiply cost without sharpening the contrast, because the contrast is between graphs rather than between models. The remaining architecture-by-family cells would add roughly 130 configurations and 1300 GPU-hours without altering any claim made here, so that budget was allocated instead to the controls, which do change what can be concluded.

Which architecture carries the depth experiments. A3TGCN is used, for one practical reason and one that protects the conclusion. Practically, it is the smallest model in the study on DSA at 424k parameters, so a fixed budget buys more configurations on it than on any alternative. Substantively, its provenance gap is among the *smallest* of the five architectures. Measured on the correlation family, which is complete for every architecture, replacing rule-assigned weights with data-estimated ones at forty-five nodes is worth 10.0 % on A3TGCN against a five-model mean of 11.0 % and a maximum of 12.0 % on GConvGRU. Running the depth experiments on A3TGCN therefore *understates* the effect they establish, whereas a choice that flattered the hypothesis would have used GConvGRU.

The random-topology control is treated differently. The obvious objection to it is that A3TGCN’s attention gate might mask a topology effect, so it is replicated on GConvLSTM, chosen precisely for being architecturally the most unlike A3TGCN of the five, with Chebyshev spectral convolution inside an LSTM and no attention anywhere. Replicating on a maximally different architecture tests that objection directly, whereas a third mechanistically similar cell would add cost without adding evidence.

Scope difference between the phases. The correlation family is narrower in Phase 2 by design. Phase 1 sweeps five threshold policies, each in three forms, being predefined, SVD-seeded adaptive and elementwise adaptive, giving fifteen correlation modes per model. Phase 2 retains all three forms but only the $\geq \text{mean}|r|$ threshold, giving three modes. A single DSA configuration costs eight LOSO folds against minutes for a Phase-1 fixed-split run, so Phase 2 carries all three forms at the one threshold that answers the transfer question and spends the remaining budget on the controls. The $\geq \text{mean}|r|$ policy was selected because it is data-driven, requiring no hand-tuned constant, and was among the strongest Phase-1 policies. Comparisons of the correlation family across phases therefore hold the threshold policy fixed and vary only the dataset.

5.4 Mapping Research Questions to Experiments

Table 5.2 states which experiments answer which research question, and the paragraphs that follow explain why each set of experiments is the appropriate evidence.

Chapter 6 reports the outcomes as eight named experiments, run in an order that narrows the question at each step.

Experiments A and B establish whether the graph helps at all, and answer RQ1. Experiments C and D decompose *which* property of the graph is doing the work, changing one property at a time, and answer RQ2. Experiments E, F, G and H ask whether the domain knowledge in that property is what matters or whether the same structure can be obtained without it, and answer RQ3.

Table 5.2: Which experiments answer which research question. The experiments themselves are listed above and reported in Chapter 6. Every one is a paired contrast in which all factors except the named one are held fixed, which is what allows the answer to be attributed to that factor.

RQ	Experiments	What settles it
RQ1	A and B, read with Section 6.4	The graph models against LSTM, TCN and CNN–BiGRU at both dimensionalities, scored on parameter count as well as accuracy, and deliberately evaluated under more than one graph so that the conditionality clause can be answered rather than assumed. Section 6.4 supplies the model-side reading, in which the architecture proves the smaller axis of variation.
RQ2	C and D, read across the two phases	The coarse-against-fine contrast, which holds the dataset, architecture, training procedure and anatomical rule fixed while changing only the node count, and the correlation-against-fine contrast, which holds the node count at 45 and changes only where the weights come from. The transfer clause is settled by reading the correlation family’s disagreement between the two phases against those two quantities.
RQ3	E, with F, G and H	The random-topology control, which holds node count, edge count, normalisation and training fixed and removes only the domain knowledge, evaluated in both roles a prior can play and replicated on a second architecture. Experiments F, G and H then ask whether the structure has to be supplied by hand at all.

Three choices in that mapping are worth defending, because a reader could reasonably have expected otherwise.

The first is that RQ3 rests on a control rather than on a comparison. Every other row of the table contrasts two graph *constructions*, and every such contrast changes at least two things at once, so none of them can attribute an outcome to the domain knowledge as opposed to the structure that carries it. Only a condition that holds the construction fixed and removes the knowledge alone can do that, which is what Experiment E is and why RQ3 could not have been folded into RQ2.

The second is that RQ2’s transfer clause is not a research question of its own. It could have been, and an earlier arrangement of this study made it one, but the experiments show that the property which governs performance at a given dimensionality and the property which governs whether a conclusion survives a change in dimensionality are the same property. Asking about them separately would present one result as two and would obscure the connection that makes it informative.

The third is that RQ2 turns on the coarse-against-fine contrast rather than on any comparison between datasets. That pair holds the dataset, the architecture, the training procedure and the anatomical rule fixed while changing only the node count, which no cross-dataset comparison can do, since two datasets differ in every respect at once. The cross-phase comparison is then read *against* that measured quantity rather than being asked to establish one.

5.5 Statistical Comparison

Differences between graph strategies on DSA are tested with the two-sided Wilcoxon signed-rank test, paired across the eight LOSO folds. Four properties of the protocol dictate the choice.

At start, every configuration is evaluated on the same eight held-out subjects, so fold k of strategy A and fold k of strategy B share a test subject. Subject difficulty is the dominant variance component here, since a single configuration ranges from roughly 62 % to 96 % across folds, far exceeding the differences between strategies. An unpaired test would charge that shared variance to noise and forfeit nearly all statistical power. Including a non-parametric form, as eight observations cannot support the normality assumption behind a paired t -test, and the fold accuracies are visibly skewed by a small number of difficult subjects. The Wilcoxon test ranks the paired differences rather than using their magnitudes [33], so it remains valid without a distributional assumption. It is the test Demšar [34] recommends for comparing two classifiers over a set of matched evaluations.

Furthermore, the aggregation of seeds, about five seeds of one configuration on one fold are repeated measurements of a single quantity rather than independent samples. Treating them as $5 \times 8 = 40$ independent pairs would inflate the nominal sample size and understate the p -values. Seeds are therefore averaged within each fold before testing, and where a contrast spans several architectures the pairing is over (model, fold).

Reporting many pairwise tests in the appendix inflates the family-wise error rate, so Holm–Bonferroni adjustment [35] is applied across the reported family. Holm’s step-down procedure is uniformly more powerful than plain Bonferroni, and unlike false-discovery-rate procedures it bounds the probability of *any* false positive, which is the appropriate guarantee when individual claims are made about individual comparisons in the text.

As about the effect size, a p -value indicates whether a difference is detectable, not whether it is large enough to matter. Each test is therefore accompanied by the matched-pairs rank-biserial correlation r , the Wilcoxon-native effect size bounded in $[-1, 1]$, computed as the difference between favourable and unfavourable signed ranks divided by their total. Reporting both guards against the opposite errors of dismissing a real effect for lack of significance at eight folds, and of promoting a detectable but negligible difference.

5.6 Implementation and Reproducibility

Models are implemented in PyTorch [36] with PyTorch-Geometric [37] and PyTorch-Geometric-Temporal [38]. All seeds, splits and hyperparameters are fixed and logged, and each run persists its per-fold and aggregate metrics as JSON. Phase 2 is executed with data-parallel scheduling on single GPUs in the DSRI and SURF environments provided by the department. Because these recurrent graph models step through 125 time steps on small graphs they are launch-bound and individually under-utilise the device, so several experiments run concurrently to keep it saturated, and `torch.compile` is enabled for the DSA runs to reduce kernel-launch overhead.

The complete pipeline, covering preprocessing, graph construction, model definitions, training, evaluation and the analysis code that produces every table and figure reported here, is submitted alongside this thesis. Configuration files fix the optimiser, schedule and seeds for both phases, so any single reported configuration can be reproduced with one command. Neither dataset is redistributed. BASICMOTIONS and EPILEPSY are retrieved from the UEA archive on first use and DSA from the UCI repository [14].

Chapter 6

Results and Analysis

The purpose of this chapter is to determine, under one fixed protocol, what a temporal graph neural network contributes to multivariate activity recognition, and which property of the supplied graph is responsible for that contribution. The evidence is presented in the order in which it becomes interpretable, so that each comparison is read against the structure it depends on.

One comparison that contains the whole argument. Before the experiments begin it is worth isolating the single pair that motivates all of them. GConvLSTM is an LSTM in which the linear maps inside the cell have been replaced by graph convolutions and nothing else has changed (Section 4.4.2), so the LSTM-to-GConvLSTM step adds a graph and holds the temporal machinery, the data, the preprocessing and the schedule fixed. On DSA that step is worth 2.3 % when the graph supplied is the coarse anatomical one, taking 79.1 % to 81.4 %. Supplied instead with the fine anatomical graph, built by the same rule from the same anatomy over the same channels, the identical step is worth -9.4 %, taking 79.1 % to 69.7 %. On EPILEPSY the same step is worth 4.1 %, 84.9 % to 89.0 %.

One architectural change therefore spans nearly twelve points of accuracy on one dataset, and its sign is decided entirely by which graph accompanies it. Asking whether a graph inductive bias helps is consequently the wrong question, because it has no single answer. The question this chapter answers instead is which property of the graph decides the sign, and the eight experiments that follow isolate one candidate property each.

Section 6.1 reports the low-dimensional phase and answers RQ1. It opens with the graphs themselves (Section 6.1.1), reconstructing the learned adjacency from trained checkpoints and comparing it against the predefined one, because the accuracy results that follow are only interpretable once it is clear what each model was given. It then establishes how the five graph architectures compare with the three sequence baselines on BASICMOTIONS and EPILEPSY, on accuracy and on computational cost, and shows that this comparison is already conditional on the choice of graph.

Section 6.3 moves to the high-dimensional phase and addresses RQ2 and RQ3 in four steps. The granularity of the predefined graph is examined first (Section 6.3.2), since it is the largest single effect in the study and every later comparison is read against it. The provenance of the edge weights is isolated next (Section 6.3.3) by holding the node count fixed at forty-five and varying only where the weights come from. Those two sections answer RQ2. The random-topology control (Section 6.3.4) then separates the contribution of anatomical structure itself from the contribution of merely having a graph, which is RQ3. Section 6.4 finally attributes the remaining variation between the five architectures to identifiable properties of their recurrent cells, which completes the model-side half of RQ1. Every comparison reported here is additionally subjected to paired significance testing, which is reported in full in Appendix A.4.

Section 6.5 closes the chapter by collecting the findings that the preceding sections establish. Their interpretation with respect to the research questions is deferred to Chapter 7.

Scope of the reported experiments. Phase 1 comprises 1429 runs across all families. Phase 2 comprises 165 runs, each of which is eight subject-independent folds, giving 1320 trained models (Table 5.1). Five families span every architecture at five seeds, namely the no-graph baselines, predefined coarse, predefined fine, adaptive, and the data-driven correlation graph, and every effect this chapter reports rests on that balanced core. Both halves of the random-topology control are run on two architectures, A3TGCN and GConvLSTM. Two further families, DTW-kNN and kNN-correlation, are run on A3TGCN. Sample sizes are annotated throughout.

6.1 Phase 1 — Low-Dimensional BasicMotions & Epilepsy Datasets

6.1.1 The Graphs Each Phase-1 Model Receives

Before any accuracy is reported, the graphs themselves are examined, since every comparison that follows depends on which adjacency a model was given. For a representative architecture and each graph source, Figs. 6.5–6.2 trace the topology from the predefined graph through its SVD-seeded and no-SVD adaptive forms, alongside a random-topology control (node radius = weighted degree, edge labels = |weight|, with the Jaccard against the predefined seed). Two patterns recur. First, the accuracy tracks the *graph source*, not the learning. The strong kNN seed (A3TGCN, Fig. 6.2) reaches 98.8 % as a predefined graph, and pushing it through the adaptive learner *lowers* it (to 95.1 %/93.9 %), whereas

the weak binary seed (Fig. 6.1) starts low and gains little. Second, on these tiny graphs the SVD-seeded learner barely moves (Jaccard ≈ 1 on EPILEPSY’s 3-node graph, where the rank- d factorisation is effectively full-rank), so the *adaptive* advantage seen at higher dimensionality (Phase 2) has no room to appear here, so consistent with the finding that *graph construction*, not end-to-end learning, drives Phase-1 accuracy.

A third pattern is visible only once the composites are read against the correlation statistics of Fig. 3.4, and it is the sharpest thing in this section. The fixed-threshold correlation graph (Fig. 6.3) applies the $|r| \geq 0.3$ policy of Section 4.3.2 to a dataset whose three inter-axis coefficients are 0.111, 0.036 and 0.092. Every one falls below the cut, so the graph it produces is *empty*. It has three nodes and no edges whatsoever. A model given that graph has no message passing left to perform and reduces to processing each channel independently through its self-loops, which is architecturally a sequence model wearing a graph model’s wrapper. It reaches 98.1 %. The same architecture on the predefined axis graph, which does carry three anatomically motivated edges, reaches 92.6 %. Removing every edge is therefore worth 5.5 points *more* than connecting the axes as the domain suggests. Two readings follow, of unequal weight. The narrow one concerns threshold policy, and it is what motivates treating that policy as an experimental variable in Section 4.3.2, since a constant chosen without reference to the observed correlation scale can silently delete the entire graph while the pipeline continues to report a perfectly ordinary accuracy. The broader one is that a graph carrying no relational information at all has outperformed a hand-designed anatomical one, which is the direction the random-topology control takes in Section 6.3.4, arrived at here by a different route. This is one architecture on one fixed split and is not offered as a result in its own right, but it is worth noticing that the first hint of the study’s central negative finding appears this early. The remaining four composites, covering the other two EPILEPSY graph sources and both BASICMOTIONS sources, are given after them and show the same pattern.

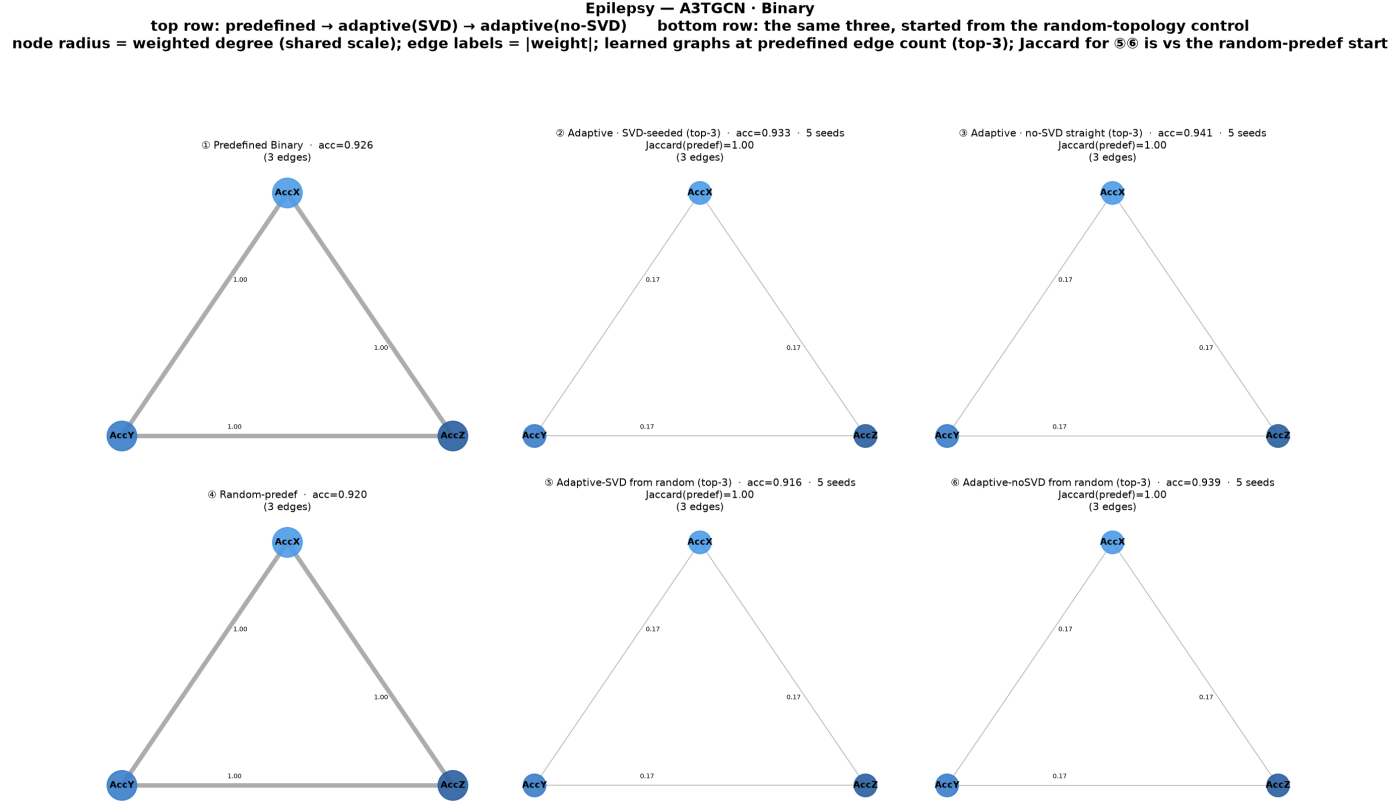


Figure 6.1: EPILEPSY (A3TGCN), **binary** (fully-connected) domain graph. The 3-node accelerometer graph. The SVD-seeded adaptive graph is essentially identical to the seed (Jaccard ≈ 1).

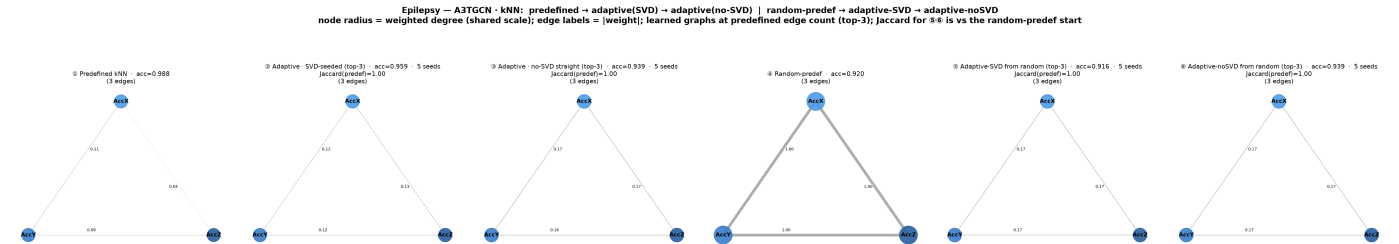


Figure 6.2: EPILEPSY (A3TGCN), **kNN** data-driven graph—the strongest EPILEPSY configuration at 98.8 % as a predefined graph. Applying the adaptive learner *reduces* accuracy, showing the good data-driven graph is best used directly.

Epilepsy — A3TGCN · Correlation, fixed $|r| \geq 0.3$
top row: predefined → adaptive(SVD) → adaptive(no-SVD) bottom row: the same three, started from the random-topology control
node radius = weighted degree (shared scale); edge labels = |weight|; learned graphs at predefined edge count (top-6); Jaccard for ⑤⑥ is vs the random-predef start

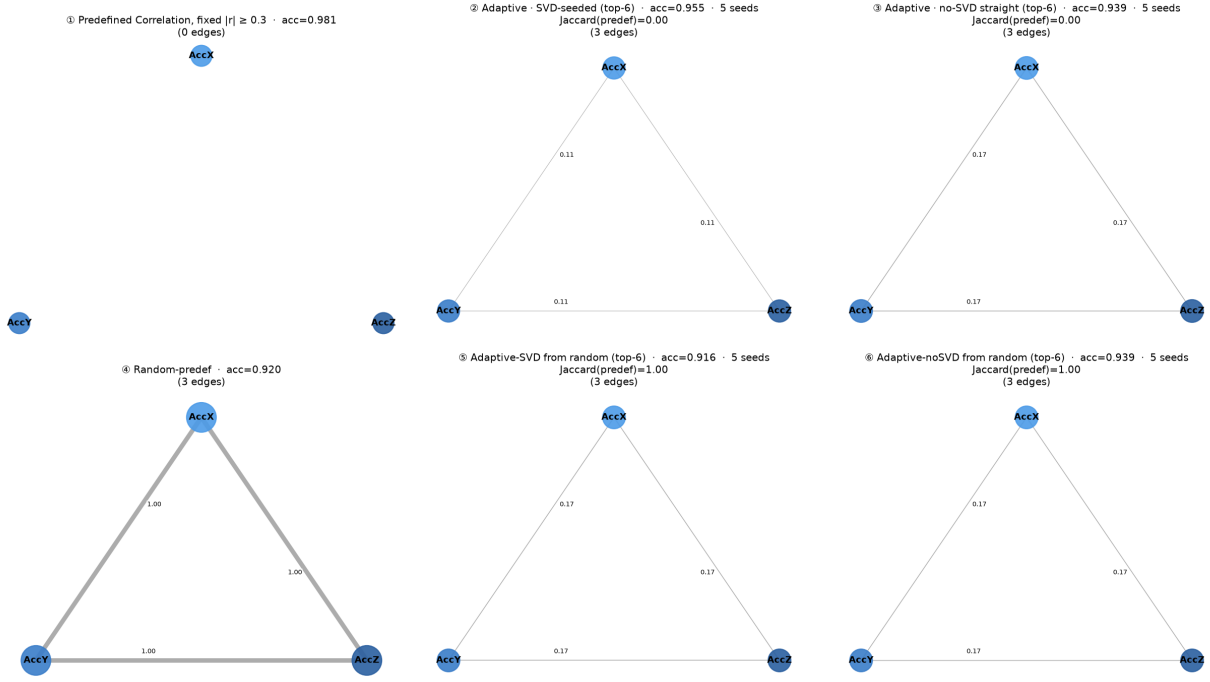


Figure 6.3: EPILEPSY (A3TGCN), the **fixed-threshold** correlation graph, meaning the $|r| \geq 0.3$ policy of Section 4.3.2 rather than one of the four data-driven thresholds. Panel ① is empty, carrying three isolated nodes and no edges, because all three of this dataset’s inter-axis coefficients fall below the threshold. It therefore has no edge labels to compare against Fig. 6.1 or Fig. 6.2, and the accuracy printed above it is what this architecture scores with no message passing at all. The surrounding text reads that number against the predefined axis graph.

Epilepsy — A3TGCN · DTW
top row: predefined → adaptive(SVD) → adaptive(no-SVD) bottom row: the same three, started from the random-topology control
node radius = weighted degree (shared scale); edge labels = |weight|; learned graphs at predefined edge count (top-3); Jaccard for ⑤⑥ is vs the random-predef start

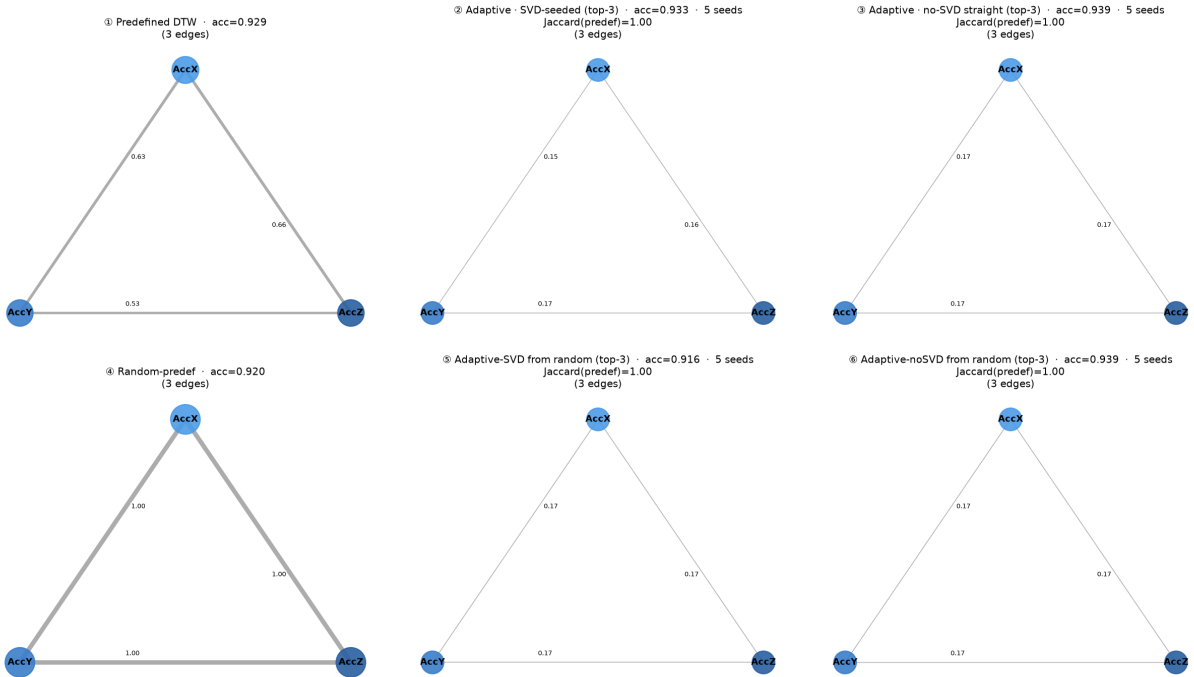


Figure 6.4: EPILEPSY (A3TGCN), **DTW-similarity** data-driven graph—the fourth graph source, completing the binary/correlation/DTW/kNN comparison for this model.

BasicMotions — GConvLSTM · Binary
top row: predefined → adaptive(SVD) → adaptive(no-SVD) bottom row: the same three, started from the random-topology control
node radius = weighted degree (shared scale); edge labels = |weight|; learned graphs at predefined edge count (top-9); Jaccard for ⑤⑥ is vs the random-predef start

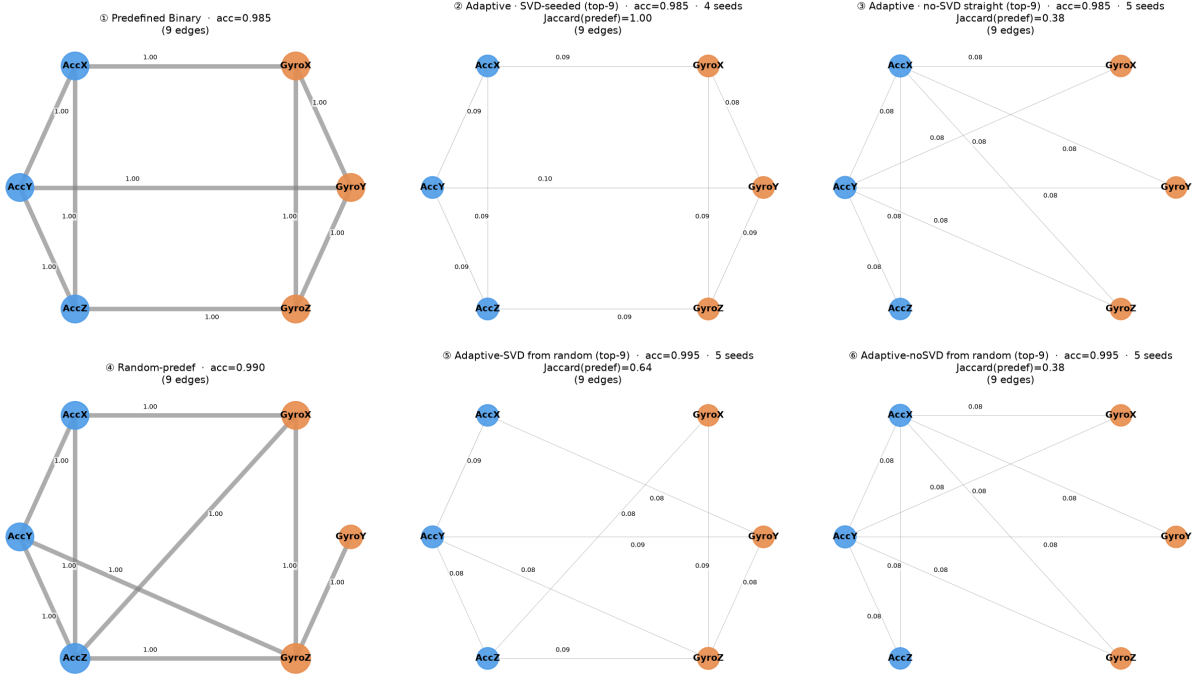


Figure 6.5: BASICMOTIONS (GConvLSTM), **binary** domain graph: predefined → adaptive(SVD) → adaptive(no-SVD), with the random-topology control. Six-node sensor-axis graph.

BasicMotions — GConvLSTM · kNN
top row: predefined → adaptive(SVD) → adaptive(no-SVD) bottom row: the same three, started from the random-topology control
node radius = weighted degree (shared scale); edge labels = |weight|; learned graphs at predefined edge count (top-11); Jaccard for ⑤⑥ is vs the random-predef start

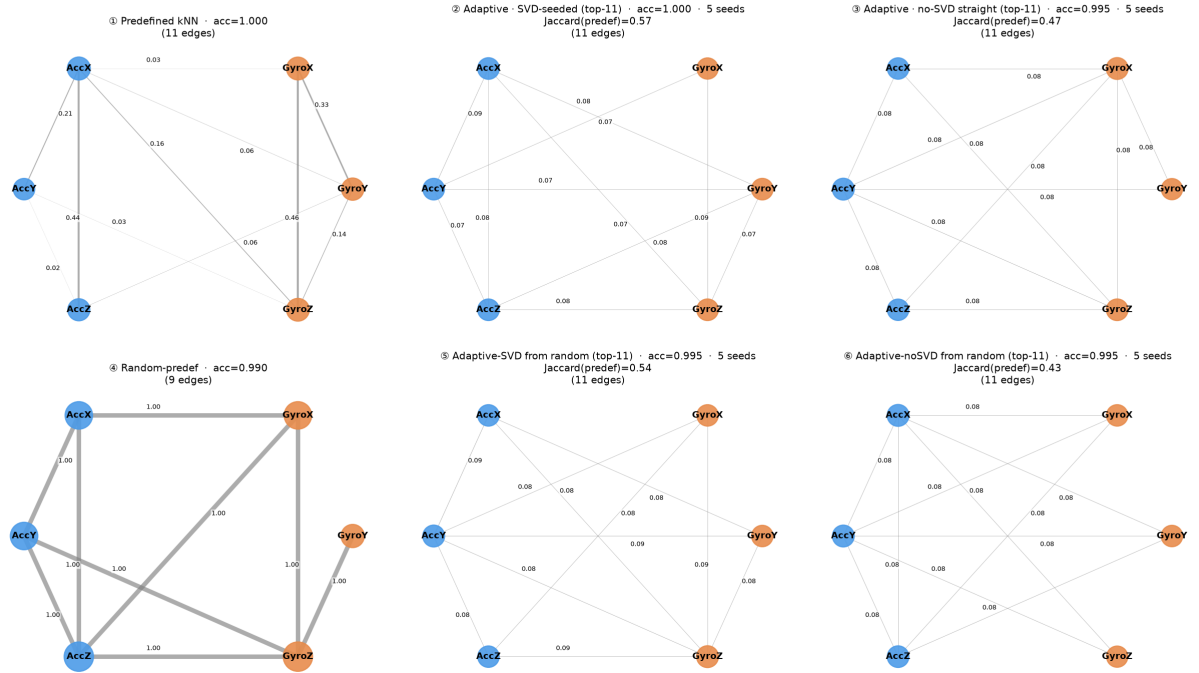


Figure 6.6: BASICMOTIONS (GConvLSTM), **kNN** data-driven graph: the same predefined-to-adaptive progression. The data-driven seed reshapes the six-node graph relative to the binary domain graph above.

6.1.2 BasicMotions

BASICMOTIONS accuracies saturate near the ceiling for every method (Table 6.1). All TGNN families exceed 99% mean accuracy, and several configurations reach 100%. The sequence baselines are slightly lower on average (95.2%), though this hides a wide spread—TCN reaches 99.5% while CNN-BiGRU manages only 91.0%. Because the dataset is tiny (80 instances) and the classes are easily separable, BASICMOTIONS has limited power to discriminate between methods and mainly serves as a sanity check that the graph models are correctly implemented and competitive. Its accuracy-versus-cost view is given in Appendix 6.2.

Table 6.1: BASICMOTIONS — performance by graph-construction approach. Each row averages over the five TGNN architectures, over every graph mode belonging to that approach, and over five seeds per mode. Best configuration is the highest single (architecture, graph mode) pair after averaging its five seeds. Configurations counts the distinct (architecture, graph mode) combinations contributing to the row, so a value of 60 means twelve graph modes evaluated on each of five architectures. It counts combinations, not training runs, and each combination was trained with five seeds. The three sequence baselines are listed individually rather than pooled, because averaging them conceals a spread of more than eight points between the weakest and the strongest. Every approach sits at the ceiling on this dataset, so it separates methods only weakly and serves as an implementation check.

Graph-construction approach	Mean over configurations		Best configuration		Conf. Combinations
	Accuracy	Macro-F1	Accuracy	Macro-F1	
Data-driven (correlation)	0.998	0.998	1.000	1.000	60
Adaptive, seeded from a random graph	0.995	0.995	1.000	1.000	50
Predefined anatomical and adaptive	0.995	0.995	1.000	1.000	45
Adaptive, no SVD seeding	0.994	0.994	1.000	1.000	20
Random-topology control	0.994	0.994	1.000	1.000	25
LSTM (no graph)	0.950	0.948	—	—	1
TCN (no graph)	0.995	0.995	—	—	1
CNN-BiGRU (no graph)	0.910	0.908	—	—	1

6.1.3 Epilepsy

EPILEPSY is more discriminative and reveals a clear ordering (Table 6.2, Fig. 6.7). The data-driven *correlation* family is strongest, averaging 95.7% across its 60 configurations, which are twelve threshold policies evaluated on each of the five architectures. Its single best configuration is A3TGCN under the top-50% threshold, at 98.7%. The predefined anatomical and adaptive graphs follow at 94.8% across 45 configurations. The sequence-baseline *family mean* is lowest (92.6%). That family average is, however, depressed by a weak LSTM (84.9%). The strongest individual baseline, CNN-BiGRU (97.2%), is in fact competitive with the best TGNN configurations and *beats* the plain predefined-graph TGNNs (~92%)—so the TGNN advantage on EPILEPSY is real only for the data-driven/adaptive graph strategies, not for the predefined graph. Within the fine-grained strategy ranking (Fig. 6.7), data-driven graphs dominate the top—correlation top-50% (0.979) and predefined-kNN (0.977)—while a plain predefined *binary* graph is the *worst* strategy (0.918), below even the random-topology control (0.927). This is the first indication (RQ3) that a naive structural prior is not automatically beneficial, and that *how* the graph is built matters more than *whether* it encodes domain knowledge. The adjacency matrices behind that ranking (Fig. A.3) explain the mechanism. The three accelerometer axes of this dataset are only weakly correlated, with a maximum absolute coefficient of 0.11 (Fig. 3.4), so a *fixed* threshold of $|r| \geq 0.3$ retains almost no edges and leaves the graph essentially empty. The *data-driven* policies, namely $\geq \text{mean}|r|$, top-50% and keep-all, instead set their cut-off from the observed scale and so retain these weak but informative couplings, which is why they head the ranking in Fig. 6.7.

Accuracy is not the whole result on this dataset, and the reason is worth establishing before the per-class comparison is read. Averaged over every configuration in the sweep the four classes are not equally difficult (Fig. 6.8). RUNNING, SAWING and WALKING sit between 0.944 and 0.994 macro-F1, whereas the EPILEPSY class sits at 0.893 F1 and 0.850 recall, the lowest of the four on both measures. The class that carries the clinical cost is therefore also the hardest to classify, which is precisely what a headline accuracy averages away, and it is the reason macro-F1 is reported beside accuracy throughout this thesis rather than as an afterthought.

The comparison is at its sharpest on that class, which is the clinically consequential one of the four and the only one on which a missed prediction carries real cost (Fig. 6.9). Overall accuracy understates the difference between the model families here. Every one of the five graph architectures recovers at least 94% of seizure windows, with A3TGCN reaching 96.5% recall and 0.976 seizure-class F1. The LSTM reaches 84.9% overall accuracy, which looks merely mediocre, yet recovers only 61.2% of seizures. The gap between a model’s overall accuracy and its seizure recall is therefore the diagnostic quantity, and it is 23.8 points for the LSTM against 2.4 points for A3TGCN, an order of magnitude apart. The two convolutional baselines sit between the extremes, with CNN-BiGRU at 90.6% seizure recall and TCN at 85.9%, so both trail every graph model on this class while remaining competitive on overall accuracy. This is the clearest single demonstration in Phase 1 that reporting accuracy alone would misrepresent what these models do.

Table 6.2: EPILEPSY — performance by graph-construction approach, using the taxonomy of Section 4.3. Aggregation is identical to Table 6.1, averaging over the five TGNN architectures, over every graph mode belonging to the approach, and over five seeds per mode. Macro-F1 tracks accuracy closely and never reorders the approaches, which confirms that this dataset’s mild class imbalance does not drive the ranking. The three sequence baselines are listed individually rather than pooled, because their spread here is 12.4 % and an averaged baseline row would misrepresent every one of them.

Graph-construction approach	Mean over configurations		Best configuration		Conf. Combinations
	Accuracy	Macro-F1	Accuracy	Macro-F1	
Data-driven (correlation)	0.957	0.955	0.987	0.987	60
Predefined anatomical and adaptive	0.948	0.946	0.988	0.989	45
Adaptive, no SVD seeding	0.940	0.937	0.971	0.970	20
Adaptive, seeded from a random graph	0.935	0.933	0.978	0.978	49
Random-topology control	0.927	0.924	0.971	0.971	25
LSTM (no graph)	0.849	0.841	—	—	1
TCN (no graph)	0.955	0.953	—	—	1
CNN-BiGRU (no graph)	0.973	0.972	—	—	1

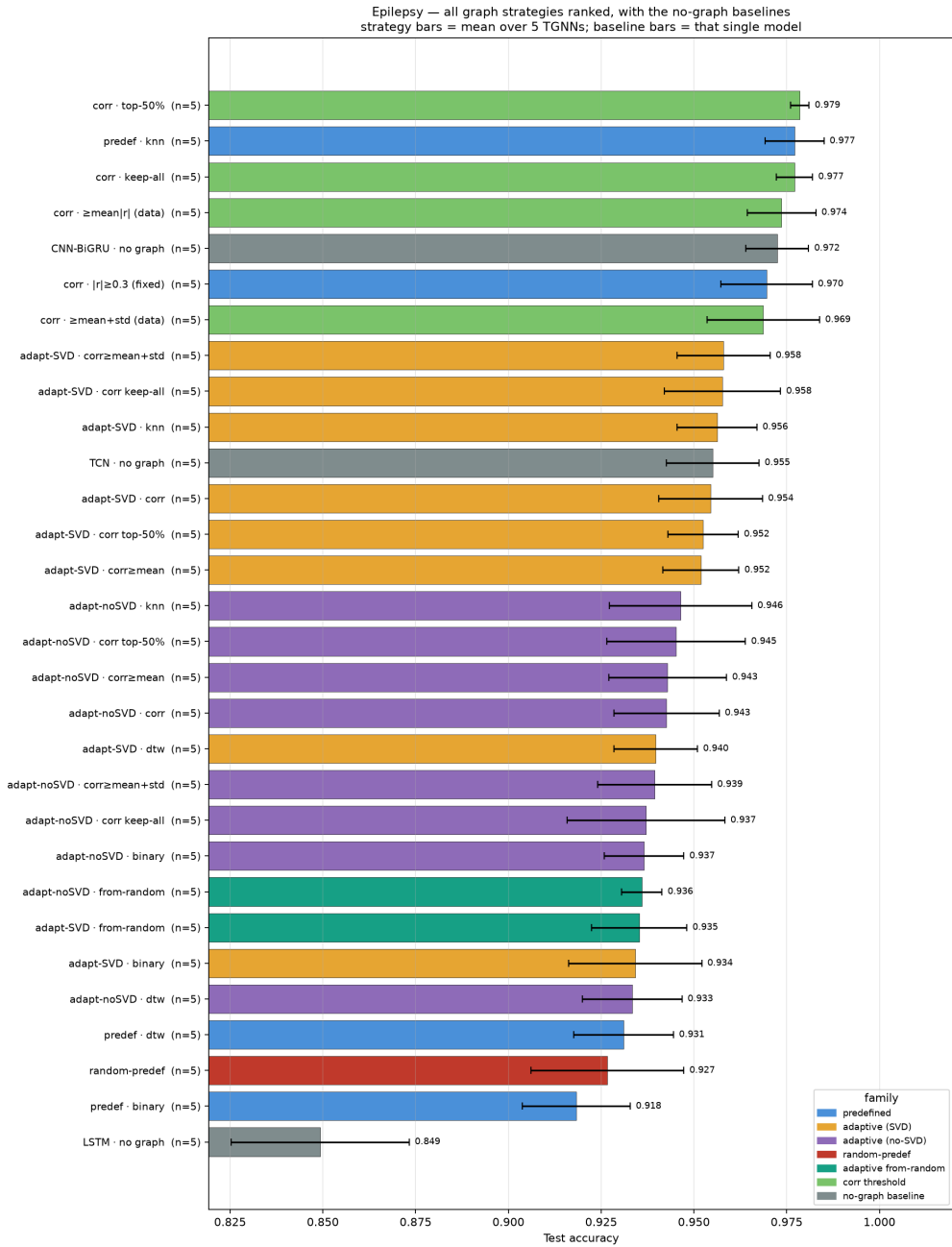


Figure 6.7: EPILEPSY — the graph-construction strategies of Section 4.3 ranked by test accuracy on the fixed archive split. Each strategy bar averages over the five TGNN architectures and over five seeds for each, so one bar summarises 25 runs, and the error bar is the standard deviation across them. The three sequence baselines carry no graph and are shown as single-architecture bars, labelled by architecture. Colour encodes the strategy family.

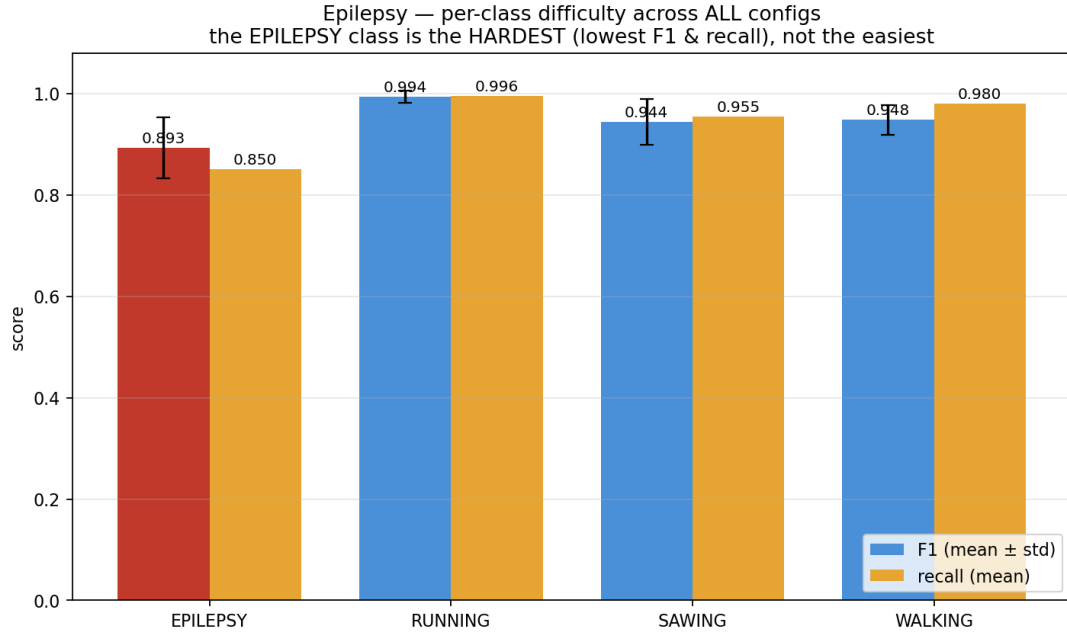


Figure 6.8: EPILEPSY — per-class difficulty averaged over every configuration in the sweep, meaning all five architectures across all 40 graph modes and five seeds each. Blue is macro-F1 with its standard deviation across configurations, orange is recall. The EPILEPSY class is the hardest of the four on both measures, at 0.893 F1 and 0.850 recall against 0.944 to 0.994 F1 for the other three. This is the opposite of what the dataset’s name might suggest, and it is the justification for the per-class analysis that follows, since a single accuracy figure averages this difficulty away entirely.

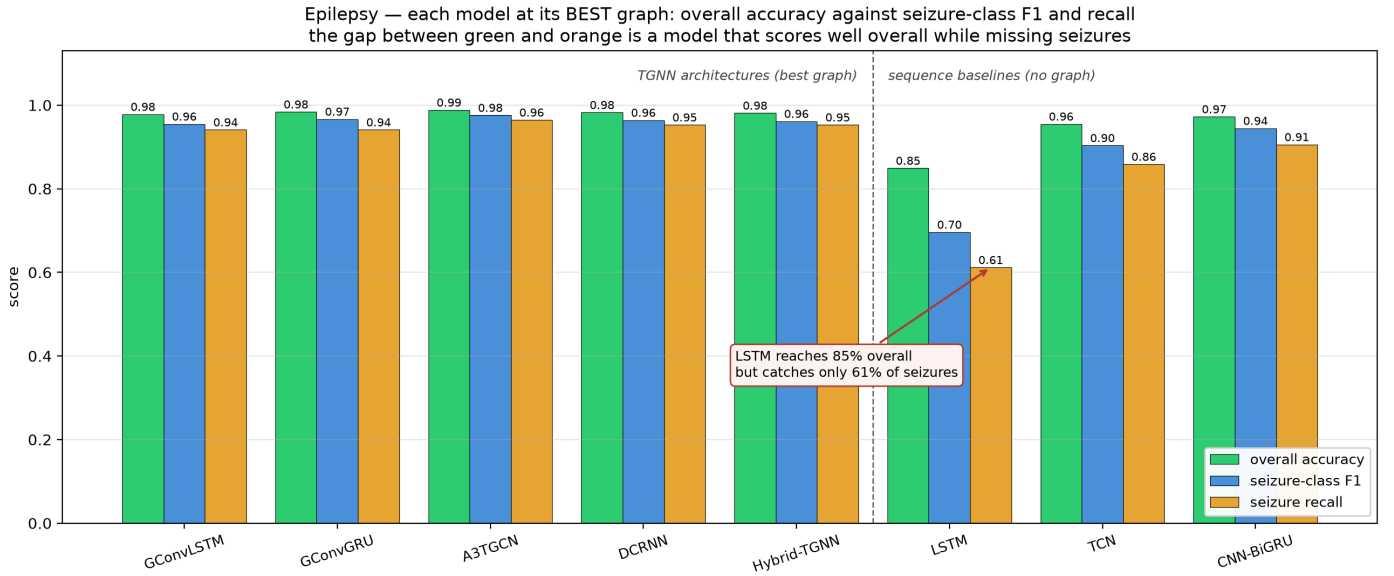


Figure 6.9: EPILEPSY — overall accuracy set against performance on the EPILEPSY (seizure) class alone. **Each TGNN is shown at its best-performing graph mode**, which is why its accuracies here are far higher than in Fig. 6.13, where every TGNN is instead pinned to the predefined axis graph. The three sequence baselines carry no graph, have therefore only one configuration, and are identical in the two figures, which is the quickest way to see that the difference is the graph and not the metric. Each bar is the mean over five seeds, and the best mode is `predefined_knn_correlation` for A3TGCN and GConvGRU, `predefined_correlation_top50` for GConvLSTM, `predefined_correlation_mean` for DCRNN and `adaptive_correlation_nothresh` for Hybrid-TGNN. The quantity to read is the distance between the green and orange bars within one model, which is how much a headline accuracy overstates that model’s performance on the class that matters clinically.

Two further views resolve that result along the axis this thesis actually varies. Fig. 6.10 groups seizure-class F1 by graph-construction strategy rather than by architecture, and shows that the data-driven correlation and kNN graphs are what lift the class above every sequence baseline, while the plain binary axis graph does not. Fig. 6.11 then removes all averaging and reports every (architecture, strategy) cell individually. Two patterns in it are worth stating. The predefined correlation and kNN columns are the strongest, reaching 0.98 for A3TGCN under the kNN graph, and the adaptive columns are close to flat within each row, since the learner converges to a similar graph whichever seed it is given. On the

class that carries the clinical cost, therefore, it is the supplied structure that helps rather than the adaptivity, which is the same conclusion Section 6.1.4 reaches on overall accuracy, arrived at here on the metric that matters.

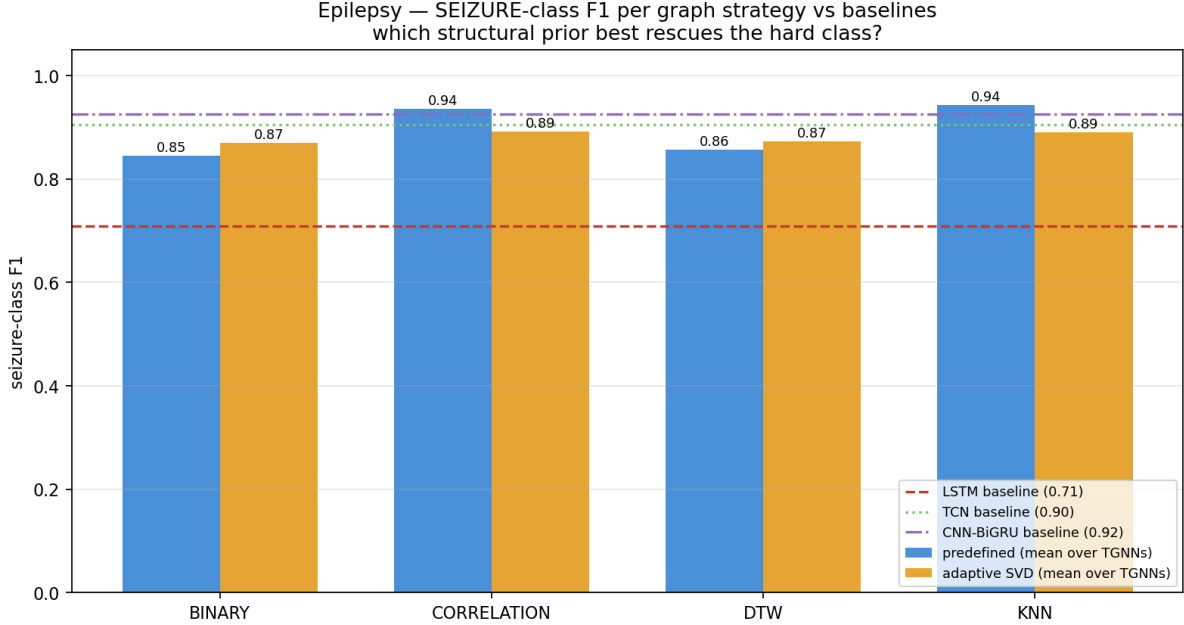


Figure 6.10: EPILEPSY — seizure-class F1 resolved by graph-construction strategy, with the three no-graph baselines drawn as horizontal reference lines at 0.71 for the LSTM, 0.90 for TCN and 0.92 for CNN-BiGRU. Solid bars use the graph directly as a predefined adjacency and hatched bars supply the same graph to the SVD-seeded adaptive learner, each averaged over the five TGNN architectures. The correlation and kNN families clear every baseline at 0.94, whereas the binary axis graph reaches only 0.85 and the DTW graph 0.86. The conditionality established for overall accuracy in Section 6.1 therefore holds on the clinically consequential class as well.

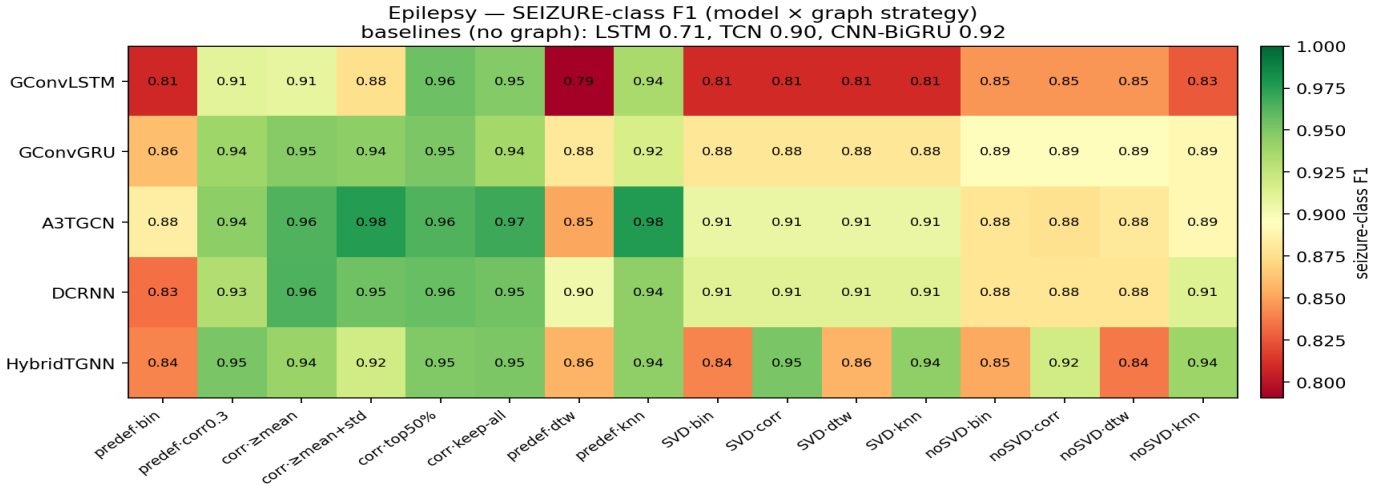


Figure 6.11: EPILEPSY — seizure-class F1 for every (architecture, graph mode) pair, with no averaging across either axis. Rows are the five TGNN architectures and columns the 16 correlation, DTW, kNN and binary modes in their predefined, SVD-seeded and no-SVD forms. The three no-graph baselines are stated in the figure title rather than drawn, since they carry no graph and would occupy a single column. Colour encodes seizure-class F1 on the fixed archive test split. The best cell is A3TGCN under the predefined kNN graph at 0.98, and the adaptive blocks on the right are visibly flatter within each row than the predefined blocks on the left, which is the per-class form of the finding that graph construction rather than graph learning is what moves performance.

6.1.4 Correlation Threshold Policies

The correlation family tops the EPILEPSY ranking (Table 6.2), and resolving it by threshold policy identifies which member of that family is responsible (Fig. 6.12). Across all five architectures the data-driven threshold policies used as a *predefined* graph ($\geq \text{mean}|r|$, top-50%, and even the fixed $|r| \geq 0.3$, shown as solid bars) are consistently the strongest (A3TGCN top-50% 98.6%, GConvGRU top-50% 97.7%), whereas pushing the *same* correlation graphs through the adaptive learner (hatched bars) generally *lowers* accuracy. At this dimensionality a good data-driven graph is therefore best used directly

rather than as a seed for further learning—a concrete instance of the broader conclusion that graph construction, not end-to-end learning, is what moves accuracy.

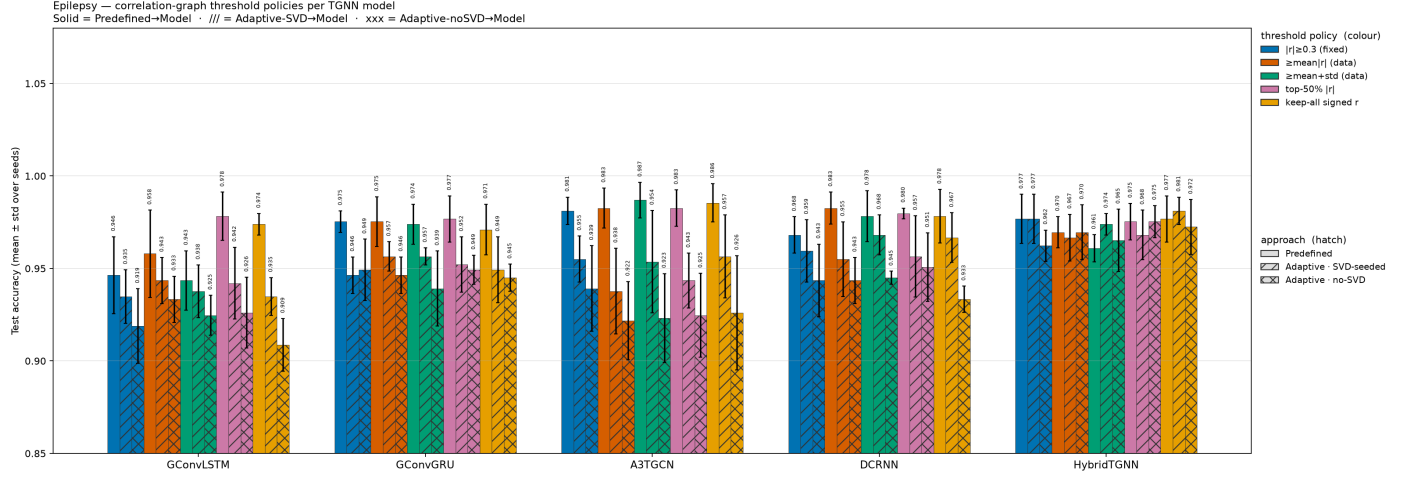


Figure 6.12: EPILEPSY — test accuracy for each correlation-graph threshold policy, resolved by TGNN architecture. Bars are means over five seeds for one (architecture, graph mode) pair, so no averaging across architectures is performed. Solid bars use the correlation graph directly as a predefined graph. Hatched bars supply the same graph to the adaptive learner, either SVD-seeded (///) or elementwise without SVD ($\times \times \times$). Threshold policies are the five of Section 4.3.2, comprising the fixed $|r| \geq 0.3$, the two data-driven thresholds, top-50% and keep-all.

6.1.5 Experiment A — Do TGNNs beat the sequence baselines at low dimensionality?

RQ1 has two axes—accuracy and computational efficiency—and on EPILEPSY they point in opposite directions for the predefined-graph TGNNs. Fig. 6.13 shows both. *Accuracy* (panel a). On saturated BASICMOTIONS the TGNNs are at the ceiling (Fig. 6.15). On the harder EPILEPSY the predefined-graph TGNNs ($\sim 89\%$ – 93%) trail the sequence baselines. Throughout this thesis the *strongest baseline* means the one with the highest mean accuracy over its five seeds, and on EPILEPSY that is CNN-BiGRU at 97.2%, followed by TCN at 95.5% and the LSTM at 84.9%. Cost, however, reverses the ordering (Fig. 6.13, panel b). The recurrent baselines are far heavier—LSTM 565 k and CNN-BiGRU 554 k parameters versus only ~ 105 k for GConvGRU, A3TGCN and Hybrid-TGNN, a $\sim 5\times$ difference (only TCN, at 203 k, is in the TGNN size range). The baselines therefore obtain their accuracy advantage at roughly five times the parameter cost, while the TGNNs come within a few points of them at a fraction of that size. Cost is reported here as parameter count rather than wall-clock training time. The runs behind this study were executed on several machines over several months, and their recorded training times are not mutually comparable, so quoting them would present hardware differences as differences between architectures. Parameter count is a property of the model itself and is unaffected. On that measure the efficiency half of RQ1 favours the TGNNs decisively. Moreover, the accuracy gap in Fig. 6.13(a) is a property of the *predefined* graph rather than of the architectures. On EPILEPSY the predefined graph connects the three axes of a single wrist sensor, so it encodes almost no relational structure. With it the five TGNNs average 91.8% against CNN-BiGRU’s 97.2%, a deficit of 5.4 points that is consistent across all 25 seed-runs and larger than the seed-to-seed spread of either group. Substituting a data-driven correlation graph (Section 6.1) and changing nothing else lifts the same five architectures to 97.4%, statistically indistinguishable from the strongest baseline, at roughly a fifth of its parameters. Fig. 6.14 shows the full distribution behind those two numbers. It differs from the ranking in Fig. 6.7 in what it resolves. Fig. 6.7 averages over the five architectures to order the graph strategies against one another, whereas Fig. 6.14 performs no averaging at all and shows every architecture-by-strategy cell individually, which is what allows the variation along the two axes to be compared. It establishes that the correlation result is a property of the graph family rather than of one fortunate configuration, since 18 of the 25 data-driven correlation configurations sit above CNN-BiGRU. Two features of that figure carry the argument. The predefined axis graph, in the second column, is among the weakest choices for every one of the five architectures, whereas the correlation and kNN columns reach 97% to 99% and bracket the CNN-BiGRU baseline. And the architecture rows vary considerably less than the strategy columns, which is the first direct evidence in this thesis that the choice of graph moves accuracy more than the choice of temporal operator. Individual configurations reach 98.7%, but that figure is the maximum over many configurations evaluated on the same fixed test split and is therefore optimistically biased. The family-level comparison is the defensible one. The honest conclusion is parity rather than victory on raw accuracy—which, combined with the five-fold parameter advantage, still resolves RQ1 in the TGNNs’ favour on the joint criterion. The role of the graph-construction strategy is taken up as RQ2 in Chapter 7.

RQ1 — Epilepsy: TGNN architectures vs sequence models (accuracy vs model size)

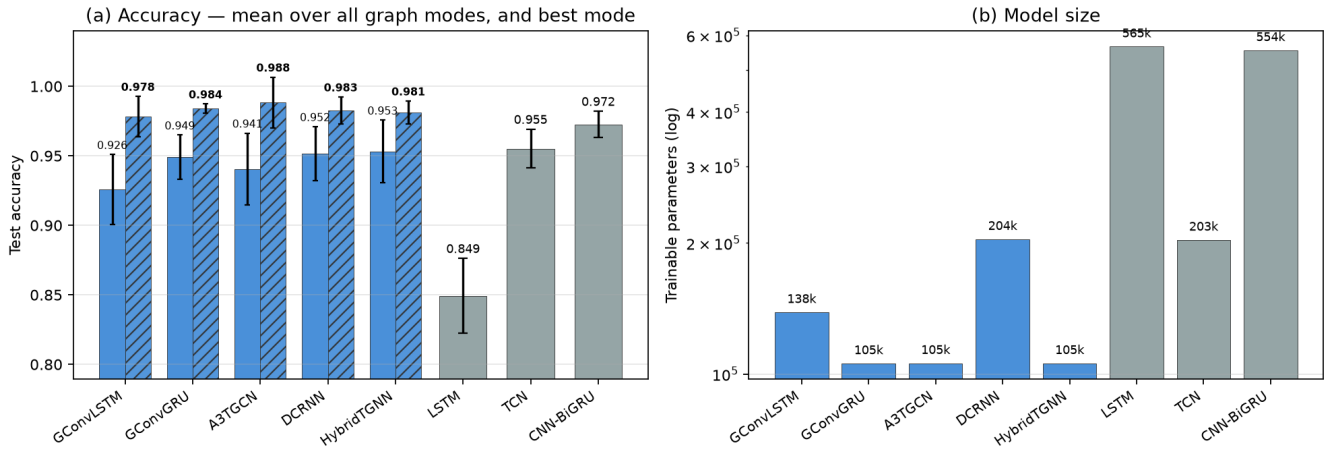


Figure 6.13: EPILEPSY — RQ1 accuracy versus computational cost. **Every TGNN here is pinned to the predefined axis graph**, which is EPILEPSY’s weakest choice, so panel (a) is a lower bound on what these architectures reach rather than their best. Fig. 6.9 shows the same architectures at their best graph mode, where they reach 97.8% to 98.8%. The three baselines carry no graph and are identical in both figures. (a) The predefined-graph TGNNs trail the strong baselines in raw accuracy, but (b) the LSTM and CNN-BiGRU baselines are $\sim 5\times$ larger (554–565k vs ~ 105 k parameters). The TGNNs reach near-baseline accuracy at a fraction of the parameters. Panel (a) fixes the graph at the predefined axis graph, which is EPILEPSY’s weakest choice. With a data-driven graph the same architectures reach parity with the strongest baseline on this dataset, CNN-BiGRU at 97.2% mean accuracy over five seeds (Fig. 6.14).

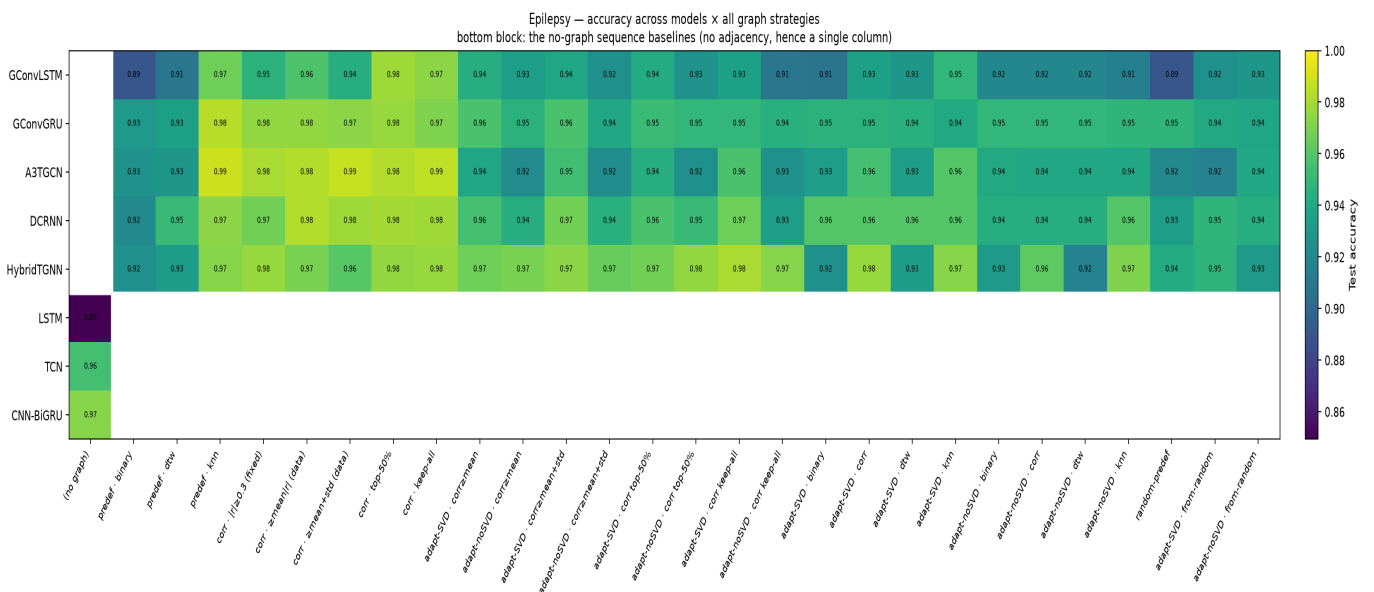


Figure 6.14: EPILEPSY — test accuracy for every (architecture, graph-construction strategy) pair, with the three no-graph sequence baselines in the bottom block. Each cell is the mean over five seeds for one pair, with no averaging across architectures or strategies. Rows are the five TGNN architectures and columns are the 40 graph modes of Section 4.3, ordered by family. Colour encodes accuracy on the fixed archive test split.

6.2 BasicMotions Accuracy versus Cost

BASICMOTIONS saturates near the ceiling for every method and is reported in the main text only as a correctness check (Section 6.1). Its accuracy-versus-cost view is given here for completeness.

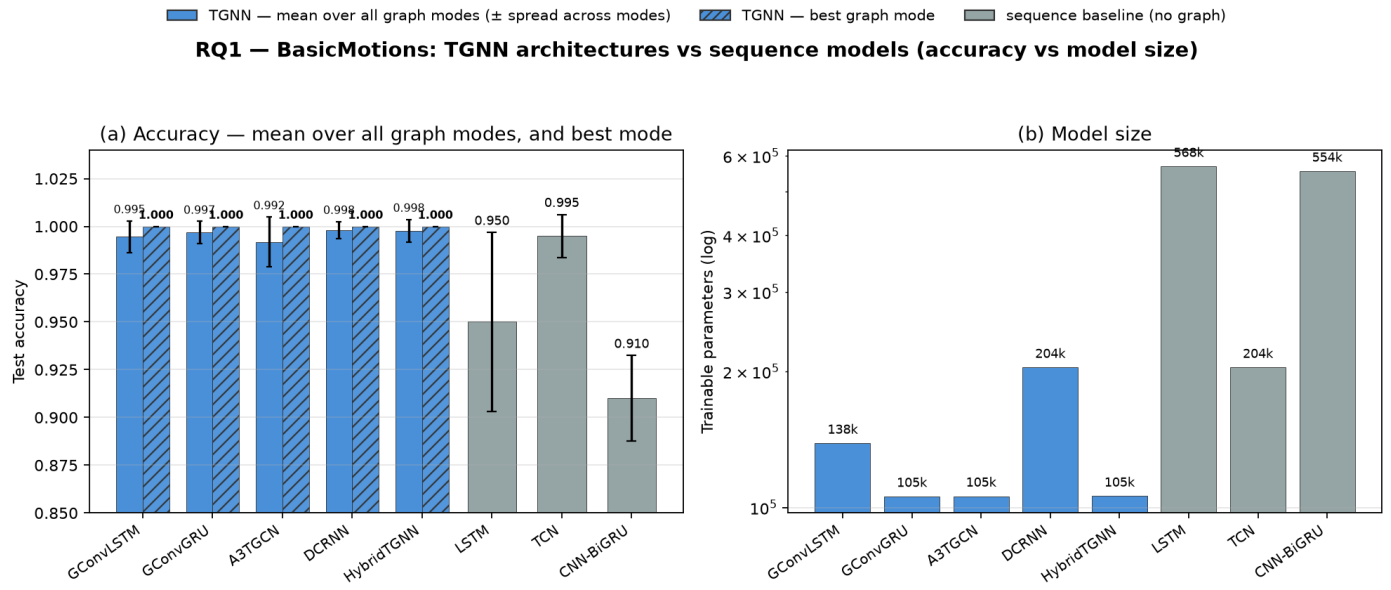


Figure 6.15: BASICMOTIONS — RQ1 accuracy against model size. Accuracies saturate near the ceiling. The TGNNs match or exceed the baselines at a small fraction of their parameter count.

6.3 Phase 2 — High-Dimensional DSA Dataset

Table 6.3 reports the DSA results by model and graph mode. The seven experiments that follow read it in one order, and each belongs to one research question. Experiment B asks whether the graph models beat the sequence baselines at all at 45 channels, completing RQ1. Experiments C and D then decompose *which* property of a graph is responsible, by varying node granularity and edge provenance one at a time, which answers RQ2. Experiment E removes the anatomical knowledge while holding the construction fixed, which answers RQ3, and Experiments F, G and H extend that question by asking whether the structure can be learned from data instead of supplied as a prior, once at each granularity, and what such learning depends on.

Table 6.3: DSA — LOSO accuracy and macro-F1 by model and graph mode (mean over folds and seeds). n is the number of seeds per cell. The baselines and the coarse, fine and adaptive graph families span $n = 5$ for every model, as are the DTW graph and the random-topology control for A3TGCN, and the correlation family for all five models. The DTW and kNN-correlation rows cover A3TGCN only. The random-topology row summarises Table 6.6, where its five replicates are independent random *graphs* rather than training seeds.

Model	Graph mode	Accuracy	Macro-F1	n
<i>Baselines (no graph)</i>				
LSTM	none	0.791	0.751	5
TCN	none	0.789	0.753	5
CNN-BiGRU	none	0.829	0.796	5
<i>TGNN — predefined coarse (5-node)</i>				
A3TGCN	coarse	0.837	0.815	5
DCRNN	coarse	0.846	0.823	5
GConvGRU	coarse	0.831	0.804	5
GConvLSTM	coarse	0.814	0.782	5
Hybrid-TGNN	coarse	0.832	0.805	5
<i>TGNN — predefined fine (45-node)</i>				
A3TGCN	fine	0.661	0.620	5
DCRNN	fine	0.718	0.685	5
GConvGRU	fine	0.677	0.638	5
GConvLSTM	fine	0.697	0.661	5
Hybrid-TGNN	fine	0.683	0.645	5
<i>TGNN — adaptive, a 5-node graph learned end-to-end from a random initialisation</i>				
A3TGCN	adaptive (5n)	0.867	0.847	5
DCRNN	adaptive (5n)	0.838	0.819	5
GConvGRU	adaptive (5n)	0.827	0.800	5
GConvLSTM	adaptive (5n)	0.818	0.791	5
Hybrid-TGNN	adaptive (5n)	0.867	0.845	5
<i>TGNN — correlation, data-driven $r \geq \text{mean} r$</i>				
A3TGCN	corr	0.761	0.729	5
DCRNN	corr	0.814	0.787	5
GConvGRU	corr	0.797	0.772	5
GConvLSTM	corr	0.814	0.787	5
Hybrid-TGNN	corr	0.799	0.770	5
<i>TGNN — other data-driven 45-node graphs (A3TGCN)</i>				
A3TGCN	DTW-kNN	0.746	0.719	5
A3TGCN	kNN-corr	0.765	0.734	5
<i>Random-topology control — 5-node (Table 6.6)</i>				
A3TGCN	random-topology	0.841	0.820	5 graphs
A3TGCN	adaptive-from-random	0.859	0.841	5 graphs

6.3.1 Experiment B — Do TGNNs beat the sequence baselines at 45 channels?

The best TGNN configurations (adaptive A3TGCN and Hybrid-TGNN, both 86.7 %, and coarse DCRNN at 84.6 %) clearly exceed every sequence baseline, namely LSTM (79.1 %), TCN (78.9 %), and CNN-BiGRU (82.9 %). The margin over the strongest baseline (CNN-BiGRU) is modest ($\sim 4\%$ for the best TGNN), and it is achieved *only* with a well-chosen graph. The coarse and adaptive graphs win, while the fine graph (Fig. 6.17) falls to 68 %, below every baseline. The graph inductive bias therefore helps at high dimensionality, but the benefit is conditional on graph construction rather than automatic.

That raises the obvious follow-up, namely whether the data-driven graphs that were strongest on EPILEPSY widen the margin here as well. They do *not*. At 76.1 % to 81.8 % across the five architectures (Section 6.3.3) they fall below both the coarse prior and the no-graph baselines. Their value to the study is diagnostic rather than competitive, and Experiment D returns to them for that purpose. Fig. A.5 summarises the best configuration each architecture reached over the whole sweep.

The separation is not an artefact of averaging over seeds. The weakest seed of A3TGCN’s and Hybrid-TGNN’s best configuration still lands above the strongest seed of CNN-BiGRU, and the seed distributions behind these means are given in Appendix A.3.

Efficiency at high dimensionality, where the TGNNs dominate outright. The Phase-1 efficiency argument (Section 6.1) was that the TGNNs buy *parity* at a fifth of the parameters. On DSA the trade disappears and the comparison becomes one-sided (Fig. 6.16). A3TGCN is the *smallest* model in the study at 424k parameters and reaches the highest accuracy at 86.7 %. CNN-BiGRU, the strongest baseline, carries 2.23M parameters—5.3 times as many—and reaches 82.9 %. The LSTM is larger still at 2.33M for 79.1 %. There is no accuracy-for-size trade-off to negotiate here. The best graph model is simultaneously the smallest and the most accurate, and every one of the five TGNN architectures is smaller than every recurrent baseline.

Table 6.4 gives the full ordering, and two rows in it carry more weight than A3TGCN’s. DCRNN is the *heaviest* graph model in the study at 1.21M parameters, and it still beats CNN-BiGRU at a little over half its size, which shows the efficiency result is a property of the graph parameterisation rather than of one unusually compact architecture. GConvLSTM is the honest exception. At 81.8 % it clears both recurrent baselines but not CNN-BiGRU, so the claim this thesis makes is that six of the ten configurations in that table beat the strongest baseline and that every one of the six is smaller than it, not that every graph model beats every baseline.

The reason is structural rather than incidental. A sequence baseline must map a 45-channel input vector through fully connected recurrent weights, so its parameter count grows with the product of channel count and hidden width. A TGNN on the coarse graph sees five nodes carrying nine features each and shares one set of convolution weights across all nodes, so widening the input costs it almost nothing. Higher dimensionality therefore favours the graph parameterisation, which is why the efficiency margin is larger on DSA ($5.3\times$) than on EPILEPSY ($5\times$ at parity rather than at an advantage). This is the clearest sense in which the Phase-1 relationships not only transfer to DSA but strengthen, and it is a second axis—alongside accuracy—on which the transfer clause of RQ2 resolves affirmatively.

Table 6.4: DSA — accuracy against model size for every architecture at its best graph, ordered by accuracy. The horizontal rule marks the strongest sequence baseline. Six TGNN configurations sit above it, and every one of them is smaller than it. DCRNN is the informative case, since it is the *heaviest* graph model in the study and still beats the strongest baseline at little more than half its size. GConvLSTM is the one architecture that does not clear CNN-BiGRU, although it clears both recurrent baselines.

Model	Graph	Accuracy	Parameters	Size vs CNN-BiGRU
Hybrid-TGNN	adaptive (5n)	0.867	620 245	3.6 $\times$ smaller
A3TGCN	adaptive (5n)	0.867	423 734	5.3 $\times$ smaller
DCRNN	coarse (5n)	0.846	1 209 748	1.8 $\times$ smaller
A3TGCN	coarse (5n)	0.837	423 574	5.3 $\times$ smaller
Hybrid-TGNN	coarse (5n)	0.832	620 245	3.6 $\times$ smaller
GConvGRU	coarse (5n)	0.832	619 924	3.6 $\times$ smaller
<i>CNN-BiGRU</i>	<i>none</i>	<i>0.829</i>	2 226 195	—
GConvLSTM	adaptive (5n)	0.818	816 948	2.7 $\times$ smaller
LSTM	none	0.791	2 334 739	larger
TCN	none	0.789	831 379	2.7 $\times$ smaller

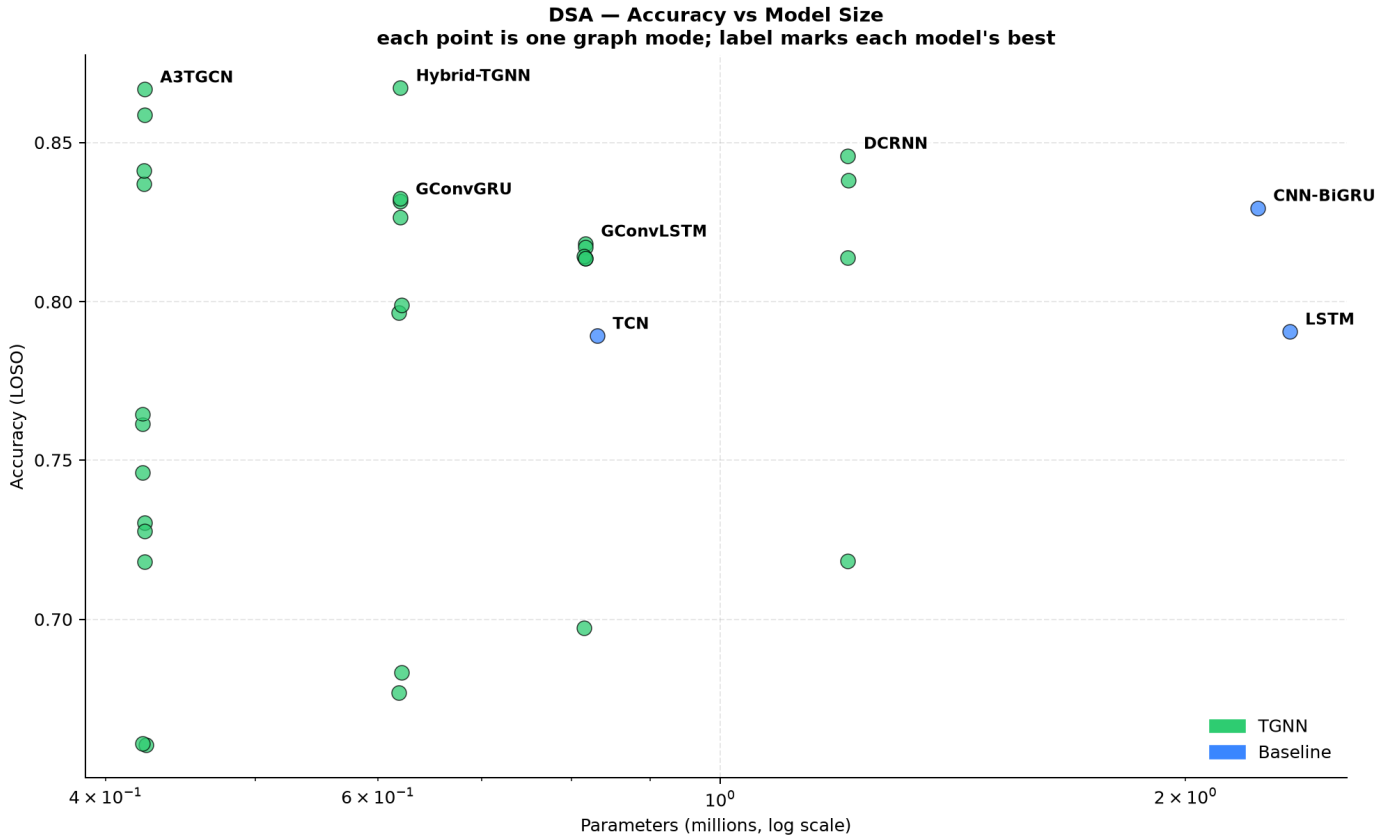


Figure 6.16: DSA — accuracy against model size, with one point per graph mode and the label marking each model’s best. The horizontal axis is logarithmic. A3TGCN (424k) sits at the top-left corner: smallest and most accurate. The recurrent baselines (CNN-BiGRU 2.23M, LSTM 2.33M) sit at the right-hand edge and below. The vertical spread within each model is the graph effect of Section 6.4—the same architecture at the same parameter count spanning 12 % to 21 % of accuracy—which is why no point on this plot can be read as a property of the architecture alone.

6.3.2 Experiment C — Does node granularity govern performance?

The predefined *fine* 45-node graph is markedly worse than the *coarse* 5-node graph across every model (Fig. 6.17). Averaged over models, coarse reaches 83.2 % and adaptive 84.3 %, whereas fine collapses to 68.7 %—a 14.5 % gap. Strikingly, the fine graph even falls *below* the no-graph baseline mean (80.3 % over LSTM/TCN/CNN-BiGRU). Adding a dense, high-resolution graph is worse than adding no graph at all. The natural first explanation is dilution—the fine graph’s 234 nearly uniform undirected edges averaging away discriminative channel structure, whereas the five-node anatomical graph aggregates the nine channels of each body segment into a coherent node signal. Section 6.3.3 tests that explanation against a graph with the *same* 45 nodes and *more* edges, and finds it insufficient. Edge density is not what sinks the fine graph.

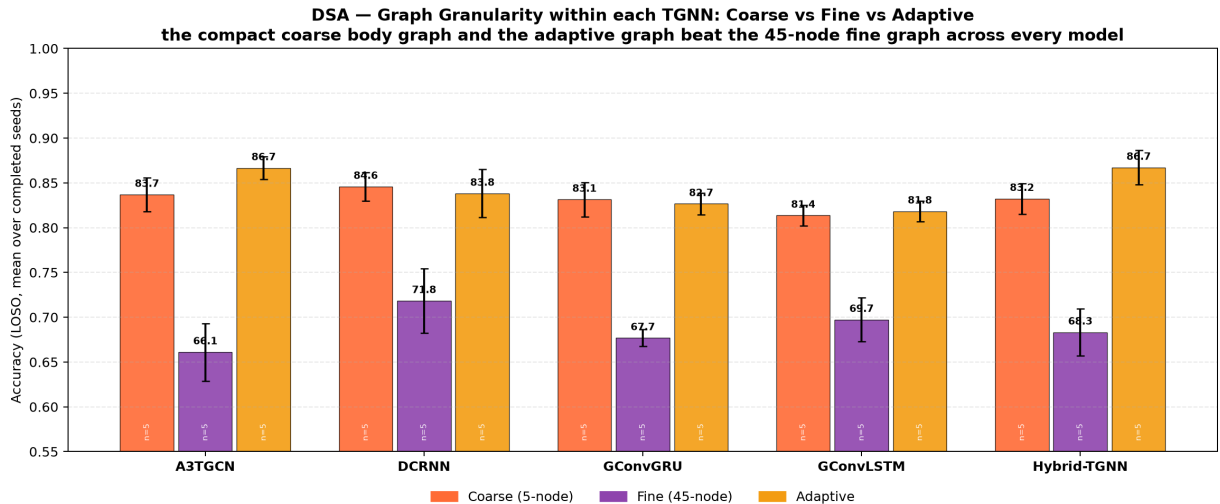


Figure 6.17: DSA — graph granularity within each TGNN. Coarse (5-node) and adaptive (learned) graphs consistently beat the fine (45-node) graph. The fine graph underperforms the no-graph baselines.

6.3.3 Experiment D — Does edge provenance govern performance independently of granularity?

The data-driven correlation graph was the *strongest* strategy in Phase 1 (EPILEPSY, 95.7 % family mean, best configuration 98.7 %). On DSA it is not. With the same data-driven threshold ($|r| \geq \text{mean}|r|$) it reaches 76.1 % for A3TGCN, 81.4 % for DCRNN and 79.7 % for GConvGRU (all $n = 5$), against coarse-graph values of 83.7 %, 84.6 % and 83.1 %. It lands *between* the fine and coarse graphs, and—importantly—below the no-graph baseline mean of 80.3 %.

This family is what makes the DSA results diagnostic rather than merely descriptive, because the correlation graph is a controlled variation on the fine graph. Both are built over all 45 sensor channels, so they hold node granularity fixed and vary only where the edge weights come from. All five TGNN architectures are evaluated in all four modes at $n = 5$, so Table 6.5 is a balanced comparison over the whole architecture set.

Table 6.5: DSA — graph strategies with node granularity and edge provenance separated. All five TGNN architectures at $n = 5$ seeds, 25 seed-runs per row. Edge counts are undirected and exclude self-loops. The correlation graph is re-estimated on each training fold, hence a range.

Graph mode	Nodes	Edges	Edge weights	Accuracy
Adaptive (learned)	5	learned	learned	0.843
Predefined coarse	5	6	uniform prior	0.832
(no graph)	—	—	—	0.803
Correlation, mean- $ r $	45	310 to 336	data-estimated	0.797
Predefined fine	45	234	uniform prior	0.687

Two effects separate cleanly. Holding edge weights uniform and varying granularity, coarse beats fine by 14.5 %. Holding granularity fixed at 45 nodes and varying only provenance, the correlation graph beats the fine graph by 11.0 %. Granularity is the larger effect, but provenance is of the same order—so the dominant-factor claim that a granularity-only reading would license is not supported.

The comparison also refutes dilution as the mechanism behind the fine graph’s collapse. If the fine graph failed because too many near-uniform edges average away discriminative structure, then the correlation graph—which is *denser*, carrying 310 to 336 undirected edges per fold against the fine graph’s 234—should fail at least as badly. It does not. It is 11.0 % better. Edge count and accuracy move in opposite directions here, so density cannot be the operative variable. What distinguishes the two is that the correlation weights are estimated from the training folds and are therefore graded and informative, whereas the fine graph’s weights take three fixed values assigned by anatomical rule. A 45-node graph is recoverable if its weights carry data. It is not recoverable merely by being sparse.

The granularity effect survives this correction, and Phase 1 explains why it did not appear there. The correlation graph operated over 3–6 channels, where the induced node set is already coarse and no aggregation problem exists. The same construction rule wins at low dimensionality and loses at high dimensionality, not because correlation is a poor statistic but because at 45 channels it induces too fine a node set—one that even good edge weights only partly rescue, leaving it below the no-graph baseline.

Finally, the two 5-node rows differ by only 0.6 %. Once granularity is chosen well, learning the adjacency adds little over a sensible anatomical prior—a point developed in Section 6.3.5.

Two further 45-node graphs confirm the ordering. If provenance rather than density is what separates the correlation graph from the fine graph, then *any* 45-node construction with data-estimated weights should beat the fine graph—including sparse ones, where a dilution account predicts the opposite. Two independent constructions test this. The **DTW-kNN** graph is built from dynamic-time-warping distances between channel prototypes and sparsified to mutual $k = 3$ neighbours, retaining roughly 98 to 105 undirected edges per fold, about *half* the fine graph’s 234. The **kNN-correlation** graph applies the same mutual- k sparsification in correlation space rather than DTW space, so it shares the sparsification rule with one and the similarity statistic with the other. Both are complete at $n = 5$ for A3TGCN.

All four 45-node graphs order by provenance and not by density (Fig. 6.18), as the following shows.

45-node graph	Undirected edges	Weights	A3TGCN acc.	n
Predefined fine	234	uniform, by rule	0.661 ± 0.032	5
DTW-kNN	98 to 105	data-estimated	0.746 ± 0.021	5
kNN-correlation	95	data-estimated	0.765 ± 0.011	5
Correlation	310 to 336	data-estimated	0.761 ± 0.010	5

DSA · A3TGCN · Node granularity held fixed at 45: rule-based vs data-estimated edge weights
Block structure = sensor units. Edge count varies 3.4-fold across the data-driven graphs, accuracy does not.

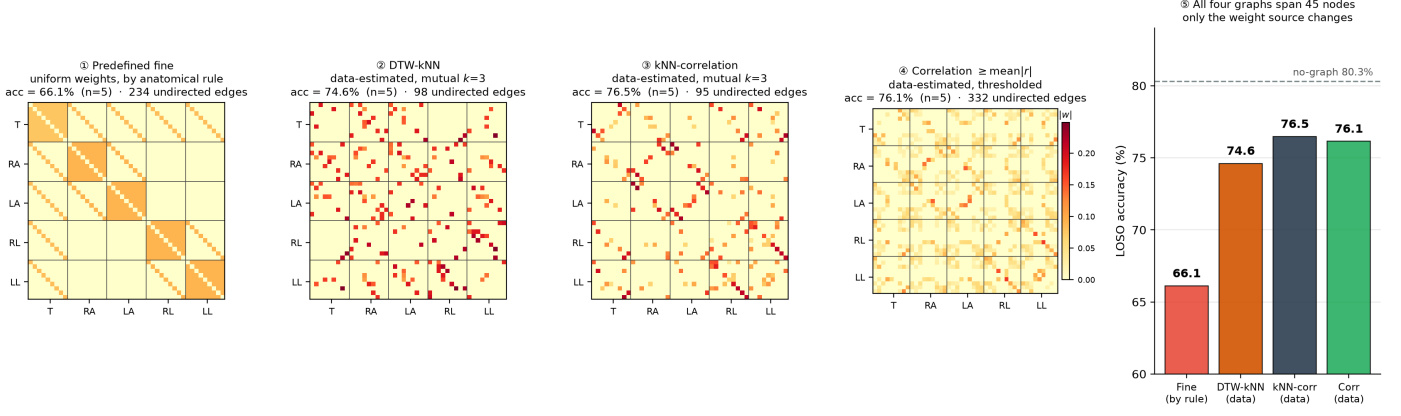


Figure 6.18: DSA — the four 45-node graphs, with node granularity held fixed and only the source of the edge weights varying. Panels – are adjacency matrices (self-loops excluded from the colour scale, which they would otherwise dominate). The 9×9 blocks are the five sensor units. The rule-based fine graph is visibly *uniform*—dense within each unit, one fixed value per edge class—whereas the data-driven graphs – are sparse and selective, keeping only channel pairs that actually covary. Panel shows the consequence. All three data-driven graphs sit 8.5 % to 10.4 % above the rule-based one despite spanning a 3.4-fold range in edge count, and all four remain below the no-graph baseline because 45 nodes is the wrong granularity regardless (Section 6.3.4). The data-driven graphs are re-estimated on each training fold.

What the data-driven graphs actually connect. The adjacency matrices of Fig. 6.18 show *that* the data-driven graphs are selective without showing *what* they select. Collapsing each 45-node graph onto the five body units answers that directly (Fig. 6.19), and the contrast is stark. The rule-based fine graph spends 77 % of its edges *inside* units (180 of 234), because each unit is a 9-clique by construction. Those edges connect an accelerometer axis to a gyroscope axis on the same sensor, which is a co-location relation rather than a movement one. Its 54 between-unit edges are exactly the coarse anatomical graph replicated nine times, once per channel type.

The three data-driven graphs invert this allocation, spending only 15 % to 25 % of their edges within units and the rest joining body segments to one another. More striking is *which* segments they join. In all three, the single heaviest between-unit link is a bilateral one—RightArm–LeftArm for DTW (14 channel pairs), RightLeg–LeftLeg for kNN-correlation (17) and for correlation (33)—and both bilateral pairs rank in the top three of the ten possible unit pairs in every case. Bilateral symmetry is precisely what the anatomical coarse graph encodes by hand, and three independent statistics recover it without being told.

This sharpens the provenance account into a mechanism. The data-driven graphs are not better because they are sparser or denser. They are better because they spend their edge budget on the between-segment relations that distinguish activities, whereas the rule-based graph spends three quarters of its budget on within-sensor relations that do not. It also explains why none of them reaches the coarse graph. They identify the right structure but express it over 45 nodes, so the aggregation that the coarse graph performs for free is never carried out.

DSA · The four 45-node graphs collapsed onto the five body units

Node size = channel pairs joined *within* that unit. Edge label = channel pairs joining the two units. Self-loops excluded.

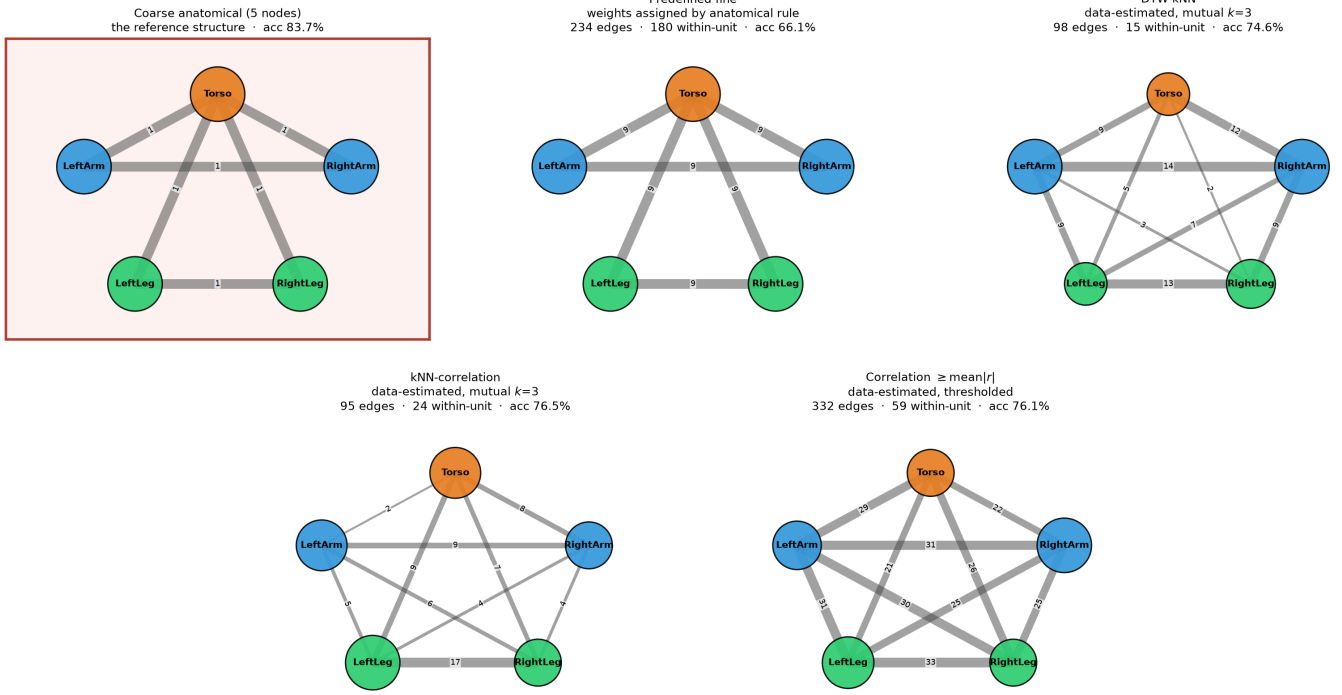


Figure 6.19: DSA — the four 45-node graphs collapsed onto the five body units, with the coarse anatomical graph boxed for reference. Node size is the number of channel pairs joined *within* that unit. Edge labels count the channel pairs joining two units. Self-loops are excluded, being an artefact of the normalisation rather than a discovered relation. A node-link view of the graphs at full 45-node resolution is unreadable at 95 to 336 edges, which is why Fig. 6.18 draws them as matrices. The collapse recovers the topology that matters for the argument.

Returning to density, this is the most direct evidence for the provenance effect, because the three data-driven graphs bracket the rule-based one from both directions on density. DTW carries roughly *half* the fine graph’s edges and correlation roughly *1.4 times* as many, yet all three beat it by 8.5 % to 10.4 %. They are also indistinguishable from *each other* despite a 3.4-fold spread in edge count. Correlation versus DTW gives a paired per-fold difference of -0.2% with no directional tendency whatever (Table A.2)—and kNN-correlation lands between them at 76.5 %. Edge count does not order these graphs at all. Where the weights come from does. The effect is consistent across three different similarity statistics (Pearson correlation, DTW distance) and two different sparsification rules (global threshold, mutual k -nearest-neighbours), which makes it a property of *using the data* rather than an artefact of any one construction recipe.

Moreover, the scope of this comparison is that the three data-driven rows are evaluated within a single architecture by design. What they isolate is a property of the *graph*, whether data-estimated weights beat rule-assigned ones at fixed granularity, so holding the architecture constant is what makes the contrast clean, and the correlation family, which is complete at $n = 5$ across all five architectures, already establishes that the provenance effect is not architecture-specific. The choice of A3TGCN for the depth families is moreover conservative rather than favourable. On the complete correlation family its provenance gap (10.0 %) is the second smallest of the five, against a mean of 11.0 % (Section 5), so these rows understate the effect a differently chosen architecture would have shown. Two points nevertheless bound the strength of the conclusion. DTW’s advantage over the fine graph is large and near-unanimous, with every fold but one favouring DTW, though at eight folds and one architecture the margin is not resolvable to the standard the balanced families meet. These two families therefore speak to the mechanism, while the generality of the provenance effect across architectures is established separately by the complete correlation family.

6.3.4 Experiment E — Does anatomical structure itself contribute anything?

Every result so far compares graphs that differ in more than one respect. The random-topology control of Section 4.3.4 removes that ambiguity. It holds the node set, the edge count, the normalisation and the training procedure fixed, and varies only *which* nodes are connected. If anatomical structure is the source of the coarse graph’s advantage, replacing it with random topology should cost accuracy. Table 6.6 reports the outcome for A3TGCN, the strongest model on DSA.

Table 6.6: DSA random-topology control (A3TGCN). The random graphs match the anatomical coarse graph in node count (5) and edge count (6), and differ from it only in which nodes are joined. Every row averages eight LOSO folds per run. The replicates are five independently drawn random graphs for the random-topology row and five training seeds elsewhere. The decisive comparison is the middle pair.

Graph	Node partition / topology	Accuracy	Replicates
Adaptive (5-node)	5 nodes, topology learned	0.867 $\pm$ 0.013	5 seeds
Random 5-node	5 nodes, topology random	0.841 $\pm$ 0.014	5 graphs
Anatomical coarse (5-node)	5 nodes, topology anatomical	0.837 $\pm$ 0.019	5 seeds
No graph (<i>baselines</i>)	no partition, no edges	0.803 $\pm$ 0.021	15 runs

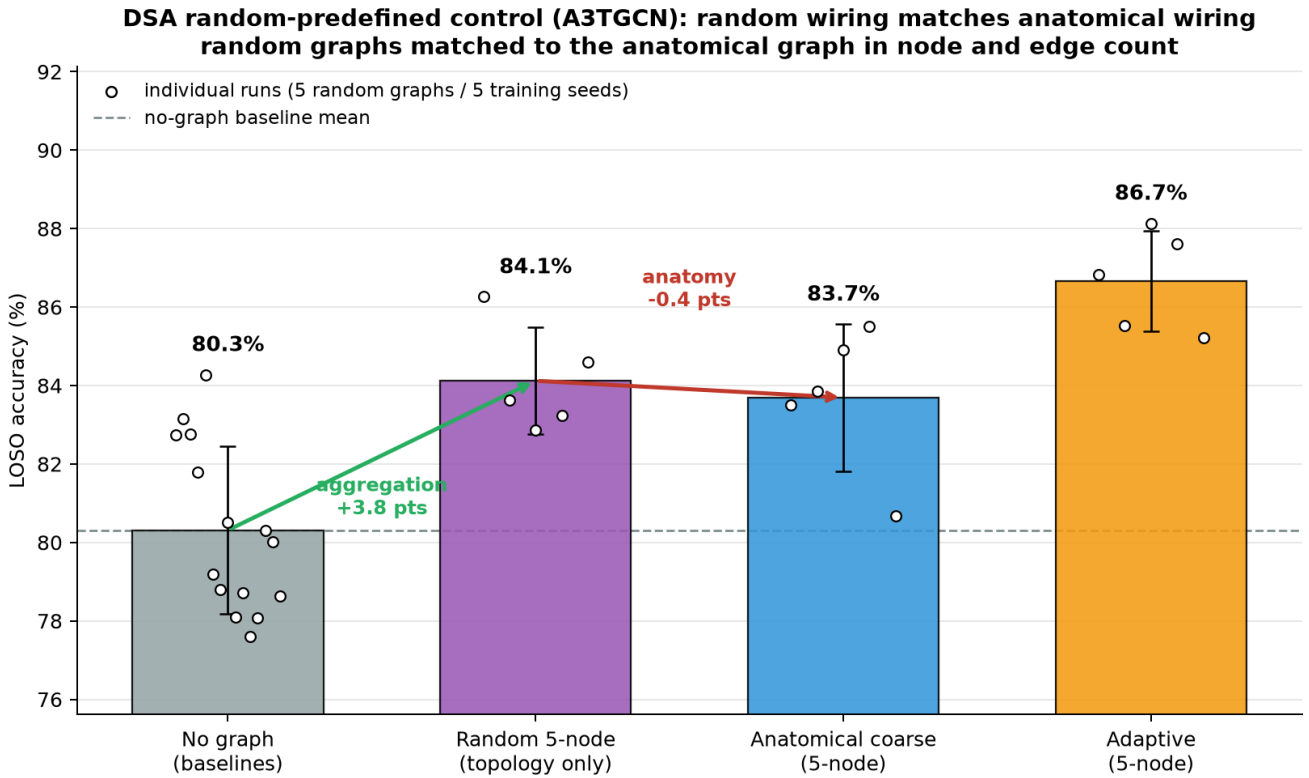


Figure 6.20: DSA random-topology control (A3TGCN). Bars are means. White points are individual runs—the five random graphs, and five training seeds for the anatomical and adaptive conditions. Aggregating 45 channels into 5 nodes is worth 3.8%. Making that aggregation anatomically correct is worth −0.4%. Learning the 5-node topology adds a further 3.0%, which is the largest remaining step but the one that does not survive the family-wise correction (Table A.2). The anatomical graph sits *inside* the distribution of random topologies, not above it.

The random predefined graph reaches 84.1% against the anatomically designed graph’s 83.7% (Fig. 6.20). The difference is not statistically detectable (Table A.2). The anatomical prior is not merely unnecessary—it is not even nominally ahead. Individual random draws span 82.9% to 86.3%, and the best of them (86.3%) essentially matches the learned adaptive graph, which suggests the 5-node aggregation admits many near-equivalent topologies rather than one privileged anatomical solution.

This decomposes the route from the no-graph baseline to the best DSA configuration into three steps of very unequal size.

- **Aggregating 45 channels into 5 body-segment nodes:** 80.3 % $\rightarrow$ 84.1 %, a gain of 3.8 %, obtainable with *any* 5-node topology.
- **Making that topology anatomically correct:** 84.1 % $\rightarrow$ 83.7 %, a change of -0.4 %, i.e. nothing.
- **Letting the topology be learned rather than fixed:** 83.7 % $\rightarrow$ 86.7 % starting from anatomy, a gain of 3.0 %, or 84.1 % $\rightarrow$ 85.9 % starting from a random topology, a gain of 1.8 %.

The three steps together carry A3TGCN from 80.3 % to 86.7 %, and they are not on equal evidential footing. The first is large and is what the control was built to isolate. The second is the one the control was built to test, and it is absent. The third is the largest remaining gain but rests on a single architecture, and the five-model average of the same contrast is only 1.1 %. The defensible reading is therefore that aggregation is what reliably buys the improvement, that anatomy buys none of it, and that learning the adjacency on top is suggestive rather than demonstrated at this sample size. Furthermore, the second step is the one this control exists to test, so it is worth stating what supports it. The 0.5 % favouring the random graph is smaller than the seed-to-seed spread of either condition, which is 1.4 and 1.9 points respectively, and a paired test over the eight folds does not separate them (Appendix A.4). The same holds on GConvLSTM, where the difference is -0.01 %. As a result, the consequence for RQ3 is direct. Elsewhere in this study the claim that structural prior knowledge fails to help rests on comparing different graph *constructions*. Here it rests on a control in which the construction is held fixed and only the domain knowledge is removed. The domain knowledge contributes nothing measurable. What the coarse graph supplies is a sensible *aggregation*—nine co-located sensor channels averaged into one node with a coherent signal—and any graph over the same five nodes supplies it equally well. In conclusion, the result admits a structural explanation, and Fig. 6.21 makes it concrete by drawing the graphs themselves. With five nodes there are only $\binom{5}{2} = 10$ possible undirected edges, and the anatomical graph uses six of them. A random six-edge draw therefore overlaps the anatomical graph substantially by construction. The five drawn here share between two and four of their six edges with it, against an expectation of 3.6. All five remain connected, and no node is more than three hops from any other, against an expectation of two for the anatomical graph. The topologies are genuinely different—seed 789 leaves the Torso with a single neighbour, discarding the hub structure entirely, and still reaches 84.6 %—but at this granularity there is not much topological information for any of them to encode, so the graph functions mainly as an aggregation scheme. The finding should not be over-generalised to graphs large enough for topology to carry real information. On DSA, at the granularity that actually works, it holds.

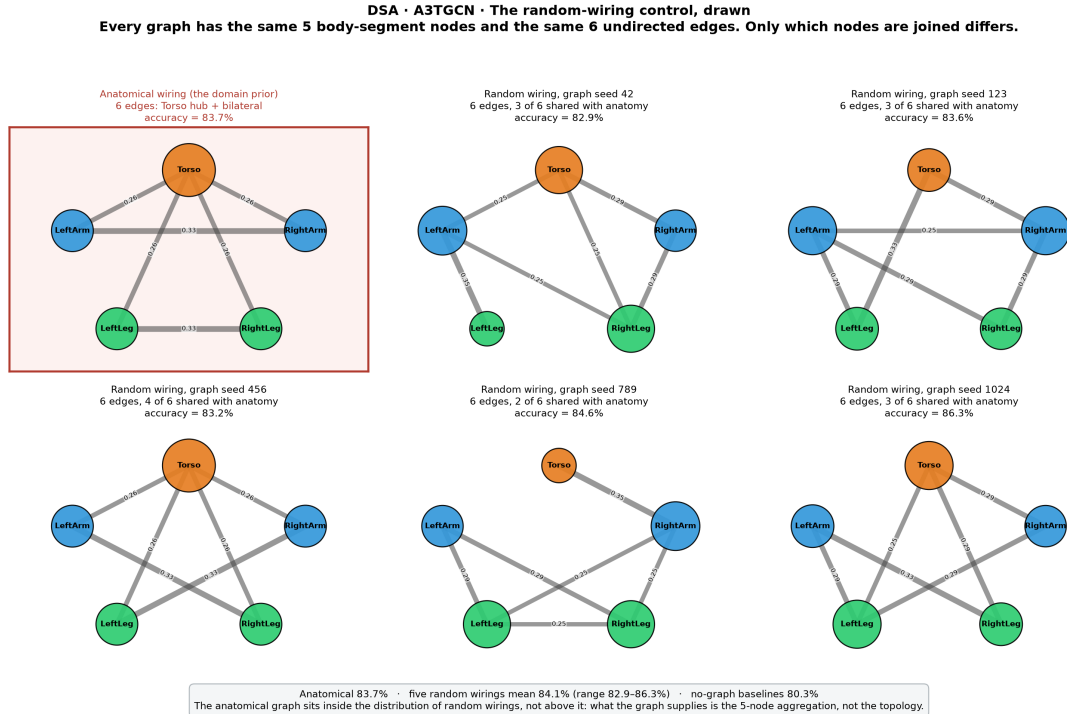


Figure 6.21: DSA — the random-topology control, drawn (A3TGCN). The anatomical graph is boxed. All six graphs have the same five body-segment nodes and the same six undirected edges, and differ only in which nodes are joined. Edge labels are the GCN-normalised weights, which follow from the topology. The topologies are regenerated from the graph seeds the experiments used, so these are the exact topologies that were trained. Seed 456 keeps the Torso hub and replaces the bilateral links with crossed ones. Seed 789 abandons the hub altogether, leaving the Torso with one neighbour. Neither is worse than anatomy. The anatomical topology sits *inside* this distribution rather than above it, which is the visual form of the claim that the graph contributes an aggregation scheme rather than topological knowledge.

DSA · GConvLSTM · The random-wiring control, drawn
Every graph has the same 5 body-segment nodes and the same 6 undirected edges. Only which nodes are joined differs.

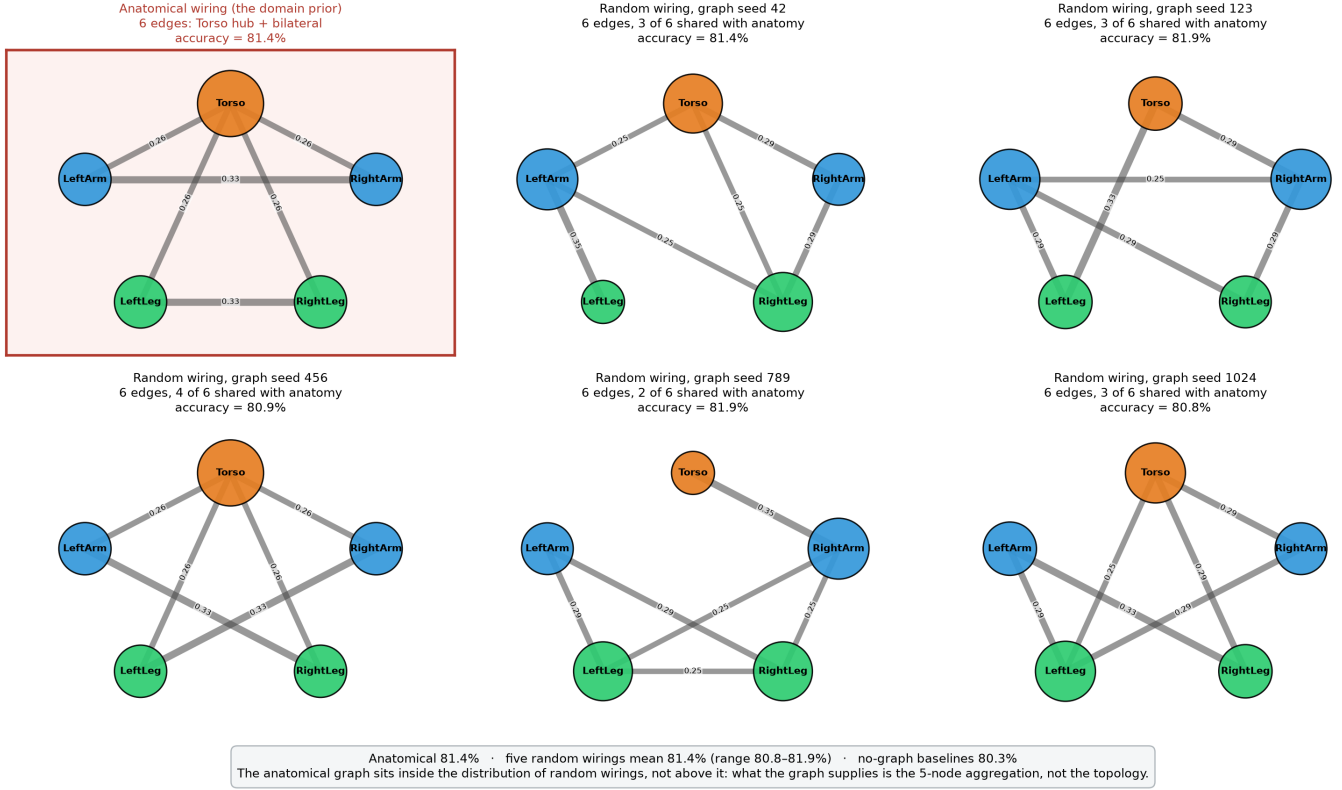


Figure 6.22: DSA — the random-topology control on GConvLSTM. The same five random topologies as Fig. 6.21, with GConvLSTM’s accuracies. The spread across topologies is narrower here (80.79 % to 81.86 % against 82.87 % to 86.26 % for A3TGCN), consistent with GConvLSTM being the architecture that responds least to its graph (Section 6.4). The anatomical graph is again inside the distribution, not above it.

The prior is not a useful starting point either. A remaining objection to the above is that the anatomical graph might still earn its place as an *initialisation*. Perhaps a predefined random graph performs comparably, yet SVD-seeding the learner from anatomy reaches a solution that a random seed cannot. The *adaptive-from-random* condition tests exactly that over five independently drawn graphs. Table 6.7 gives the resulting 2×2 .

Table 6.7: DSA random-topology control, both halves on both architectures (8 LOSO folds per run). Columns vary *what the graph is*. Rows vary *whether it is adaptive*. Random replicates are five independently drawn graphs. Anatomical replicates are five training seeds. All eight cells differ by less than one percentage point within a column pair.

Architecture	Adjacency	Anatomical topology	Random topology	difference
A3TGCN	Predefined	0.837 ± 0.019	0.841 ± 0.014	random +0.4
	Adaptive from it	0.867 ± 0.013	0.859 ± 0.007	anatomical +0.8
	<i>gain from adaptive</i>	+3.0	+1.8	
GConvLSTM	Predefined	0.814 ± 0.012	0.814 ± 0.005	random −0.0
	Adaptive from it	0.818 ± 0.012	0.817 ± 0.012	anatomical +0.1
	<i>gain from adaptive</i>	+0.5	+0.4	

Neither column difference is statistically detectable, in either role (Table A.2). Learning is worth 1.7% to 3.0% from *either* origin, and the origin itself makes no measurable difference. The anatomical graph is therefore neither a better predefined graph nor a better initialisation—the two roles in which a structural prior could plausibly have paid off.

Replication on a second architecture. A natural objection to a single-architecture control is that A3TGCN’s temporal attention gate might compensate for a poorly chosen topology, masking a topology effect that a plainer model would expose. The *entire* control—both the predefined and the adaptive-from-random halves, five random graphs each—was therefore repeated on **GConvLSTM**, which is Chebyshev-spectral convolution inside an LSTM, with no attention anywhere. It reproduces the same absence of effect in both roles. As a predefined graph, random topology reaches 81.35% (five graphs spanning 80.79% to 81.86%) against anatomical 81.36%, a difference of -0.01% (the same five topologies, carrying GConvLSTM’s accuracies, are drawn in Fig. 6.22). As an initialisation for the adaptive graph it reaches 81.72% against 81.83%, a difference of -0.11% . Neither is statistically detectable, and both effect sizes are indistinguishable from zero (Table A.2).

Across two architectures with entirely different spatial operators, then, anatomical topology is worth under one percentage point in either role, with effect sizes indistinguishable from zero on the non-attention model. The finding is a property of the graph, not of A3TGCN’s attention gate. The two architectures do differ in how much they gain from making the adjacency adaptive at all—1.7% to 3.0% for A3TGCN against 0.4% to 0.5% for GConvLSTM—which is consistent with the per-architecture pattern in Section 6.3.5, where the two models carrying an extra selection mechanism are the ones that benefit from learning the graph.

Both architectures are run at training seed 42, and the control’s replicates vary the *graph* whereas the anatomical rows vary the training seed. The means are comparable as expectations, but the two dispersions measure different sources of variation and should not be read as interchangeable. Both conditions are evaluated on both architectures.

6.3.5 Experiment F — Is a learned graph better than a good prior at 5 nodes?

Across all five architectures at $n = 5$, learning the adjacency helps in a minority of cases. For three of the five models the compact anatomical graph and the learned adaptive graph are within noise of each other, or the prior is ahead. GConvLSTM (coarse 81.4% vs adaptive 81.8%), GConvGRU (83.1% vs 82.7%) and DCRNN (84.6% vs 83.8%, the one model where the learned graph is measurably behind its prior). Averaged over all five models the two strategies differ by 1.1% (coarse 83.2% vs adaptive 84.3%), an order of magnitude smaller than the 14.5% granularity gap, and of the same order as the seed-to-seed variation within a single configuration. On DSA the learned and anatomical 5-node graphs therefore perform equivalently at the resolution this design can resolve.

The two models that gain from learning the graph are A3TGCN (coarse 83.7% $\rightarrow$ adaptive 86.7%) and Hybrid-TGNN (83.2% $\rightarrow$ 86.7%), and they now sit joint-first overall. Their means differ by 0.0005 against seed standard deviations of 0.013 and 0.019 respectively, so they should be read as tied rather than ranked. That the explicitly blended Hybrid model converges on the same accuracy as the purely learned A3TGCN graph is itself informative—two different routes to a graph reach the same ceiling. Topology reconstruction (Fig. 6.27) shows the learned adaptive graph recovers the hand-designed body-segment structure only partially (edge-set Jaccard ≈ 0.50 against the coarse graph), yet reaches comparable accuracy—the model finds an equally effective but not identical relational structure. Figs. 6.24 and 6.23 give the per-model view for GConvLSTM, whose learned graph barely improves on its prior, and for A3TGCN, the strongest adaptive model of the five.

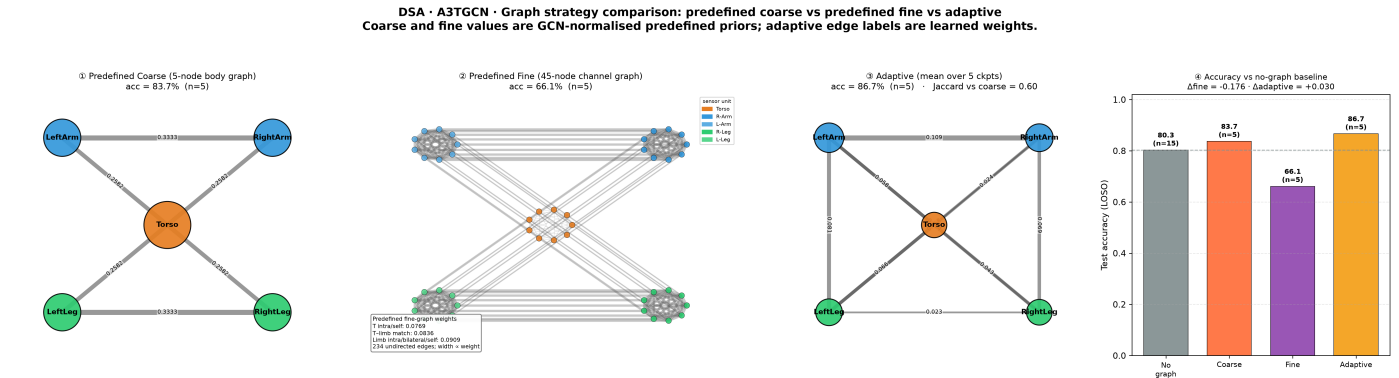


Figure 6.23: DSA — per-model comparison (A3TGCN, the strongest adaptive model at 86.7%): predefined coarse, predefined fine, and learned adaptive graphs against the no-graph baseline. The learned adaptive graph is A3TGCN’s best configuration and clearly exceeds all sequence baselines.

DSA · GConvLSTM · Graph strategy comparison: predefined coarse vs predefined fine vs adaptive
Coarse and fine values are GCN-normalised predefined priors; adaptive edge labels are learned weights.

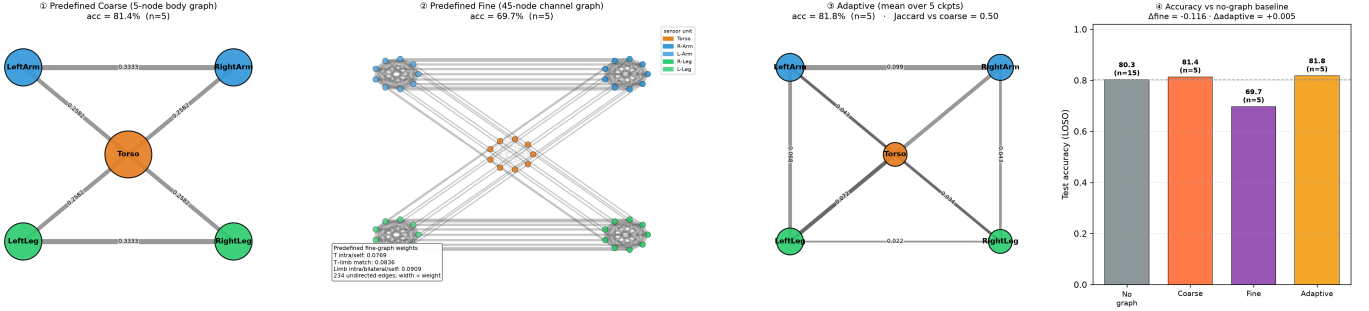


Figure 6.24: DSA — GConvLSTM: predefined coarse, predefined fine, and learned adaptive graphs against the no-graph baseline. Coarse and fine are predefined, GCN-normalised anatomical priors: the coarse panel labels every edge, whereas the fine panel reports its three exact edge-weight classes (Section A.2). Adaptive labels are learned weights. The fine graph loses to the baseline. Coarse and adaptive marginally exceed it.

DSA · GConvGRU · Graph strategy comparison: predefined coarse vs predefined fine vs adaptive
Coarse and fine values are GCN-normalised predefined priors; adaptive edge labels are learned weights.

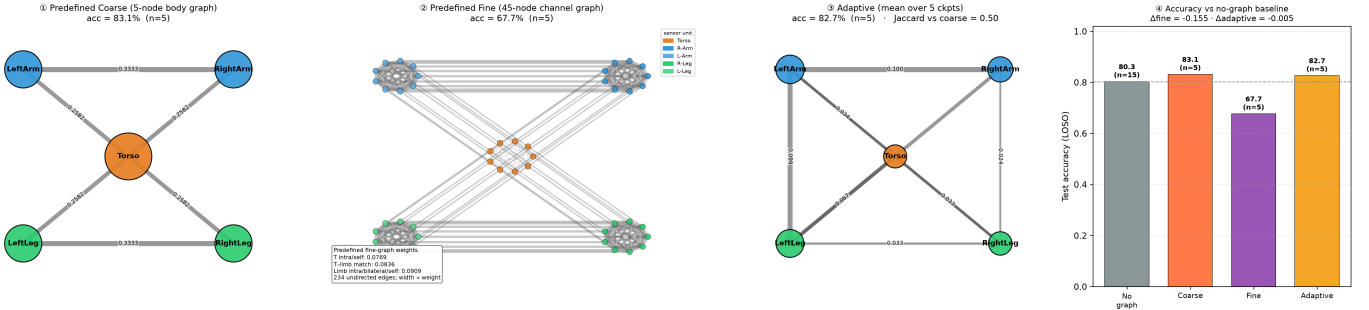


Figure 6.25: DSA — GConvGRU, $n = 5$: coarse 83.1 %, fine 67.7 %, adaptive 82.7 %. Like DCRNN, its learned graph does not quite reach the anatomical prior, and both sit close to the 80.3 % no-graph mean.

DSA · DCRNN · Graph strategy comparison: predefined coarse vs predefined fine vs adaptive
Coarse and fine values are GCN-normalised predefined priors; adaptive edge labels are learned weights.

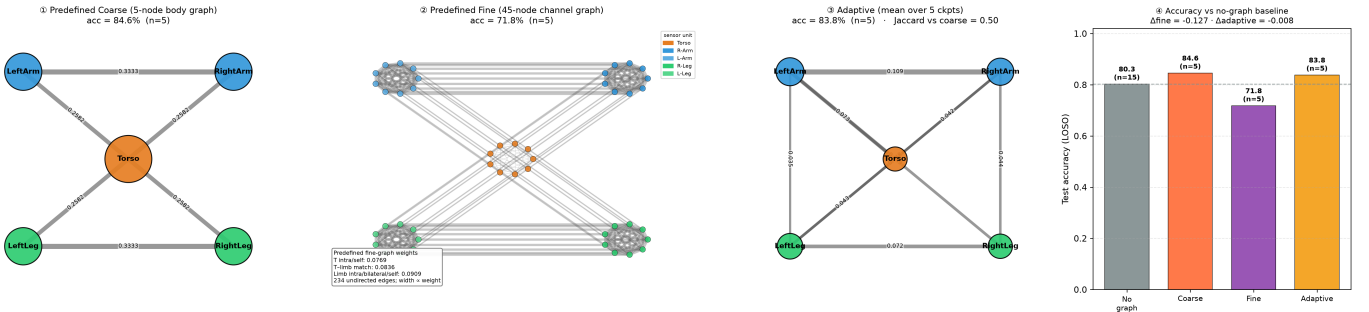


Figure 6.26: DSA — DCRNN, $n = 5$: coarse 84.6 %, fine 71.8 %, adaptive 83.8 %. The one model whose learned graph does not reach its anatomical prior, despite recovering that prior to the same degree as the gated cells (edge-set Jaccard 0.50). Its 71.8 % is the best fine-graph result of any architecture (Section 6.4).

DSA — Graph Topology: Predefined Coarse vs Predefined Fine vs Adaptive (GConvLSTM)
Node radius = weighted degree · edge labels = |weight| · the adaptive graph recovers the hand-designed body-segment structure

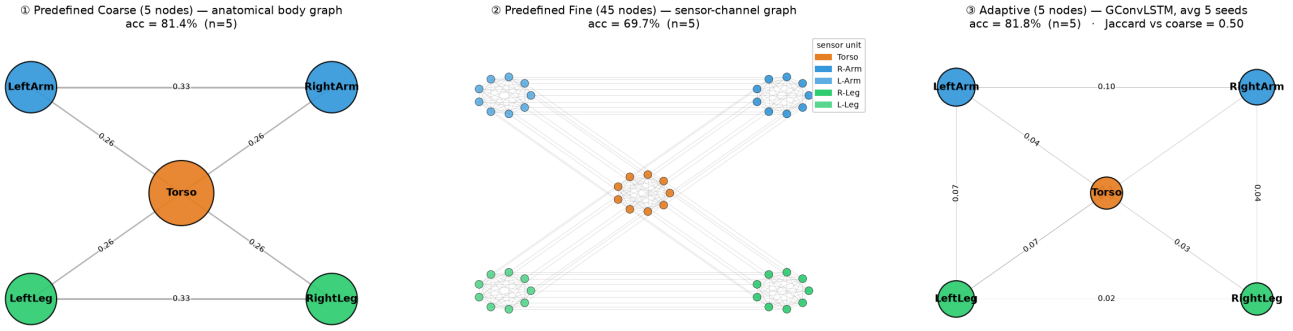


Figure 6.27: DSA — graph topology: predefined coarse (5-node anatomical) vs predefined fine (45-node channel) vs learned adaptive (GConvLSTM). Node radius $\propto$ weighted degree. The learned adaptive graph partially recovers the body-segment structure (Jaccard $\approx$ 0.50).

DSA · Hybrid-TGNN · Graph strategy comparison: predefined coarse vs predefined fine vs adaptive
Coarse and fine values are GCN-normalised predefined priors; adaptive edge labels are learned weights.

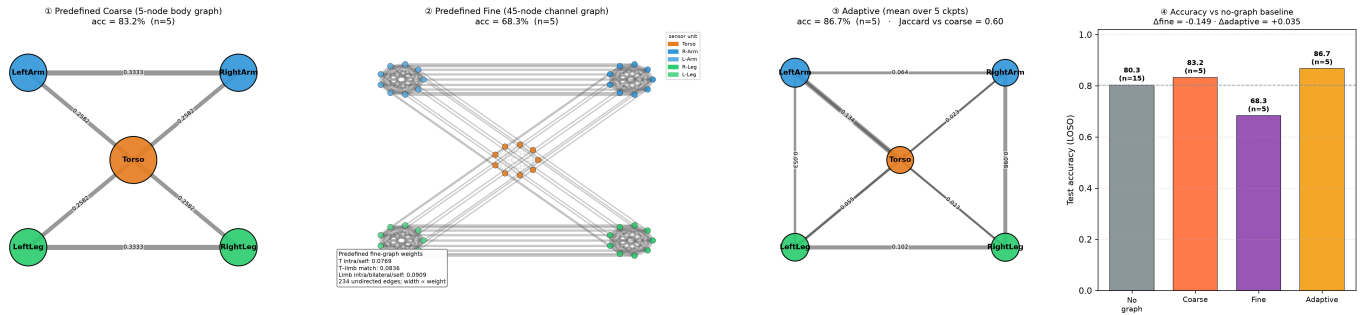


Figure 6.28: DSA — Hybrid-TGNN, $n = 5$: coarse 83.2%, fine 68.3%, adaptive 86.7%. Its learned graph attains the highest edge-set agreement with the anatomical prior of any model (Jaccard 0.60, tied with A3TGCN and against 0.50 for the three recurrent cells), consistent with the hybrid’s blended adjacency being anchored to that prior by construction (Section 4.4.4).

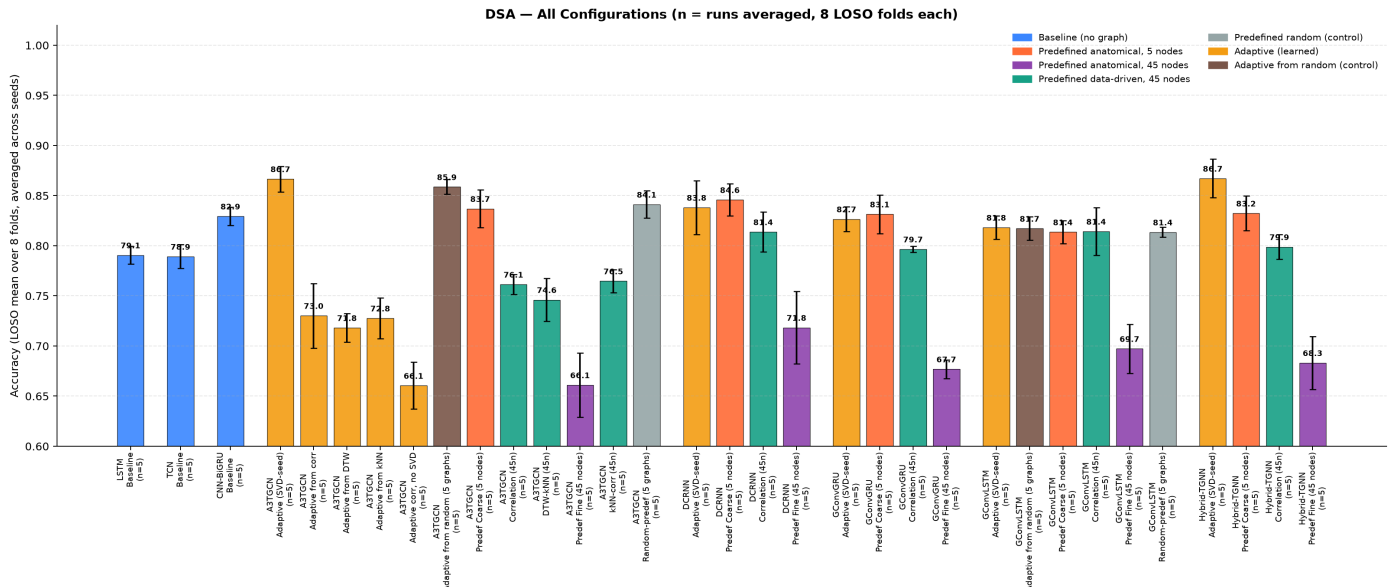


Figure 6.29: DSA — every configuration in the sweep, grouped by architecture and coloured by graph family, being the complete inventory behind Table 6.3. Colour encodes edge *provenance* rather than node count, so the adaptive category (orange) deliberately spans both granularities and the 45-node anatomical graph (purple) is kept separate from the 45-node data-driven ones (green). The two halves of the random-topology control are drawn as separate categories rather than pooled, the predefined half in grey and the adaptive-from-random half in brown, so that each can be read against its anatomical counterpart. The surrounding text reads the three patterns this arrangement is built to expose.

The remaining four architectures show the same coarse/fine/adaptive ordering, and their panels follow (Figs. 6.24–6.28). Section 6.4 analyses what separates them, and the two predefined structures themselves are shown in Fig. 4.3.

Fig. 6.29 collects every configuration of the sweep into a single view, and three things in it are worth reading off directly. The first concerns the adaptive family, which the figure colours by provenance rather than by node count so that both of its granularities land in one category. For A3TGCN that category runs from 86.7% at five nodes down to 66.1% to 73.0% at forty-five. One strategy therefore spans twenty points on granularity alone, which is the Phase-2 finding stated from the strategy’s own side rather than the graph’s, and it shows that learning the adjacency does not rescue the 45-node representation. Experiment G takes that up directly.

The second is that the 45-node anatomical graph sits at or near the bottom of every architecture’s group without exception, and the third is that replacing it with a data-driven graph over the same 45 nodes recovers most of that deficit. Together those two are the provenance effect of Experiment D, visible across all five architectures at once rather than one contrast at a time. Finally, the two halves of the random-topology control are drawn as separate categories rather than pooled, so the comparison they exist to make stays legible. Used directly as a predefined graph the control sits with the 5-node anatomical graph, and used as an initialisation for the learner it sits with the adaptive graph seeded from anatomy. Neither is displaced, which is Experiment E in inventory form.

6.3.6 Experiment G — Does the benefit of learning reverse at 45 nodes?

Experiment F found that learning a 5-node adjacency helps some architectures and not others. This experiment asks the same question at 45 nodes, and the answer turns out to depend on granularity in a way that neither result shows on its own. Where Experiments C to E vary which graph the model is given, this one varies only whether that graph is *predefined* or *adaptive*. Four paired comparisons share node set, training folds and initial adjacency within each pair, and differ solely in whether the adjacency is held at its initialisation or optimised end-to-end with the classifier (Table 6.8, Fig. A.4). The 5-node row of Table 6.8 repeats Experiment F’s result so that the sign change is visible in one place.

Table 6.8: DSA — A3TGCN. Effect of *training* the adjacency, holding the node set and the initial graph fixed within each row. $n = 5$ seeds per cell, eight LOSO folds each. Paired tests over the folds are reported in Appendix A.4, where none of the 45-node rows is separable.

Nodes	Initial graph	Predefined (%)	Adaptive (%)	Δ accuracy
5	anatomical coarse	83.69	86.66	+2.97
45	correlation $\geq \text{mean} r $	76.14	73.03	−3.12
45	DTW–kNN	74.61	71.82	−2.79
45	kNN–correlation	76.47	72.77	−3.70

The sign of the effect is determined by granularity. At five nodes, learning adds 2.97% over the anatomical prior it starts from. At 45 nodes it subtracts between 2.79 and 3.70 points, and it does so for *three independently constructed graphs*—a thresholded correlation graph, a DTW-based mutual- k graph, and a correlation-based mutual- k graph—whose edge counts differ more than threefold. Pooling the three gives a mean penalty of −3.20% over 24 paired fold observations, with the learned arm lower in 16 of 24—a consistent direction, but not a detectable difference (Table A.2).

These results do not establish that an adaptive adjacency *harms* performance at 45 nodes, since no individual family and no pooled comparison separates the two arms. What the data support is the weaker claim that *there is no evidence that training a 45-node adjacency improves on leaving it predefined at a data-estimated initialisation*, with a consistent point estimate of roughly three points in the opposite direction across three independent constructions. Given that a learned 45-node adjacency introduces 2025 additional free parameters, the absence of any measurable return is itself the finding.

This reconciles two results that would otherwise appear to conflict. The adaptive family is the best-performing graph strategy on DSA (Table 6.5), yet learning is unhelpful at 45 nodes. Both are true because the adaptive family is a *5-node* learned graph. What makes it strong is the granularity it inherits, not the learning applied on top. The learning component itself is worth +1.13 points over the coarse graph as a difference of means, which is within the run-to-run variation of the experiment.

6.3.7 Experiment H — Does where the learned graph starts matter more than the learning itself?

Experiments F and G both varied whether the adjacency is trained. This one varies **where its training begins**, holding everything else fixed, and it produces the largest effect of the three.

The adaptive graph does not hold an adjacency directly. It holds two low-rank node-embedding tables whose product forms one, so a prior graph reaches the learner only through a truncated SVD of that graph (Section 4.3.3). Removing the SVD seeding while changing nothing else—same node set, same seed graph, same folds, same schedule—costs 6.98% of accuracy, 73.03% with the seeding against 66.05% without, at $n = 5$ each.

That number is worth setting against every other learning effect measured in this thesis (Table 6.9). Training the adjacency at five nodes is worth 2.97%, training it at forty-five costs −3.20%, and choosing the prior it starts from is

worth -0.4% . The initialisation of the low-rank factorisation is worth more than any of them, and more than twice the largest.

Table 6.9: Every learning-related effect measured in this thesis, ranked by magnitude. All are DSA, A3TGCN except the fourth row, which averages five architectures. Every row is a difference of means, so the fourth differs from the median paired difference the test in Appendix A.4 operates on. The initialisation of the low-rank parameterisation is the largest of them, which is the finding of this experiment. Paired tests for all rows are in Appendix A.4.

What is varied, everything else held fixed	Nodes	Δ accuracy
SVD seeding of the embeddings, on against off	45	+6.98
Training the adjacency against leaving it fixed	5	+2.97
Training the adjacency against leaving it fixed	45	-3.20
Learned adjacency against the coarse prior, five models	5	+1.13
Anatomical against random topology	5	-0.40

The direction is what Section 4.3.3 predicts. By the Eckart–Young–Mirsky theorem the rank- d truncation is the closest the factorisation can come to the seed graph, so seeding places the optimiser at the best available representation of the prior, whereas a random start places it somewhere the prior never reaches. The gradient signal available afterwards is evidently not strong enough to recover the difference. In this parameterisation, where the optimiser starts constrains what it can reach, and it does so more firmly than the subsequent training changes it.

Two qualifications belong with the result. It is measured on one architecture at eight folds, and the paired test over those folds does not separate the two arms once every comparison in the study is accounted for (Appendix A.4), so the effect is reported as the largest and most consistent of the learning effects rather than as an established one. The practical reading is nevertheless unambiguous, since a practitioner choosing to learn an adjacency has to initialise it somehow, and this is the only evidence in the study bearing on that choice.

Phase 2 is not the only place the comparison exists. Phase 1 runs the same contrast across thirteen seed graphs on both low-dimensional datasets, and there it does nothing at all. Across 129 paired comparisons the mean gain is 0.19% , the median is exactly zero, and 48 of the 129 pairs return identical outcomes (Table 6.10). That is what the mechanism predicts. At three and six nodes there is almost nothing for a rank- d truncation to discard, so the seeded and unseeded starts very nearly coincide and the initialisation cannot matter. The effect appears only where the node set is large enough for the truncation to be lossy, which places this experiment on the same granularity axis as Experiments C and D rather than beside them.

Table 6.10: SVD seeding of the learned adjacency, measured separately in each phase. The effect is confined to the high-dimensional node set. Phase 1 pairs every seeded mode with its unseeded counterpart over two datasets, five architectures and thirteen seed graphs, whereas Phase 2 has one such pair, A3TGCN with the correlation-mean seed, over five seeds.

	Phase 1, 3 and 6 nodes	Phase 2, 45 nodes
Paired comparisons	129	5 seeds, A3TGCN
Mean gain from seeding	+0.19	+6.98
Median gain	0.00	+6.98
Better / worse / identical	52 / 29 / 48	5 / 0 / 0
Spread across seeds	—	+4.65 to +11.12

6.4 Which Architectural Properties Matter

The results so far treat the five TGNNs as interchangeable carriers of a graph. They are not. This section reads Table 6.3 down the other axis. For a fixed graph, what separates the architectures, and what property of each cell explains its behaviour. The conclusion is that the architectures differ systematically and interpretably—but by much less than the graph does, which is itself the study’s central result stated from the model side.

The architecture is the smaller axis of variation. Reading Table 6.3 across graphs within one architecture, and across architectures within one graph, gives two very different pictures (Table 6.11). Holding the architecture fixed and changing the graph moves accuracy by 12.1% to 20.6% . Holding the graph fixed and changing the architecture moves it by 3.2% to 5.7% . The graph is worth roughly three to four times the architecture on this dataset. Every per-model observation below therefore describes a real and reproducible difference, but one that operates inside a band narrower than the one the graph controls. This is the quantitative form of the thesis’s main claim, and it is why graph construction rather than operator choice is where the study locates the decisive design decision.

Table 6.11: DSA — the two axes along which LOSO accuracy varies, computed from Table 6.3 ($n = 5$ throughout). Left: holding the architecture fixed, the range in accuracy across the four graph modes. Right: holding the graph mode fixed, the range in accuracy across the five architectures. The accuracy range column is the highest LOSO accuracy in that row minus the lowest, so A3TGCN’s 20.6 is $86.7 - 66.1$. That the two ranges do not overlap is the finding.

<i>Vary the graph, fix the architecture</i>			<i>Vary the architecture, fix the graph</i>		
Architecture	Accuracy range	best $\rightarrow$ worst	Graph mode	Accuracy range	best $\rightarrow$ worst
A3TGCN	20.6	86.7 $\rightarrow$ 66.1	Coarse (5n)	3.2	84.6 $\rightarrow$ 81.4
Hybrid	18.4	86.7 $\rightarrow$ 68.3	Adaptive (5n)	4.9	86.7 $\rightarrow$ 81.8
GConvGRU	15.4	83.1 $\rightarrow$ 67.7	Correlation (45n)	5.3	81.4 $\rightarrow$ 76.1
DCRNN	12.8	84.6 $\rightarrow$ 71.8	Fine (45n)	5.7	71.8 $\rightarrow$ 66.1
GConvLSTM	12.1	81.8 $\rightarrow$ 69.7			

A3TGCN, the most graph-sensitive cell. A3TGCN is GConvGRU with a per-node sigmoid attention gate added on top (Section 4.4.2). That one addition makes it both the best and the worst model in the study, depending entirely on what graph it is given. With a learned 5-node graph it is the joint best configuration on DSA (86.7%). With the 45-node anatomical graph it is the *worst* of all five (66.1%), and it carries the largest range of any architecture (20.6%). The mechanism is consistent with what the gate does. Attention re-weights each node’s contribution per time step, so it amplifies whatever discriminative structure the graph already exposes. Over five body-segment nodes with a well-formed adjacency there is a clear signal to amplify. Over 45 channels connected by three fixed weight values there is not, and a gate with 45 near-indistinguishable nodes to arbitrate between adds a selection problem instead of solving one. A3TGCN is accordingly the right architecture when the graph is known to be good and the wrong one when it is not, and it is the reason the random-topology control needed a replication on a cell without attention (Section 6.3.4).

GConvGRU, the reference cell, and what attention is worth. GConvGRU is the same GRU-based graph cell without the gate, which makes the A3TGCN-versus-GConvGRU contrast the cleanest available test of what temporal attention contributes—the two differ in exactly one component. It is worth +0.6 points on the coarse graph, -1.6 on the fine graph, and +4.0 on the adaptive graph. Attention therefore does not pay a fixed dividend. It pays when the graph is good, and costs when the graph is poor, which is the same conditional pattern the thesis reports at the level of the graph itself. GConvGRU on its own is the median model in every mode, never best, never worst, and the smallest and cheapest of the five at 106 k parameters.

DCRNN, the most robust to a bad graph and the least helped by a good one. DCRNN replaces Chebyshev filtering with diffusion convolution—a truncated bidirectional random walk with separate learnable weights per hop (Section 4.4.1). It behaves differently from the other four in two respects, and both follow from that substitution. It is the strongest model on both *bad* graphs, reaching 71.8% on the fine graph, 5.7% clear of A3TGCN at the bottom, and 81.4% on the correlation graph, the only architecture to keep a 45-node graph at baseline level. And it is the only model whose learned adjacency ends up *behind* its anatomical prior (-0.8 points). Per-hop weights give the cell an internal mechanism for re-weighting the graph’s influence. It can learn to discount a hop that carries noise, which partially repairs a poorly constructed adjacency without touching the adjacency itself. The corollary is that it has correspondingly less to gain when the adjacency is already good, which is exactly the pattern observed. The robustness is not free. DCRNN carries 204 k parameters, roughly double the gated cells, and was the slowest model to train in Phase 1 at a median 1616 s against 858 s for A3TGCN. It is the choice when the graph cannot be trusted and the budget allows it.

GConvLSTM, the flattest responder, and why it was the right control. GConvLSTM computes all four LSTM gates from a *single* Chebyshev convolution over $[x_t, h_{t-1}]$, whereas GConvGRU needs two because its reset gate must modulate h_{t-1} before the candidate is formed. One convolution per step is the least spatial mixing of any cell here, and the consequences are visible throughout. GConvLSTM has the lowest ceiling on DSA (81.8%, the weakest best-configuration of the five), the narrowest range across graphs (12.1%), and the smallest gain from learning the adjacency (+0.4 points). It is also the weakest TGNN on EPILEPSY (93.1% mean against 95.9% for Hybrid-TGNN), so the pattern holds across both phases and both dimensionalities. Consistency, rather than performance, is what makes it valuable here. Because GConvLSTM responds least to the graph, it is the architecture on which a topology effect would be hardest to mask, which is why the random-topology control was replicated on it (Section 6.3.4). That the control returns the same absence of effect on the most graph-sensitive cell (A3TGCN) and the least (GConvLSTM) is what makes that finding a property of the graph rather than of one architecture.

Hybrid-TGNN, reaching the same ceiling by a different route. The hybrid blends the predefined and learned adjacencies with a single learnable scalar (Section 4.4.4), initialised to trust the prior. It reaches 86.7%, tying A3TGCN for best on DSA, and it gains the most of any model from making the adjacency adaptive (+3.5 points). Its interest is not its rank but its topology. It attains the highest edge-set agreement with the anatomical prior of any model (Jaccard 0.60), consistent with an adjacency anchored to that prior by construction. Two architectures therefore arrive at the same accuracy from opposite directions—A3TGCN by learning a 5-node graph freely and departing from anatomy (Jaccard 0.50 for the plain recurrent cells), the hybrid by starting from anatomy and being pulled only partway away. That both

converge on 86.7% supports the reading developed in Section 6.3.4. At five nodes many topologies are near-equivalent, so what the graph supplies is the aggregation and not the topology. The hybrid’s blend weight is the study’s most direct readout of that, and it is why the model earns its place despite adding no accuracy over A3TGCN.

What generalises across the five. Three regularities hold for every architecture without exception. All five rank the graphs in the same order, coarse and adaptive far above correlation, and correlation far above fine. All five place the 45-node anatomical graph last, below every no-graph baseline. And no architecture recovers a poorly constructed graph. The best fine-graph result of any model (71.8%) remains 8.5% below the worst coarse-graph result (81.4%). The two architecture-dependent effects are narrower and both concern *selection*. Thhe two models carrying an extra selection mechanism (A3TGCN’s attention gate, the hybrid’s learned blend) are the two that gain from learning the adjacency, and the model with an internal per-hop re-weighting (DCRNN) is the one most able to tolerate a bad one. Choice of temporal operator is therefore a second-order decision that adjusts how sharply a model responds to its graph, not one that changes which graph it should be given.

6.5 Summary of Findings

Experiments A to H establish six findings, collected in Table 6.12. Each is stated as a difference in accuracy, each names the experiment that produced it, and each is subjected to paired significance testing in Appendix A.4. Chapter 7 carries the same letters throughout, so any claim made there can be traced to the experiment behind it without searching. Their interpretation with respect to the research questions is the subject of Chapter 7.

Table 6.12: Summary of findings. Each row states one finding, the evidence that establishes it, and where it is reported and tested. Every finding is stated as a difference in accuracy. All six are subjected to paired significance testing in Appendix A.4, which is where the statistical machinery and the adjusted values are reported.

Finding	Evidence	Experiment	Tested in
1. The graph matters more than the architecture. Changing the graph moves accuracy 12% to 21% within a fixed architecture. Changing the architecture moves it 3% to 6% within a fixed graph.	DSA, five architectures $\times$ four graphs, all at $n = 5$	§6.4	Table A.2
2. Node granularity governs performance. A 5-node graph beats a 45-node graph by 14.5%, with every one of 40 paired observations agreeing.	coarse vs fine, weights held uniform in both	C (§6.3.2)	Table A.2
3. Edge provenance governs performance independently of granularity. Data-estimated weights beat rule-assigned ones by 11.0% at the same 45 nodes—and the data-driven graph is <i>denser</i> , so this is not a density effect.	correlation vs fine, granularity held at 45	D (§6.3.3)	Table A.2
4. Structural prior knowledge contributes nothing measurable. Random topology matched in node and edge count is within 0.5% of anatomy, in both roles a prior could play.	random-topology control, five random graphs, two architectures	E (§6.3.4)	Table A.2
5. What transfers across dimensionality is granularity, not the construction rule. The correlation graph wins at 3–6 channels and loses at 45, because the same rule induces a far finer node set.	EPILEPSY 95.7% vs DSA 76.1% to 81.8%	A, D (§6.3.3)	Table A.2
6. Where a learned graph starts matters more than the learning. Removing the SVD seeding of the embeddings costs 6.98%, more than twice any other learning effect measured here.	SVD on vs off, node set and seed graph held fixed	H (§6.3.7)	Table A.2

Chapter 7

Discussion of Experimental Results

Chapter 6 opened with a single pair, LSTM against GConvLSTM, in which one architectural change was worth 2.3 % under one graph and −9.4 % under another. The eight experiments then decomposed that span into its causes. This chapter reads the decomposition back against the three research questions. The short form of what follows is that the graph decides the outcome, that two measurable properties of the graph decide what the graph does, and that the domain knowledge conventionally assumed to be the valuable part of a sensor graph is not among them.

7.1 RQ1: Does the graph help, and is the answer conditional on which graph?

Answered by Experiment A (Section 6.1) at low dimensionality and Experiment B (Section 6.3) at 45 channels. Cost is read from Table 6.4, and the model-side reading from Section 6.4.

RQ1 resolves in favour of the temporal graph models on the joint criterion it poses—accuracy *and* computational cost—and its second clause resolves affirmatively. The improvement is conditional, and this study is able to state the condition precisely rather than leaving it implicit. The graph supplied to the model must carry relational information at a granularity suited to the channel count. The remainder of this section establishes both halves of that answer and identifies the condition exactly.

BASICMOTIONS is saturated and therefore uninformative beyond confirming correctness. On EPILEPSY the graph inductive bias is *not* sufficient on its own. With the domain predefined graph the strongest baselines—CNN-BiGRU (97.2 %) and TCN (95.5 %)—outperform the predefined-graph TGNNs (91.8 %), a deficit of 5.4 points ($p < 0.001$). The TGNNs close that gap only once a well-constructed graph is used. With a data-driven correlation graph the same five architectures average 97.4 %, a difference from CNN-BiGRU indistinguishable from zero ($p = 0.65$), with the largest gains on the hardest class. The honest reading is parity rather than victory—the best single configuration reaches 98.7 %, but that is a maximum over many configurations scored on one fixed test split. Efficiency, however, is decisively in the TGNNs’ favour and is the second half of RQ1. The recurrent baselines carry $\sim 5\times$ the parameters (LSTM 565 k, CNN-BiGRU 554 k vs ~ 105 k for the lightest TGNNs), so the TGNNs match or approach them at a small fraction of the model size (Fig. 6.13). The answer to RQ1 is therefore that TGNNs are markedly more parameter-efficient and—with a well-chosen graph—at least as accurate. Only the naive predefined graph leaves them behind the strongest baselines, and even then at a fifth of the parameters.

The contribution of this answer is therefore not the ranking but the *condition*. A benchmark that had evaluated the TGNNs only with the domain graph would have concluded that they lose to CNN-BiGRU. A benchmark that had reported only their best configuration would have concluded that they win. Both conclusions are artefacts of a graph choice left uncontrolled. By varying the graph while holding architecture, protocol and seeds fixed, this study shows that the same five architectures move from a reliable 5.4-point deficit to statistical parity on the strength of the adjacency alone.

The same point holds at 45 channels and can there be quantified against the alternative explanation. Read down the architecture axis rather than the graph axis, holding the graph fixed and changing the temporal operator moves accuracy by 3.2 % to 5.7 %, whereas holding the operator fixed and changing the graph moves it by 12.1 % to 20.6 % (Table 6.11). The two ranges do not overlap. Whatever a practitioner gains by selecting among GConvLSTM, GConvGRU, A3TGCN, DCRNN and the hybrid is bounded above by roughly a quarter of what the same practitioner gains or loses by selecting the graph.

Which architectural properties matter, and how much. That the architecture is the smaller axis does not make the architectures interchangeable, and the differences between them are systematic rather than incidental (Section 6.4). They track one identifiable property of each cell, namely how the architecture *selects* among the signals the graph gives it. Cells carrying an extra selection mechanism, being A3TGCN’s per-node attention gate and the hybrid’s learned blend, respond most strongly to the graph in both directions. They are the two best models on a good graph (86.7 % each) and, in A3TGCN’s case, the worst on a bad one (66.1 %), because a gate amplifies whatever structure it is given and there is none to amplify at 45 near-uniform nodes. They are also the only two that gain from learning the adjacency. At the

opposite extreme, DCRNN’s per-hop diffusion weights let it discount an uninformative hop internally, which makes it the most tolerant of a badly constructed graph, best of all five on both the fine graph (71.8 %) and the correlation graph (81.4 %), and correspondingly the least helped when the graph is already good, the one model whose learned adjacency ends up behind its prior. GConvLSTM, which derives all four gates from a single graph convolution and so performs the least spatial mixing per step, responds least to the graph in either direction and has the lowest ceiling in both phases.

The architectural half of RQ1 is therefore a trade-off rather than a ranking. Selection capacity buys accuracy when the graph is trustworthy and costs it when the graph is not, and internal re-weighting buys robustness at the price of upside. Both effects are real and reproducible across phases, and both are second-order. No architecture recovers a poorly constructed graph, since the best fine-graph result of any model remains 8.5 % below the worst coarse-graph result. The architectural choice adjusts the outcome and the graph determines it, which is why graph construction, examined next as RQ2, is the decisive variable rather than the presence of a graph or the choice of temporal operator.

7.2 RQ2: Which properties govern performance, what is each worth, and do they transfer?

Answered by Experiment C (Section 6.3.2) and Experiment D (Section 6.3.3), read across the two phases for the transfer clause.

RQ2 is answered with two measured quantities rather than with a ranking of strategies. Node *granularity* governs performance and is worth 14.5 %. Edge *provenance*, meaning whether the weights are estimated from data or assigned by rule, governs it independently and is worth 11.0 %. Both survive the move to 45 channels, but only one of them survives in the form a low-dimensional experiment would have predicted, and that asymmetry is the transfer answer.

Granularity. At low dimensionality (EPILEPSY, Experiment A) data-driven correlation and kNN graphs are the strongest strategies and a plain predefined-binary graph is the weakest. At high dimensionality (DSA, Experiment C) the picture reorganises around granularity. A compact anatomical prior over five body-segment nodes is excellent, whereas a *denser* predefined prior over forty-five channel nodes is catastrophic, falling below the no-graph baselines. More structure, or higher-resolution structure, is therefore not better. What the graph has to supply is aggregation at the right level, and the coarse-against-fine contrast measures exactly that, holding the dataset, the architecture, the training procedure and the anatomical rule fixed while changing only the node count. Every one of the 40 paired observations favours the five-node graph, at $r = 1.00$ and $p < 0.001$ after correction (Table A.2).

Provenance. The second property is isolated by the correlation graph, which is a controlled variation on the fine graph over the same 45 nodes (Experiment D, Section 6.3.3). It is 11.0 % more accurate. The obvious explanation, that the fine graph fails through dilution by too many near-uniform edges, is ruled out by the same pair, because the correlation graph carries *more* edges, being 310 to 336 undirected against 234. Edge count and accuracy move in opposite directions here, so density cannot be the operative variable. What separates them is that the correlation weights are estimated from the training folds and are therefore graded and informative, whereas the fine graph’s take three fixed values assigned by rule. Two further 45-node constructions, one built from DTW distances and one from mutual nearest neighbours in correlation space, agree despite spanning a 3.4-fold range in edge count, which makes the effect a property of using the data rather than an artefact of one recipe.

Transfer. The two quantities behave differently when dimensionality changes, and the difference is instructive. The correlation graph is the *winner* on EPILEPSY at 95.7 % and falls to 76.1 % to 81.8 % on DSA, below the no-graph baselines. The construction rule did not change and correlation did not stop being a sound statistic. What changed is the node set the rule induces. Over three to six channels a correlation graph is already coarse and no aggregation problem exists, whereas over forty-five channels the same rule induces a node set that good edge weights only partly rescue. The property that fails to transfer is therefore not the rule but the granularity it implies at the new channel count, which is the same property Experiment C measures directly. This is why the transfer question belongs to RQ2 rather than standing alone. The answer to what governs performance and the answer to what governs transfer are one answer, and it is a property that can be checked in advance rather than discovered after the experiment.

The practical consequence is a reordering of where design effort belongs. Against these two quantities the predefined-versus-learned dichotomy that organises much of the literature accounts for 1.1 % of accuracy across five architectures and cannot be distinguished from no difference at all ($p_{\text{Holm}} = 0.531$, Table A.2). Choosing the level at which to aggregate channels, and estimating the edge weights from data, is worth an order of magnitude more than choosing whether the adjacency is supplied or learned.

7.3 RQ3: Does the domain knowledge contribute anything beyond the aggregation?

Answered by Experiment E (Section 6.3.4), which is a control rather than a comparison, with Experiments F, G and H (Sections 6.3.5–6.3.7) asking whether the structure has to be supplied by hand at all.

The answer is that it does not. Anatomical topology contributes nothing measurable, in either of the two roles a structural prior could play, and the result is the outcome of a control designed to produce it rather than an absence of evidence.

Every argument in the preceding section infers the value of domain knowledge by contrasting graphs that differ in several respects at once, and no such contrast can separate the value of the anatomy from the value of having a graph at all. The random-topology control removes every difference but one. It holds the node count at five, the edge count at six, the normalisation and the training procedure fixed, and varies only which nodes are joined (Section 4.3.4). As a predefined graph, random topology reaches 84.1 % against anatomy’s 83.7 % ($p_{\text{Holm}} = 1.000$). As the initialisation the learner departs from, it reaches 85.9 % against 86.7 % ($p_{\text{Holm}} = 1.000$). Both differences are under one percentage point, smaller than the seed-to-seed spread of either condition, and the entire control repeated on GConvLSTM, chosen for being architecturally the most unlike A3TGCN of the five, agrees to within 0.1 %. Learning is worth 1.7 % to 3.0 % from either starting point. The starting point is worth nothing.

What a predefined sensor graph actually supplies. This reframes the object rather than merely scoring it. What the coarse graph contributes is not the adjacency but the *partition*, the decision that these nine channels belong to one body segment and should be summarised together. The route from the no-graph baseline to the best configuration decomposes accordingly, into 3.8 % for aggregating 45 channels into 5 nodes, which any five-node topology supplies, and -0.4% for making that aggregation anatomically correct. Once the partition is fixed, how its parts are connected to one another is close to irrelevant, at least at a granularity where every node is within a hop or two of every other. The recommendation is therefore more specific than to use a domain graph. Expert effort is repaid in deciding what to group, and not in drawing edges between the groups.

Can the structure be obtained without the prior? If the knowledge is not what matters, the natural question is whether the graph has to be supplied by hand at all, and Experiments F, G and H answer it in three parts. At five nodes a learned adjacency is not reliably better than the anatomical prior. It exceeds it for A3TGCN and Hybrid-TGNN, the two cells carrying an extra selection mechanism, ties for GConvGRU and GConvLSTM, and falls 0.8 % behind for DCRNN, with the five-model mean not separable from the prior (Experiment F). At forty-five nodes the sign reverses, and training the adjacency is worth -3.20% across three independently constructed seed graphs, so learning does not rescue a granularity that was chosen badly (Experiment G). And where the learning begins matters more than whether it happens, since removing the SVD seeding of the embeddings costs 6.98 %, more than twice any other learning effect measured here (Experiment H). Both observations point the same way as the control. The learned graph gains little over the anatomical one because both operate on the same partition, and the partition is what was doing the work.

The correlation family sharpens the point from the other side. As Section 6.1.4 establishes, a data-driven correlation graph is most effective when used *directly* as a predefined graph, and pushing the same graph through the adaptive learner generally lowers accuracy. A well-estimated graph is not improved by further end-to-end learning, which is a concrete instance of the broader conclusion that graph construction rather than graph learning is what moves accuracy.

7.4 Threats to Validity

Internal. The DSA grid allocates breadth and depth according to what each claim requires (Section 5), which bounds what the single-architecture families can be asked to support. The baselines and the coarse, fine, adaptive and correlation families span all five architectures at $n = 5$, and every claim reported as significant in Section A.4 rests on those balanced families. The DTW-kNN and kNN-correlation families, and the four adaptive-from-data-driven families, are evaluated for A3TGCN. Their position in the 45-node ordering is therefore consistent with the provenance account rather than independent confirmation of it, and within those families an architecture-specific effect would be indistinguishable from a general one.

Statistical power. The tests operate on eight LOSO folds. That is adequate for the large construction effects, which reach the maximum attainable effect size ($r = 1.00$), but weak for differences of one or two points. With $n = 8$ the smallest attainable two-sided p is 0.0078, so a genuine small effect can easily go undetected. Two comparisons here reach nominal significance and fail after Holm adjustment—DTW-versus-fine ($p = 0.016$, $p_{\text{Holm}} = 0.188$, $r = 0.94$) and adaptive-versus-random ($p = 0.039$, $p_{\text{Holm}} = 0.430$, $r = 0.83$)—and both carry large effect sizes. These should be read as *not established*, not as shown to be absent. The random-topology conclusion is stated accordingly in Section 6.3.4, as no evidence of an anatomical advantage, in a design with demonstrated sensitivity to the effects that do matter, rather than proof of equivalence.

Control scope. The random-topology control covers two architectures in both roles—A3TGCN and GConvLSTM, predefined and adaptive-from-random—which addresses the concern that the result might depend on A3TGCN’s attention mechanism. All four comparisons agree to within one percentage point. Because GConvLSTM was selected as the architecture least like A3TGCN, that agreement is the decisive test rather than an exhaustive one. Extending the control

to the three remaining cells would broaden the base without changing what the two most dissimilar architectures already establish. Both architectures are run at a single training seed, and the control’s replicates vary the random graph whereas the anatomical comparison’s vary the training seed, so the two dispersions are not interchangeable.

External. Two small UEA datasets and one larger HAR dataset may not generalise to other MTS domains. The random-topology finding in particular is established at a granularity (five nodes, six edges) where the space of possible topologies is small. It should not be extrapolated to graphs large enough for topology to carry substantial information.

Construct. Accuracy and macro-F1 capture predictive quality but not calibration or robustness. Variance is mitigated with five seeds and subject-independent LOSO, and edge density is controlled by the random-topology control.

Chapter 8

Conclusion and Future Work

8.1 Answers to the Research Questions

This thesis benchmarked five TGNN architectures against three sequence baselines across low- and high-dimensional HAR datasets under a controlled protocol, systematically varying the graph-construction strategy over 165 DSA configurations and 1429 Phase-1 runs.

RQ1. The graph inductive bias is established as an opportunity whose realisation depends on a stated condition, and the conditionality is the finding rather than a caveat on it. On the UEA datasets the TGNNs match the strongest sequence baseline at roughly a fifth of its parameters, and on DSA the best configurations reach 86.7% against CNN-BiGRU’s 82.9%. The same five architectures, given a poorly constructed graph, fall to 68.7%, below every baseline. The width of that range, spanned by architectures held otherwise identical, locates the decisive design choice in the graph rather than in the model, and the two axes of variation do not overlap. Changing the graph within a fixed architecture moves accuracy by 12.1% to 20.6%, whereas changing the architecture within a fixed graph moves it by 3.2% to 5.7%. The architectures nonetheless differ systematically, and along one identifiable property, namely how each cell selects among the signals its graph supplies. Cells with an extra selection mechanism, being A3TGCN’s attention gate and the hybrid’s learned blend, respond most strongly to the graph in both directions and are the only two that gain from learning it, whereas DCRNN’s per-hop diffusion weights buy tolerance of a bad graph at the cost of upside. The choice of temporal operator therefore adjusts how sharply a model responds to its graph and does not change which graph it should be given. The efficiency relationship is the one that transfers most strongly. On EPILEPSY the TGNNs bought parity at a fifth of the parameters, whereas on DSA the smallest model in the study, A3TGCN at 424k, is also the most accurate, against 2.23M parameters for the strongest baseline.

RQ2. The study identifies the two properties of a constructed graph that govern performance and quantifies each. Node *granularity* is worth +13.4 points and edge *provenance*, meaning whether weights are estimated from data or assigned by rule, is worth +11.2 points, both as median paired differences over the evaluation folds, and both established with every paired observation favouring the same direction. As differences of means the same two effects are 14.5% and 11.0%. Set against these, the predefined-versus-learned distinction that organises much of the literature accounts for +1.5 points and cannot be distinguished from no difference at all.

The transfer clause resolves through the same two quantities. The ordering in which graph models beat sequence baselines holds for the coarse and adaptive graphs and inverts for the fine graph. The sharpest case is the correlation graph, which is the strongest strategy at low dimensionality and among the weakest at high dimensionality, not because correlation is a poor statistic but because over 45 channels the same rule induces a far finer node set. What fails to transfer is therefore not the construction rule but the granularity it implies at the new channel count, which is a property that can be checked in advance rather than discovered after the experiment. The property that governs performance and the property that governs transfer are the same property, which is why the two are answered together.

RQ3. Structural prior knowledge contributes nothing measurable, and the evidence is a control rather than a comparison. A random-topology five-node graph matched to the anatomical graph in node and edge count performs indistinguishably from it as a predefined graph (84.1% against 83.7%) *and* as a seed for the learned adjacency (85.9% against 86.7%). Neither difference survives correction, both are smaller than the run-to-run variation of the experiment, and the entire control repeated on a second and deliberately dissimilar architecture agrees to within 0.1%. The coarse graph’s advantage over the no-graph baseline decomposes accordingly, into +3.8 points for *aggregating* 45 channels into 5 nodes, which any five-node topology supplies, and −0.4 points for making that aggregation anatomically correct.

What a predefined sensor graph contributes on this dataset is therefore a partition and not a topology, a decision about which channels to summarise together rather than a claim about how the resulting parts relate. Asking whether the structure can instead be obtained without the prior gives a consistent answer. At five nodes a learned adjacency is not reliably better than the anatomical one, at forty-five nodes learning is worth −3.20% across three independent seed graphs, and where the learning begins matters more than whether it happens at all, since removing the SVD seeding

costs 6.98 %. The learned graph gains little over the anatomical one because both operate on the same partition, and the partition is what was doing the work.

In conclusion, the consistent thread is that graph *construction* dominates every other design decision examined here—more than the temporal operator, and more than whether the adjacency is learned. For practitioners the implication is concrete. Effort spent choosing the level at which to aggregate channels, and estimating edge weights from data, is better repaid than effort spent encoding domain structure or learning the adjacency end-to-end.

8.2 Future Work

A controlled timing study. The efficiency side of RQ1 is answered here on parameter count alone, because the recorded wall-clock training times in this study span several machines and several revisions of the harness and are not comparable with one another. Recovering that axis needs no new experiments and no new data, only a single controlled measurement pass: fix one device and one software stack, and time a fixed number of forward-backward-step iterations for each of the eight architectures at each of the three input shapes used here. That would turn training cost from an unreported quantity into a second measured axis alongside model size, and it would establish whether the parameter advantage the graph models hold translates into a throughput advantage or is offset by the sparse-operator overhead their convolutions carry.

This study closes one question and opens a sharper one. What a graph contributes here is an *aggregation*, a decision about which channels to summarise together, rather than a topology, and the level at which that aggregation happens governs accuracy more than any other design choice examined. Having located the decisive variable, the useful continuation is to study it directly rather than to compare further construction recipes. They are ranked below by expected value.

The first is to map the granularity curve. This thesis measures granularity at two points, five nodes and forty-five, and finds 14.5 % between them. Where the optimum lies is unknown, and it is the quantity a practitioner needs. Intermediate partitions of the DSA channels—by modality (fifteen nodes), by axis, or by unsupervised clustering of the training correlations, would trace accuracy against node count with everything else held fixed. The study motivates but cannot test the expectation that the curve is non-monotone and peaks nearer five, because the benefit at five nodes comes from averaging nine channels into a coherent per-segment signal, and that averaging weakens as the partition refines.

The second follows from the same result and is the more interesting of the two. If the graph supplies a partition, the open problem is obtaining one where no anatomy is available. DSA is unusually generous in handing the partition over for free, and most multivariate settings are not. Replacing the fixed partition with a differentiable pooling layer that assigns channels to super-nodes end-to-end would let the question this thesis asks of the adjacency be asked of the partition. Does it recover the anatomical grouping, and does it need to? The random-topology control suggests not, which would be a useful negative result for domains without a physical prior. That is where the finding becomes a method rather than an observation about HAR.

Two controls deserve widening first, because both bound how far the present conclusions travel. The random-topology result is established at five nodes and six edges, a space too small for topology to carry much information, so it demonstrates the irrelevance of topology at that granularity rather than in general. A 45-node random graph matched to the fine graph would test the opposite regime. Similarly, the depth families run on one architecture, and although that choice is conservative rather than flattering (Section 5), replicating them on a second cell would convert a consistent observation into an established one. Neither is expensive, since both extend pipelines already in the submitted code.

Statistically, several comparisons here are reported as undetected rather than absent. Where the argument depends on equivalence—most importantly the anatomical and random topologies—a two-one-sided-test procedure against a pre-specified margin would let that claim be made positively rather than as a failure to reject. A dataset with more subjects would serve the same end, and would also give the granularity curve usable confidence intervals.

Two smaller additions would improve the benchmark without changing its conclusions. Classical anchors—DTW-1NN, Euclidean-1NN and a ROCKET-family method—would situate the comparison against the non-deep reference points of the time-series literature. They are the natural extension of the benchmark as an artefact, and sit outside the graph-construction question it was built to answer. And the efficiency analysis (Fig. 6.16) is expressed in parameters, which is a proxy for deployment cost rather than a measurement of it. On-device latency, memory and energy per window decide whether a wearable model ships, and the recurrent cells unroll sequentially over 125 time steps, so their fivefold parameter advantage may narrow or reverse on hardware that rewards parallelism.

These amount to one recommendation. The comparative phase of this question is finished, and this thesis is evidence that ranking predefined graphs against learned ones answers the wrong question. What is needed is a treatment of channel aggregation as a first-class modelling decision, measured across its range and learned where no prior is available.

Appendix A

Supporting Material

This appendix holds material that supports the results of Chapter 6 without being needed to follow the argument. These are the exact edge weights of the predefined graphs, the seed distributions behind the DSA comparison, the paired significance tests, the comparison-validity table, and the reason computational cost is reported as parameter count. Each is referenced from the point in the main text where it is relevant.

A.1 What Each Comparison Isolates

Table A.1 states, for every comparison in Table A.2, which properties are held fixed and which vary. A comparison in which exactly one property differs *isolates* that property, so its result may be attributed to it. A comparison in which two properties differ is a comparison of *strategies*. It remains informative about which strategy to choose while attributing nothing to either property alone. Every result this thesis reports as significant comes from a single-property row.

Table A.1: What each comparison isolates. Here *fixed* means the property is identical in both arms, so it cannot contribute to the difference. Rows marked *one property* support attribution to that property. Rows marked *two* compare whole strategies and are reported as such, not as evidence about a single factor.

Comparison	Nodes	Edge weights	Adjacency	Isolates
<i>One property differs — attribution is licensed</i>				
Coarse vs Fine	5 vs 45	fixed (rule)	fixed	node granularity
Correlation vs Fine	fixed (45)	data vs rule	fixed	edge provenance
DTW vs Fine	fixed (45)	data vs rule	fixed	edge provenance
Correlation vs DTW	fixed (45)	both data	fixed	edge density
Random vs Coarse	fixed (5)	random vs anatomical	fixed	domain knowledge
Adaptive vs adaptive-from-random	fixed (5)	random vs anatomical seed	fixed (learned)	value of the prior
Adaptive vs predefined (45n)	fixed (45)	fixed (same seed graph)	learned vs fixed	training the adjacency
SVD vs no-SVD seeding	fixed (45)	fixed	initialisation	where the optimiser starts
Adaptive vs Coarse	fixed (5)	learned vs rule	learned vs fixed	learned vs predefined adjacency
<i>Two or more differ — reported as a strategy comparison</i>				
Coarse vs Correlation	5 vs 45	rule vs data	fixed	— (two properties)
Adaptive (5n) vs Fine (45n)	5 vs 45	learned vs rule	learned vs fixed	— (three properties)
Adaptive vs Random 5n	fixed (5)	learned vs random	learned vs fixed	— (two properties)

A.2 Exact Weights of the Predefined DSA Graphs

Section 4.3 states that the predefined topologies are binary and that all numerical weights follow from the GCN renormalisation of Eq. 4.1. The resulting values are given here in full, so that either graph can be reconstructed exactly.

In the **coarse** graph the Torso has $\tilde{d}_T = 5$ and every limb has $\tilde{d}_L = 3$, so a Torso–limb edge is $1/\sqrt{5} \cdot 3 = 0.2582$ and a bilateral limb edge is $1/\sqrt{3} \cdot 3 = 0.3333$.

In the **fine** graph every channel has eight within-unit neighbours and a self-loop. A Torso channel also has four matching-channel links to limb units, giving $\tilde{d}_T = 1 + 8 + 4 = 13$. A limb channel has the Torso and its bilateral counterpart as two matching-channel neighbours, giving $\tilde{d}_L = 1 + 8 + 2 = 11$. The only non-zero off-diagonal weights are therefore $1/13 = 0.0769$ for two Torso channels in the same unit, $1/\sqrt{13} \cdot 11 = 0.0836$ for a matching Torso–limb pair, and $1/11 = 0.0909$ for two channels in a limb unit or a bilateral matching-channel pair. Self-loops carry $1/13$ on Torso channels and $1/11$ on limb channels. The fine adjacency thus has 234 undirected off-diagonal links plus 45 self-loops, represented as 513 non-zero entries in the symmetric 45×45 matrix. Fig. A.1 maps each edge type to its value.

That the fine graph takes only *three* distinct off-diagonal values, spanning 18% end to end, is the concrete sense in which its weights carry no information about which channel pairs actually covary—the property Section 6.3.3 identifies as the cause of its collapse.

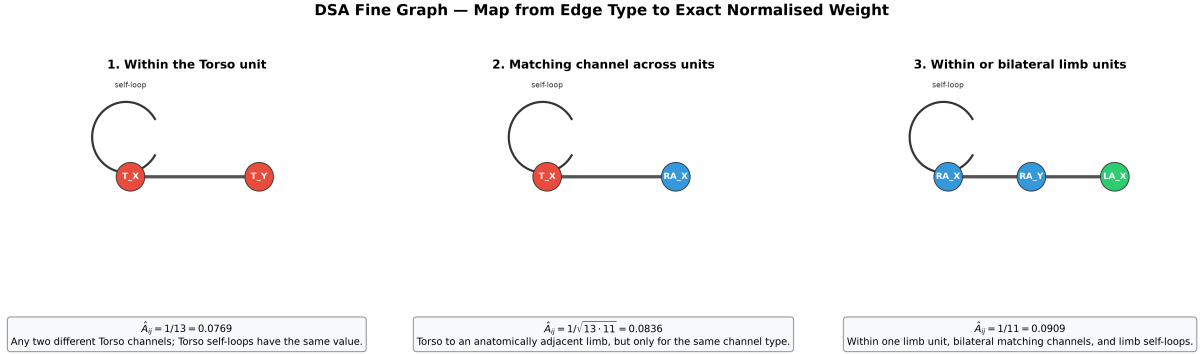


Figure A.1: DSA fine graph: mapping from an edge type to its exact normalised weight. The examples identify which channel pairs receive each value. A loop around a node denotes its self-loop. It receives the same normalised value as a within-unit edge for that node’s body unit.

A.3 Seed Distributions Behind the DSA Comparison

Fig. A.2 shows the seed distributions behind those means rather than the means alone. The separation is not an artefact of averaging. The weakest seed of A3TGCN’s and Hybrid-TGNN’s best configuration still lands above the strongest seed of CNN-BiGRU.

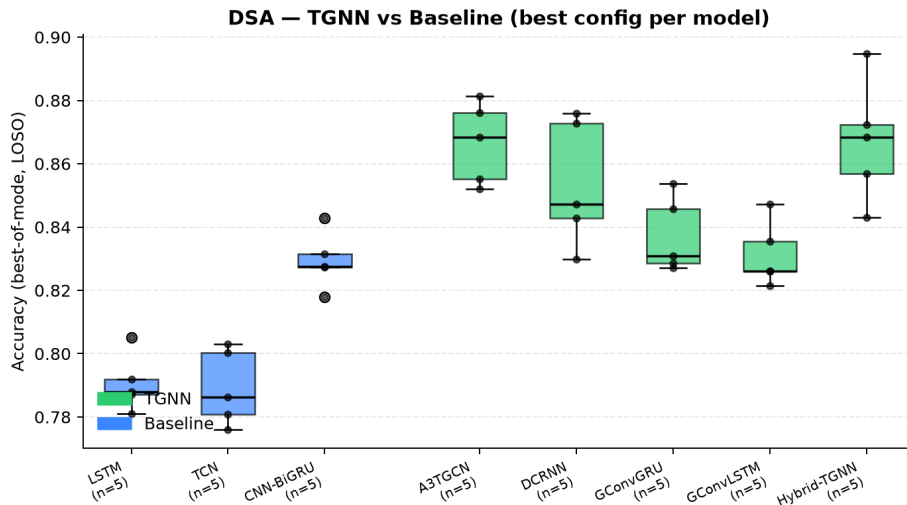


Figure A.2: DSA — distribution over the five seeds of each model’s best graph mode. Boxes span the interquartile range, points are individual seeds. Selecting the best mode per model is a maximum over modes and is therefore optimistically biased for any *single* model, but the bias applies equally to the baselines (which have one mode each) and the TGNNs, and the separation between the two groups exceeds the within-model seed spread.

A.4 Paired Significance Tests

The comparisons above are differences of means. Table A.2 subjects them to the paired Wilcoxon signed-rank tests specified in Section 5.5, with Holm–Bonferroni adjustment across all sixteen tests. The outcome divides the findings cleanly into those the data support and those it does not.

Each comparison below holds every property fixed except the one it is designed to test, so its result may be attributed to that property. Table A.1 in Appendix A.1 states this row by row, and identifies the three comparisons that vary more than one property and are therefore reported as strategy comparisons rather than as evidence about a single factor. Every result reported as significant comes from a single-property comparison.

Table A.2: DSA — paired Wilcoxon signed-rank tests over LOSO folds, seeds averaged within fold, Holm-adjusted across all sixteen tests. n is the number of paired observations (model $\times$ fold, or fold alone where a single model is involved). r is the matched-pairs rank-biserial effect size. $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

Comparison (A vs B)	n	A (%)	B (%)	Δ	p_{Holm}	r
<i>Graph construction — the large effects</i>						
Coarse (5n) vs Fine (45n)	40	83.2	68.7	+13.4	<0.001***	+1.00
Adaptive (5n) vs Fine (45n)	40	84.3	68.7	+14.9	<0.001***	+1.00
Correlation (45n) vs Fine (45n)	40	79.7	68.7	+11.2	<0.001***	+0.97
Coarse (5n) vs Correlation (45n)	40	83.2	79.7	+1.5	0.156	+0.45
<i>Learning and prior knowledge — the small effects</i>						
Adaptive (5n) vs Coarse (5n)	40	84.3	83.2	+1.5	0.531	+0.34
Adaptive vs Random 5n	8	86.7	84.1	+2.6	0.430	+0.83
<i>The random-topology control — both halves</i>						
Random 5n vs Coarse 5n (A3TGCN)	8	84.1	83.7	+0.5	1.000	+0.28
Adaptive vs Adaptive-from-random	8	86.7	85.9	+0.9	1.000	+0.33
Random 5n vs Coarse 5n (GConvLSTM)	8	81.4	81.4	-0.0	1.000	-0.06
Adaptive vs Adaptive-from-random (GConvLSTM)	8	81.8	81.7	+0.5	1.000	+0.11
<i>45-node ordering (A3TGCN)</i>						
DTW vs Fine	8	74.6	66.1	+9.8	0.188	+0.94
Correlation vs DTW	8	76.1	74.6	-0.2	1.000	-0.06
<i>Learning the adjacency at 45 nodes (A3TGCN)</i>						
Adaptive vs predefined: correlation	8	73.0	76.1	-3.8	1.000	-0.44
Adaptive vs predefined: DTW–kNN	8	71.8	74.6	-3.0	1.000	-0.28
Adaptive vs predefined: kNN–corr	8	72.8	76.5	-1.3	1.000	-0.39
SVD seeding: with vs without	8	73.0	66.1	+8.3	0.430	+0.83

The adjusted p and the effect size answer different questions. Two columns carry the argument of Table A.2, and they are not substitutes for one another. The adjusted p reports whether a difference could plausibly have arisen by chance once the fact that sixteen tests were performed has been accounted for. The effect size r reports how consistently the eight folds agreed, so that $r = 1.00$ indicates that every paired observation favoured the same side without exception, while r near zero indicates that the folds pointed in no particular direction. Read together they distinguish three situations. A small adjusted p accompanied by a large r is an established effect. A large adjusted p accompanied by a small r is a genuine absence of effect. A large adjusted p accompanied by a large r is an effect that this design, at eight folds, lacked the resolution to establish, which is a statement about the experiment rather than about the world and is treated as such throughout.

The reported difference is a median rather than a mean. The Δ column gives the median paired difference, which is the quantity the Wilcoxon test operates on, and it therefore differs slightly from the differences of means quoted in Section 6.3.3 (14.5% and 11.0%). Both tables cover the same five architectures at five seeds, so the two sets of numbers are consistent estimates of the same effects on the same data and differ only in the summary statistic used. The decomposition table reports means because they are more readily interpreted, whereas this table reports the medians on which the test is actually computed.

Effects that survive correction. Both effects in the granularity and provenance decomposition survive comfortably. Coarse over fine and adaptive over fine reach $r = 1.00$ —every one of the 40 paired observations favours the 5-node graph, without exception—at $p < 0.001$ after correction. Correlation over fine reaches $r = 0.97$ at $p < 0.001$. The claim that graph construction governs performance on DSA is therefore on firm statistical ground, not merely suggested by mean differences.

Effects that the design cannot resolve. The adaptive-versus-coarse difference does *not* survive correction ($p_{\text{Holm}} = 0.531$, $r = 0.34$, raw $p = 0.059$). Averaged over five models, learning the adjacency cannot be distinguished from using the anatomical prior. The difference is therefore reported as undetected, and the two strategies treated as equivalent at the resolution this design can achieve. Coarse-versus-correlation likewise does not survive ($p_{\text{Holm}} = 0.156$, $r = 0.45$). Once all five architectures are included, a 5-node anatomical graph and a 45-node data-estimated one cannot be separated, even though both decisively beat the 45-node rule-based graph.

Three further comparisons reach nominal significance but fall below threshold after the family-wise correction. These are DTW-versus-fine ($p = 0.016$, $p_{\text{Holm}} = 0.188$, $r = 0.94$), adaptive-versus-random ($p = 0.039$, $p_{\text{Holm}} = 0.430$, $r = 0.83$), and SVD-versus-no-SVD seeding ($p = 0.039$, $p_{\text{Holm}} = 0.430$, $r = 0.83$). All three carry large effect sizes and are best read as strongly suggestive but not established at $n = 8$ —the distinction between no effect and insufficient power to resolve one is material here, and Section 7.4 treats it as the latter. All sixteen comparisons form a single family for the purposes of correction. Testing a subset of them would relax the adjustment without reducing the number of claims made, so the conservative reading is the one adopted throughout.

Density does not order the data-driven graphs. The correlation-versus-DTW test is the most clear-cut negative result. The two graphs differ 3.4-fold in edge count (310 to 336 against 98 to 105 undirected) yet their paired per-fold difference is -0.2% with $r = -0.06$ —not merely non-significant but with no directional tendency whatever. Combined with both graphs decisively beating the denser fine graph, this rules out edge density as an explanatory variable at 45 nodes and leaves weight provenance as the operative factor.

Interpreting the random-topology control. The random-versus-anatomical test returns $p = 0.547$ (Holm-adjusted to 1.000) with a small effect size. A non-significant result is *not* proof of equivalence, and it would be wrong to claim the two graphs are identical in effect. Three observations nevertheless make the conclusion robust. First, the point estimate favours the *random* graph (+0.5 points), so there is not even a directional hint that anatomy helps. Second, the effect size $r = 0.28$ is small, whereas the same test at the same n detects the granularity contrasts at $r \approx 1.00$ —the design demonstrably has the power to find effects that matter, and finds nothing here. Third, five independent random topologies were tested, so the comparison is against a distribution of random topologies rather than a single lucky draw. The defensible statement is therefore that *no evidence was found that anatomical structure outperforms random topology at matched granularity and density, in a design with demonstrated sensitivity to the effects that do matter.*

A.5 Figures Supporting the Results Chapter

These figures support Chapter 6 without being needed to follow its argument. Each is referenced from the point in the main text where it is relevant.

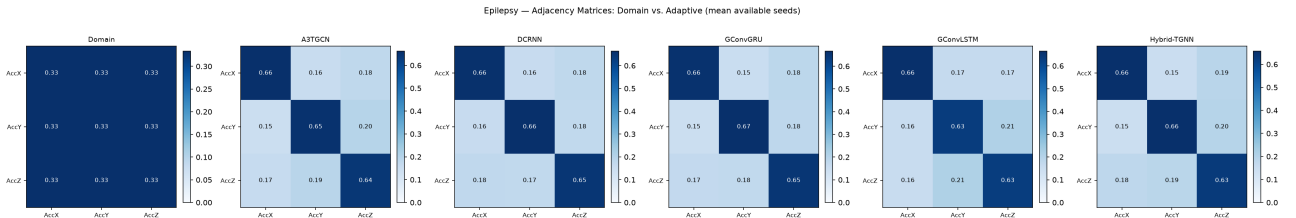


Figure A.3: EPILEPSY — adjacency matrices for the predefined graph and for the learned adaptive graph of each of the five TGNN architectures, estimated on the fixed archive training split. Each cell is one edge weight after the normalisation of Eq. 4.1, averaged over five seeds. Heatmaps rather than network drawings are used because they show *every* weight, including those a network view would threshold away.

DSA · A3TGCN · An adaptive graph helps at 5 nodes and does not at 45
Three independent 45-node constructions, all three negative, none significant

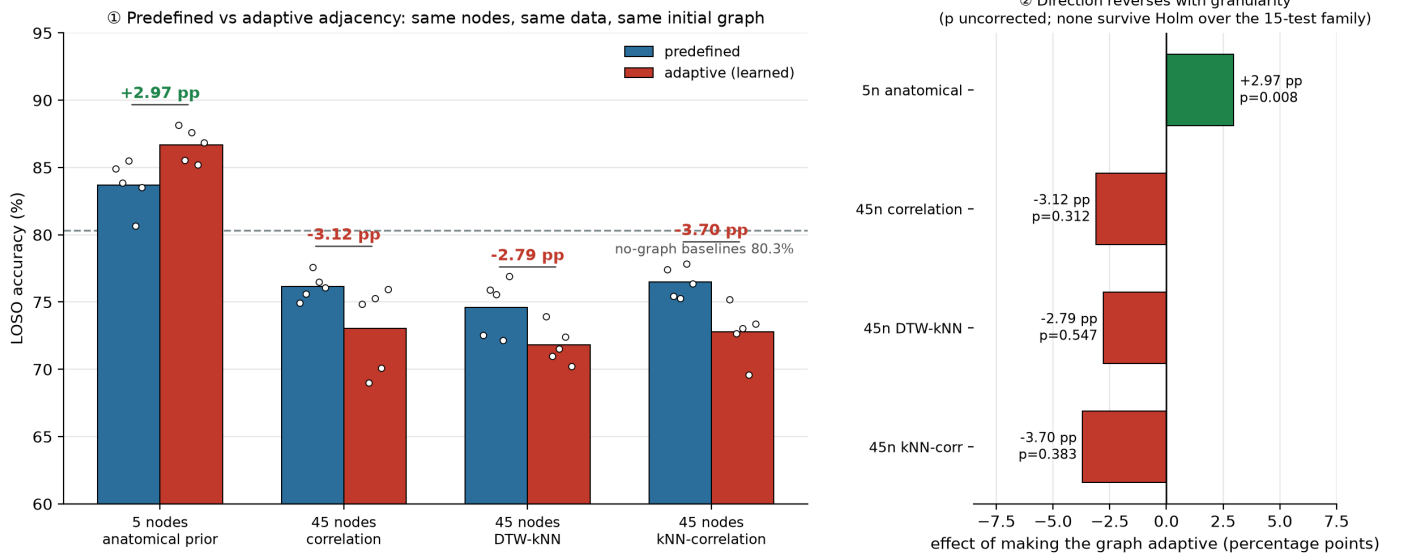


Figure A.4: DSA, A3TGCN. The effect of making the adjacency adaptive rather than predefined, holding the node set, the training data and the initial graph constant within each pair. Hollow points are individual seed means ($n = 5$ per condition, eight LOSO folds each). Left panel, accuracies. Right panel, the paired difference with uncorrected p -values. Values in the right panel are percentage points of accuracy. The Holm-adjusted values over the full sixteen-test family are in Table A.2.

DSA — Best Configuration per Model
(candidate graph strategies only; random-wiring controls excluded)

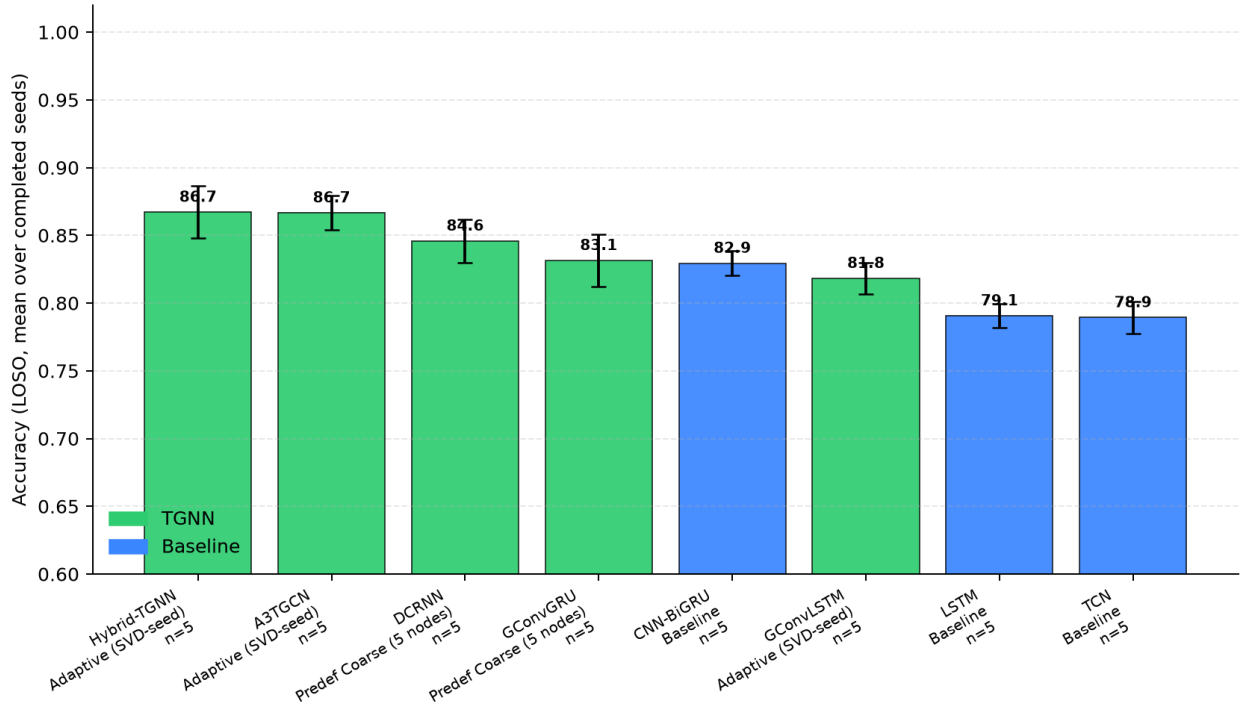


Figure A.5: DSA — the best-performing graph mode for each of the eight architectures, as LOSO accuracy averaged over eight folds and five seeds. This figure isolates one question, namely whether a graph model can beat a sequence model when it is given the graph that suits it best, and it answers that the five TGNN architectures at their best modes exceed all three sequence baselines. It does not address how much the graph choice matters, which is the separate question Fig. 6.17 answers, and the fine-graph configurations that fall below the baselines are by construction absent here.

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